Optimization — Exam Study Agenda

Statements (without solutions), pseudocode, validation tables and exam-style problems

IMCA — Master's in Mathematical and Computational Modeling

Scope: `ch1`, `ch2-v2`, `ch3`, `ch3_13-15`, `ch4`, `ch5`, `ch6`

About this document. Its objective is to prepare the written exam on ch1–ch6: it collects every proof, counterexample and exercise of the notes, together with added practice, pseudocode as implemented in real libraries, validation tables and exam-style problems, organized by unit.

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U0. Taylor's theorem

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Dependencies: none inside this unit.

T1 Proof *ch2-v2, Theorem 1.1, p. 1*

Let $f:\mathbb{R}^n\to\mathbb{R}$ be of class C^1 and $p\in\mathbb{R}^n$. Prove that

$$f(x+p)=f(x)+\int_0^1 \nabla f(x+tp)^{\mathsf T}p\,dt, \tag{T1}$$

$$f(x+p)=f(x)+\nabla f(x+tp)^{\mathsf T}p \quad \text{for some } t\in(0,1). \tag{T2}$$

If moreover f is of class C^2, prove that

$$\nabla f(x+p)=\nabla f(x)+\int_0^1 \nabla^2 f(x+tp)\,p\,dt, \tag{T3}$$

$$f(x+p)=f(x)+\nabla f(x)^{\mathsf T}p+\int_0^1 (1-t)\,p^{\mathsf T}\nabla^2 f(x+tp)\,p\,dt, \tag{T4}$$

$$f(x+p)=f(x)+\nabla f(x)^{\mathsf T}p+\tfrac12\,p^{\mathsf T}\nabla^2 f(x+tp)\,p \quad \text{for some } t\in(0,1). \tag{T5}$$

Explain also why (T3) does not admit, in general, a mean-value form analogous to (T2).

Q1 Exam problem *added, 4 pts*

Taylor bounds for an L-smooth loss. Let $f:\mathbb{R}^n\to\mathbb{R}$ be of class C^2 with $\|\nabla^2 f(x)\|_2\le L$ for all x.

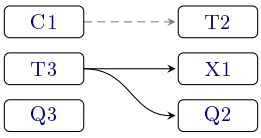
a) Using Taylor's theorem in integral form, $\nabla f(y)=\nabla f(x)+\int_0^1\nabla^2 f(x+t(y-x))(y-x)\,dt$, prove that $\|\nabla f(y)-\nabla f(x)\|\le L\|y-x\|$ for all x,y.

b) Prove the descent lemma: $f(y)\le f(x)+\nabla f(x)^{\mathsf T}(y-x)+\frac{L}{2}\|y-x\|^2$ for all x,y.

c) For the logistic loss $f(w)=\sum_{i=1}^m\big[\ln(1+e^{x_i^{\mathsf T}w})-y_i x_i^{\mathsf T}w\big]$, with $x_i\in\mathbb{R}^n$, $y_i\in\{0,1\}$ and $X=[x_1,\dots,x_m]^{\mathsf T}$, show that $\nabla^2 f(w)=X^{\mathsf T}D(w)X$ with $D(w)$ diagonal and $0\preceq D(w)\preceq\frac14 I$, and deduce that $L=\frac14\|X\|_2^2$ is a valid constant in a) and b).

U1. Convexity

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C1 Counterexample *ch1, p. 3 (version corrected in class)*

Show that the conclusion "every local minimum is global" fails if convexity of f or of the domain is removed.

a) $f(x)=x^3-3x$ on $\mathbb{R}$: prove that $x^*=1$ is a local minimum and that f has no global minimum.

b) Square donut: $\Omega=\{x\in\mathbb{R}^2:\ 1\le\|x\|_\infty\le 2\}$ and $f(x_1,x_2)=x_1$. Prove that f is convex, that Ω is not convex, that $(1,0)$ is a local minimum of f on Ω, and that it is not a global minimum.

T2 Proof *ch1, Theorem 2.1, p. 2*

Let $C \subseteq \mathbb{R}^n$ be convex and nonempty and $f : C \to \mathbb{R}$ convex. Prove that if $x^* \in C$ is a local minimum of f on C, then it is a global minimum of f on C.

T3 Proof *ch1, Proposition 2.1, pp. 3–4*

Let $C \subseteq \mathbb{R}^n$ be open and convex and $f : C \to \mathbb{R}$ of class C^1.

a) Prove that f is convex on C if and only if $f(y) \geq f(x) + \nabla f(x)^{\mathsf{T}}(y - x)$ for all $x, y \in C$.

b) If f is C^2, prove that f is convex on C if and only if $\nabla^2 f(x) \succeq 0$ for all $x \in C$.

Indicate where in the proof the openness of C is used.

X1 Proposed exercise *ch1, p. 8*

For $\delta > 0$, the Huber loss is

$$\rho_\delta(t) = \begin{cases} \tfrac{1}{2}t^2, & |t| \leq \delta, \\ \delta|t| - \tfrac{1}{2}\delta^2, & |t| > \delta. \end{cases}$$

Prove that ρ_δ is convex and of class C^1 on $\mathbb{R}$, and find $\rho_\delta'(t)$.

Q2 Exam problem *added, 4 pts*

Log-sum-exp and affine composition. Let $\operatorname{lse}(z) = \ln \sum_{i=1}^m e^{z_i}$ for $z \in \mathbb{R}^m$, and $\pi_i(z) = e^{z_i} / \sum_j e^{z_j}$.

a) Show that $\nabla \operatorname{lse}(z) = \pi(z)$ and $\nabla^2 \operatorname{lse}(z) = \operatorname{diag}(\pi) - \pi\pi^{\mathsf{T}}$, and prove $\nabla^2 \operatorname{lse}(z) \succeq 0$ by writing $v^{\mathsf{T}} \nabla^2 \operatorname{lse}(z) v$ as a variance (or with Cauchy–Schwarz).

b) Let $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$. Prove that $f(x) = \operatorname{lse}(Ax + b)$ is convex on $\mathbb{R}^n$ and write $\nabla^2 f(x)$ in terms of A and $\pi(Ax + b)$.

c) Show that lse is not strictly convex: find $v \neq 0$ with $\operatorname{lse}(z + tv)$ affine in t for every z. Give a condition on A under which f in b) is strictly convex.

Q3 Exam problem *added, 4 pts*

Operations that preserve convexity.

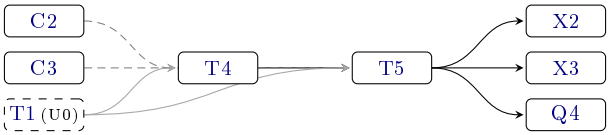
a) Let $C \subseteq \mathbb{R}^n$ be nonempty. Prove that the support function $\sigma_C(y) = \sup_{x \in C} y^{\mathsf{T}} x$ is convex on the set where it is finite.

b) Let $\mathbb{S}^n$ be the space of symmetric $n \times n$ matrices. Prove that $\lambda_{\max}(Y) = \sup_{\|v\|=1} v^{\mathsf{T}} Y v$ is convex on $\mathbb{S}^n$, and that $\lambda_{\min}$ is concave.

c) Let C be nonempty, closed and convex. Prove that the distance $d_C(x) = \min_{y \in C} \|x - y\|$ is convex, starting from the definition and the convexity of $(x, y) \mapsto \|x - y\|$ on $\mathbb{R}^n \times C$.

U2. Unconstrained optimality conditions

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C2 Counterexample *ch1, p. 6*

Show that $\nabla f(x^*) = 0$ is not sufficient for a local minimum:

a) $f(x) = x^3$ at $x^* = 0$.

b) $f(x_1, x_2) = x_1^2 - x_2^2$ at $x^* = (0, 0)$ (saddle point).

T4 Proof *ch1, Theorem 3.1 (FONC), p. 5*

Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuously differentiable on an open neighbourhood of x^*. Prove that if x^* is a local minimum of f, then $\nabla f(x^*) = 0$.

C3 Counterexample *ch1, p. 7*

Show that $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*) \succeq 0$ do not imply a local minimum:

a) $f(x) = x^3$ at $x^* = 0$.

b) $f(x) = -x^4$ at $x^* = 0$ (strict local maximum).

T5 Proof *ch1, Theorem 3.2, pp. 6–7*

Let $f : \mathbb{R}^n \to \mathbb{R}$ be of class C^2 on a neighbourhood of x^*.

a) (SONC) If x^* is a local minimum, then $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*) \succeq 0$.

b) (SOSC) If $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*) \succ 0$, then x^* is a strict local minimum.

X2 Proposed exercise *ch1, p. 8*

Let $f(x_1, x_2) = 2x_1^2 + x_2^2 - 2x_1x_2 + 2x_1 - 4x_2$.

a) Write $f(x) = \frac{1}{2}x^\mathsf{T} Ax + b^\mathsf{T} x$.

b) Compute $\nabla f(x)$ and $\nabla^2 f(x)$.

c) Find all stationary points and classify them with the second-order necessary condition (SONC: at a local minimum, $\nabla f = 0$ and $\nabla^2 f \succeq 0$) and the second-order sufficient condition (SOSC: $\nabla f = 0$ and $\nabla^2 f \succ 0$ give a strict local minimum).

X3 Proposed exercise *ch1, p. 8*

Let $f(x_1, x_2) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$ (Rosenbrock).

a) Prove that $x^* = (1, 1)^\mathsf{T}$ is the only stationary point.

b) Verify $\nabla^2 f(x^*) \succ 0$ and conclude that x^* is a strict global minimum.

c) Show that the Hessian is indefinite or singular in some regions of the plane.

Note. In b), SOSC only gives a strict local minimum; the global character needs a separate argument.

Q4 Exam problem *added, 4 pts*

Stationary points of a quartic. Let $f(x_1, x_2) = x_1^4 + x_2^4 - 4x_1x_2 + 1$.

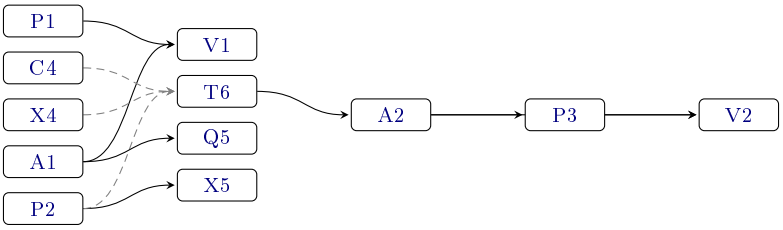
a) Find all stationary points of f.

b) Classify each one with the second-order conditions (minimum, maximum or saddle).

c) Prove that f is coercive (use $x_1^4 + x_2^4 \geq \frac{1}{2}(x_1^2 + x_2^2)^2$ and $4|x_1x_2| \leq 2(x_1^2 + x_2^2)$), deduce that a global minimizer exists, and find the global minimum value. Then show, with $g(x_1, x_2) = x_1^2 + x_2^3$ at the origin, that the second-order necessary conditions can hold at a point that is not a local minimizer.

U3. Line search: Armijo and Wolfe

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C4 Counterexample

ch2-v2, p. 4

Definitions.

- *Wolfe conditions* for $\phi(\alpha) = f(x_k + \alpha p_k)$ and $0 < c_1 < c_2 < 1$: Armijo $\phi(\alpha) \le \phi(0) + c_1 \alpha\, \phi'(0)$; weak curvature $\phi'(\alpha) \ge c_2\, \phi'(0)$; strong curvature $|\phi'(\alpha)| \le c_2 |\phi'(0)|$. Weak Wolfe = Armijo + weak curvature; strong Wolfe = Armijo + strong curvature.

Problem. Let $f(x) = x^2$, $x_0 = 2$, $p_k = -1$ and $\alpha_k = (1/2)^{k+1}$, with $x_{k+1} = x_k + \alpha_k p_k$.

a) Prove that p_k is a descent direction and that $f(x_{k+1}) < f(x_k)$ for all k.

b) Prove that $x_k \to 1$, which is not a stationary point.

c) Determine which of the conditions (Armijo, curvature) these steps violate.

X4 Proposed exercise

ch2-v2, p. 16

Definitions.

- *Wolfe conditions* for $\phi(\alpha) = f(x_k + \alpha p_k)$ and $0 < c_1 < c_2 < 1$: Armijo $\phi(\alpha) \le \phi(0) + c_1 \alpha\, \phi'(0)$; weak curvature $\phi'(\alpha) \ge c_2\, \phi'(0)$; strong curvature $|\phi'(\alpha)| \le c_2 |\phi'(0)|$. Weak Wolfe = Armijo + weak curvature; strong Wolfe = Armijo + strong curvature.

Problem. Let $f(x) = x^2$, $p_k = -\operatorname{sign}(x_k)$ and $\alpha_k = 2 + 3 \cdot 2^{-(k+1)}$.

a) Show that p_k is a descent direction for all k.

b) Prove that the sequence oscillates and does not converge to $x^* = 0$, and explain which condition fails.

Note. The original statement does not fix x_0; take $x_0 = 2$.

P1 Practice

added

Definitions.

- *Wolfe conditions* for $\phi(\alpha) = f(x_k + \alpha p_k)$ and $0 < c_1 < c_2 < 1$: Armijo $\phi(\alpha) \le \phi(0) + c_1 \alpha\, \phi'(0)$; weak curvature $\phi'(\alpha) \ge c_2\, \phi'(0)$; strong curvature $|\phi'(\alpha)| \le c_2 |\phi'(0)|$. Weak Wolfe = Armijo + weak curvature; strong Wolfe = Armijo + strong curvature.

Problem. Backtracking with Armijo. Let $f(x_1, x_2) = 10x_1^2 + x_2^2$, $x_0 = (1, 1)^{\mathsf{T}}$, $p_0 = -\nabla f(x_0)$ and $\phi(\alpha) = f(x_0 + \alpha p_0)$.

a) Write $\phi(\alpha)$ and $\phi'(0)$, and find the exact minimizer α^* of ϕ.

b) Run A1 by hand with $\alpha_{\text{init}} = 1$, $c_1 = 10^{-4}$, $\rho_{\text{lo}} = 0.1$, $\rho_{\text{hi}} = 0.5$: list each α tried, $\phi(\alpha)$, the Armijo bound, the interpolated value α_t and the safeguarded next trial.

c) Compare with plain halving ($\alpha \leftarrow \alpha/2$ from $\alpha = 1$): how many evaluations of f does each need? Explain why the interpolated value α_t is not used as it is.

d) For the accepted step, decide whether it satisfies weak Wolfe and strong Wolfe with $c_2 = 0.9$ and with $c_2 = 0.1$.

A1 Pseudocode *line search: Armijo backtracking with safeguarded interpolation*

Backtracking$(\phi;\ \alpha_{\text{init}}, c_1, \rho_{\text{lo}}, \rho_{\text{hi}}, j_{\max})$ $\quad \phi(\alpha) = f(x+\alpha p)$, $\phi'(0) = \nabla f(x)^{\mathsf{T}} p < 0$

1: $\alpha \leftarrow \alpha_{\text{init}}$; $\quad \phi_\alpha \leftarrow \phi(\alpha)$
2: **for** $j = 1, \dots, j_{\max}$ **do**
3: $\quad$ **if** $\phi_\alpha \le \phi(0) + c_1\,\alpha\,\phi'(0)$ **then return** (α, ϕ_α) (Armijo holds)
4: $\quad$ **if** $j = 1$ **then** $\alpha_t \leftarrow -\dfrac{\phi'(0)\,\alpha^2}{2\,[\phi_\alpha - \phi(0) - \phi'(0)\,\alpha]}$ (quadratic through $\phi(0), \phi'(0), \phi_\alpha$)
5: $\quad$ **else** $\alpha_t \leftarrow \dfrac{-b + \sqrt{b^2 - 3a\,\phi'(0)}}{3a}$ (cubic through $\phi(0), \phi'(0), \phi_\alpha, \phi_{\text{prev}}$)
6: $\quad \alpha_{\text{prev}} \leftarrow \alpha$; $\quad \phi_{\text{prev}} \leftarrow \phi_\alpha$; $\quad \alpha \leftarrow \min\{\max\{\alpha_t,\ \rho_{\text{lo}}\,\alpha\},\ \rho_{\text{hi}}\,\alpha\}$ (safeguard)
7: $\quad \phi_\alpha \leftarrow \phi(\alpha)$
8: **return** failure (p not a usable descent direction, or f too noisy)

Cubic coefficients (line 5), with $r = \phi_\alpha - \phi(0) - \phi'(0)\alpha$ and $r_{\text{p}} = \phi_{\text{prev}} - \phi(0) - \phi'(0)\alpha_{\text{prev}}$:

$$\begin{pmatrix} a \\ b \end{pmatrix} = \frac{1}{\alpha - \alpha_{\text{prev}}} \begin{pmatrix} 1/\alpha^2 & -1/\alpha_{\text{prev}}^2 \\ -\alpha_{\text{prev}}/\alpha^2 & \alpha/\alpha_{\text{prev}}^2 \end{pmatrix} \begin{pmatrix} r \\ r_{\text{p}} \end{pmatrix}.$$

Defaults: $c_1 = 10^{-4}$, $\rho_{\text{lo}} = 0.1$, $\rho_{\text{hi}} = 0.5$, $j_{\max} \approx 30$, and $\alpha_{\text{init}} = 1$ for Newton and quasi-Newton directions. This is the backtracking of Dennis–Schnabel (Alg. A6.3.1), also used by `BackTracking` in Julia's LineSearches. Each trial costs one evaluation of f and no gradient. The interpolation reuses values already computed, so a rejection usually costs one extra evaluation. The safeguard keeps each new trial between 10% and 50% of the previous one, so the loop ends as plain halving does (the case $\rho_{\text{lo}} = \rho_{\text{hi}} = \frac{1}{2}$). The value ϕ_α is returned so that the caller does not evaluate f again.

V1 Validation table *A1; solves P1*

$f(x) = 10x_1^2 + x_2^2$, $x = (1,1)^{\mathsf{T}}$, $p = -\nabla f(x) = (-20,-2)^{\mathsf{T}}$, $\phi(\alpha) = 10(1-20\alpha)^2 + (1-2\alpha)^2$, $\phi(0) = 11$, $\phi'(0) = -404$, $\phi'' = 8008$, so $\alpha^* = \frac{404}{8008} = \frac{101}{2002} \approx 0.050450$. A1 with $\alpha_{\text{init}} = 1$, $c_1 = 10^{-4}$, $\rho_{\text{lo}} = 0.1$, $\rho_{\text{hi}} = 0.5$.

trial	α	$\phi(\alpha)$	bound $\phi(0) + c_1\alpha\phi'(0)$	decision	next trial
1	1	3611	10.9596	reject	$\alpha_t = \frac{101}{2002} \approx 0.050450$, clipped to [0.1, 0.5]: 0.1
2	0.1	10.64	10.99596	**accept**	

Plain halving tries $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}$ and accepts $\alpha = 0.0625$ with $\phi = 1.390625$ (5 evaluations). Since ϕ is quadratic, α_t is its exact minimizer, but it is less than 10% of the rejected step, so the safeguard replaces it by 0.1. Curvature at $\alpha = 0.1$: $\phi'(0.1) = 396.8$. Weak Wolfe holds for every c_2 ($396.8 \ge c_2 \cdot (-404)$); strong Wolfe fails with $c_2 = 0.9$ ($396.8 > 363.6$) and with $c_2 = 0.1$.

What it shows. Interpolation needs 2 evaluations where halving needs 5. The lower safeguard ρ_{lo} is a deliberate trade: it gives up the exact step here, but it protects against tiny steps when the model is poor. Armijo alone does not control the curvature condition.

P2 PRACTICE *added*

Definitions.

- *Wolfe conditions* for $\phi(\alpha) = f(x_k + \alpha p_k)$ and $0 < c_1 < c_2 < 1$: Armijo $\phi(\alpha) \le \phi(0) + c_1 \alpha\, \phi'(0)$; weak curvature $\phi'(\alpha) \ge c_2\, \phi'(0)$; strong curvature $|\phi'(\alpha)| \le c_2 |\phi'(0)|$. Weak Wolfe = Armijo + weak curvature; strong Wolfe = Armijo + strong curvature.

Problem. Let $\phi(\alpha) = \phi(0) + \phi'(0)\alpha + \frac{1}{2} a \alpha^2$ with $a > 0$ and $\phi'(0) < 0$, and let α^* be its minimizer.

a) Find, in terms of c_1, c_2 and α^*, the set of $\alpha > 0$ satisfying Armijo; the set satisfying weak curvature; and the set satisfying strong curvature.

b) Deduce for which values of c_1 the exact step α^* satisfies Armijo.

c) Deduce that the set of Wolfe steps is nonempty for every $0 < c_1 < c_2 < 1$.

T6 PROOF *ch2-v2, Lemma 2.1, pp. 3–4*

Definitions.

- *Wolfe conditions* for $\phi(\alpha) = f(x_k + \alpha p_k)$ and $0 < c_1 < c_2 < 1$: Armijo $\phi(\alpha) \le \phi(0) + c_1 \alpha\, \phi'(0)$; weak curvature $\phi'(\alpha) \ge c_2\, \phi'(0)$; strong curvature $|\phi'(\alpha)| \le c_2 |\phi'(0)|$. Weak Wolfe = Armijo + weak curvature; strong Wolfe = Armijo + strong curvature.

Setting. $f : \mathbb{R}^n \to \mathbb{R}$ of class C^1, bounded below on the ray $\{x_k + \alpha p_k : \alpha > 0\}$; p_k a descent direction ($\nabla f(x_k)^{\mathsf{T}} p_k < 0$); $0 < c_1 < c_2 < 1$.

Problem. Prove that there is an interval of steps $\alpha > 0$ on which both the weak and the strong Wolfe conditions hold.

A2 PSEUDOCODE *line search: strong Wolfe, bracketing and zoom (N&W, Alg. 3.5–3.6)*

StrongWolfe$(\phi, \phi';\ \alpha_1, c_1, c_2, \alpha_{\max})$ $\quad \phi(\alpha) = f(x + \alpha p)$, $\phi'(\alpha) = \nabla f(x + \alpha p)^{\mathsf{T}} p$, $\phi'(0) < 0$

$\alpha_0 \leftarrow 0$
for $i = 1, 2, \ldots, i_{\max}$ **do** (bracketing phase)
  evaluate $\phi(\alpha_i)$
  if $\phi(\alpha_i) > \phi(0) + c_1 \alpha_i \phi'(0)$ **or** ($i > 1$ **and** $\phi(\alpha_i) \ge \phi(\alpha_{i-1})$) **then return** Zoom$(\alpha_{i-1}, \alpha_i)$
  evaluate $\phi'(\alpha_i)$; **if** $|\phi'(\alpha_i)| \le -c_2 \phi'(0)$ **then return** α_i
  if $\phi'(\alpha_i) \ge 0$ **then return** Zoom(α_i, α_{i-1})
  $\alpha_{i+1} \leftarrow \min\{2\alpha_i,\ \alpha_{\max}\}$

Zoom$(\alpha_{\mathrm{lo}}, \alpha_{\mathrm{hi}})$: $\alpha_{\mathrm{rec}} \leftarrow 0$ (α_{lo}: best Armijo point so far; $\phi'(\alpha_{\mathrm{lo}})(\alpha_{\mathrm{hi}} - \alpha_{\mathrm{lo}}) < 0$)
for $j = 0, 1, \ldots, j_{\max}$ **do**
  $[a, b] \leftarrow$ interval with endpoints $\alpha_{\mathrm{lo}}, \alpha_{\mathrm{hi}}$; $\delta \leftarrow b - a$
  if $j \ge 1$ **then** $\alpha_j \leftarrow$ minimizer of the cubic through $\phi(\alpha_{\mathrm{lo}}), \phi'(\alpha_{\mathrm{lo}}), \phi(\alpha_{\mathrm{hi}}), \phi(\alpha_{\mathrm{rec}})$
  if $j = 0$ **or** $\alpha_j \notin [a + 0.2\delta,\ b - 0.2\delta]$ **then** $\alpha_j \leftarrow$ minimizer of the quadratic through $\phi(\alpha_{\mathrm{lo}}), \phi'(\alpha_{\mathrm{lo}}), \phi(\alpha_{\mathrm{hi}})$
  if $\alpha_j \notin [a + 0.1\delta,\ b - 0.1\delta]$ **then** $\alpha_j \leftarrow \frac{1}{2}(\alpha_{\mathrm{lo}} + \alpha_{\mathrm{hi}})$ (bisection)
  evaluate $\phi(\alpha_j)$
  if $\phi(\alpha_j) > \phi(0) + c_1 \alpha_j \phi'(0)$ **or** $\phi(\alpha_j) \ge \phi(\alpha_{\mathrm{lo}})$ **then**
    $\alpha_{\mathrm{rec}} \leftarrow \alpha_{\mathrm{hi}}$; $\alpha_{\mathrm{hi}} \leftarrow \alpha_j$ (step too long)
  else
    evaluate $\phi'(\alpha_j)$; **if** $|\phi'(\alpha_j)| \le -c_2 \phi'(0)$ **then return** α_j
    if $\phi'(\alpha_j)(\alpha_{\mathrm{hi}} - \alpha_{\mathrm{lo}}) \ge 0$ **then** $\alpha_{\mathrm{rec}} \leftarrow \alpha_{\mathrm{hi}}$; $\alpha_{\mathrm{hi}} \leftarrow \alpha_{\mathrm{lo}}$ **else** $\alpha_{\mathrm{rec}} \leftarrow \alpha_{\mathrm{lo}}$
    $\alpha_{\mathrm{lo}} \leftarrow \alpha_j$
return failure

Defaults: $c_1 = 10^{-4}$; $c_2 = 0.9$ for Newton and quasi-Newton directions, $c_2 = 0.1$ for nonlinear CG; $i_{\max} = j_{\max} = 10$; $\alpha_1 = 1$ for quasi-Newton (the caller chooses α_1, see A3 and A5). This is `line_search_wolfe2` in SciPy (the fallback search of its BFGS and CG) and `StrongWolfe` in Julia's LineSearches. MINPACK-2's Moré–Thuente search (`dcsrch`), used by L-BFGS-B and by SciPy's `line_search_wolfe1`, follows the same bracket-then-zoom plan with a more elaborate interpolation. The interpolated trial is used only if it lies well inside the bracket (away from both endpoints); otherwise the interval is bisected, which guarantees that the bracket shrinks. A step that satisfies the strong Wolfe conditions also satisfies the weak ones.

P3 PRACTICE *added*

Strong Wolfe search. Let $f(x_1, x_2) = 10x_1^2 + x_2^2$, $x_0 = (1, 1)^\mathsf{T}$, $p_0 = -\nabla f(x_0)$, $\phi(\alpha) = f(x_0 + \alpha p_0)$, $c_1 = 10^{-4}$ and $c_2 = 0.9$. The strong Wolfe conditions are $\phi(\alpha) \le \phi(0) + c_1\alpha\,\phi'(0)$ (Armijo) and $|\phi'(\alpha)| \le c_2|\phi'(0)|$ (strong curvature).

a) Run A2 by hand from $\alpha_1 = 1$. For each trial record the phase (bracketing or zoom), the bracket $(\alpha_{\text{lo}}, \alpha_{\text{hi}})$, how the trial was obtained (quadratic, cubic or bisection), ϕ, ϕ' and the outcome.

b) Repeat from $\alpha_1 = 10^{-3}$. Which condition drives the search in each case?

c) Prove that, while the bracketing phase has not stopped, $\phi(\alpha_i)$ decreases and $\phi'(\alpha_i) < 0$; deduce that when Zoom is called the interval contains steps satisfying the strong Wolfe conditions (use T6).

V2 VALIDATION TABLE *A2; solves P3 a) and b)*

$f(x) = 10x_1^2 + x_2^2$, $x = (1, 1)^\mathsf{T}$, $p = (-20, -2)^\mathsf{T}$; $\phi(\alpha) = 10(1 - 20\alpha)^2 + (1 - 2\alpha)^2$, $\phi(0) = 11$, $\phi'(0) = -404$, $c_2|\phi'(0)| = 363.6$, $\phi'(\alpha) = -400(1 - 20\alpha) - 4(1 - 2\alpha)$.

phase	$(\alpha_{\text{lo}}, \alpha_{\text{hi}})$	trial from	α	$\phi(\alpha)$	$\phi'(\alpha)$	outcome
			From $\alpha_1 = 1$			
bracket		α_1	1	3611		Armijo fails: Zoom$(0, 1)$
zoom $j = 0$	$(0,\ 1)$	bisection	0.5	810		Armijo fails: $\alpha_{\text{hi}} \leftarrow 0.5$
zoom $j = 1$	$(0,\ 0.5)$	quadratic	$\frac{101}{2002} \approx 0.050450$	0.809191	≈ 0	**accept**
			From $\alpha_1 = 10^{-3}$			
bracket		α_1	0.001	10.600004	-395.992	curvature fails: double
bracket		$2\alpha_1$	0.002	10.208016	-387.984	curvature fails: double
bracket		$2\alpha_2$	0.004	9.448064	-371.968	curvature fails: double
bracket		$2\alpha_3$	0.008	8.024256	-339.936	**accept**

In zoom $j = 0$ the quadratic through $\phi(0), \phi'(0), \phi(1)$ gives $\alpha = 0.050450$, which lies within 10% of the endpoint 0 of $[0, 1]$, so it is replaced by the midpoint. At $j = 1$ the same quadratic minimizer lies inside $[0.05, 0.45]$ and is used; it is the exact minimizer of ϕ, so $\phi' = 0$. Reproduced with SciPy's `scalar_search_wolfe2`.

What it shows. From a step that is too long, Armijo drives the search and interpolation reaches the exact minimizer in two zoom steps. From a step that is too small, Armijo always holds and it is the curvature condition that forces the step to grow (by doubling).

X5 Proposed exercise *ch2-v2, p. 15*

Let $f(x_1, x_2) = 10x_1^2 + x_2^2$, $x_0 = (1, 1)^\mathsf{T}$ and $p_0 = -\nabla f(x_0)$.

a) Compute the exact step α^* minimizing $\phi(\alpha) = f(x_0 + \alpha p_0)$.

b) Prove that, for every $c_1 \in (0, 0.5)$ and $c_2 \in (c_1, 1)$, α^* satisfies the weak Wolfe conditions: $f(x_k + \alpha_k p_k) \le f(x_k) + c_1 \alpha_k \nabla f(x_k)^\mathsf{T} p_k$ and $\nabla f(x_k + \alpha_k p_k)^\mathsf{T} p_k \ge c_2 \nabla f(x_k)^\mathsf{T} p_k$, with $0 < c_1 < c_2 < 1$ with $k = 0$.

Q5 Exam problem *added, 4 pts*

Armijo steps for an L-smooth function. Let $f : \mathbb{R}^n \to \mathbb{R}$ be of class C^1 with ∇f Lipschitz with constant L, $g = \nabla f(x)$, p a descent direction ($g^\mathsf{T} p < 0$) and $c_1 \in (0, 1)$.

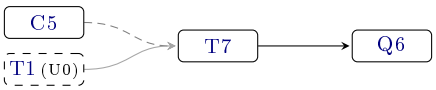
a) Using $f(x + \alpha p) \le f(x) + \alpha g^\mathsf{T} p + \frac{L}{2}\alpha^2 \|p\|^2$, prove that the Armijo condition $f(x + \alpha p) \le f(x) + c_1 \alpha g^\mathsf{T} p$ holds for every $\alpha \in (0, \bar\alpha]$, with $\bar\alpha = \dfrac{2(1 - c_1)(-g^\mathsf{T} p)}{L\|p\|^2}$.

b) Consider a backtracking search that starts at α_{init} and, after each rejection, chooses the next trial in $[\rho_{\text{lo}}\alpha, \rho_{\text{hi}}\alpha]$ with $0 < \rho_{\text{lo}} \le \rho_{\text{hi}} < 1$ (A1). Prove that it returns a step $\alpha \ge \min\{\alpha_{\text{init}}, \rho_{\text{lo}}\bar\alpha\}$, and bound the number of rejections in terms of $\alpha_{\text{init}}/\bar\alpha$ and ρ_{hi}.

c) For $f(x) = \frac{1}{2} x^\mathsf{T} Q x$ with $Q \succ 0$ and $p = -g$, compute exactly the largest α that satisfies Armijo and compare it with $\bar\alpha$ when $L = \lambda_{\max}(Q)$. When are they equal?

U4. Zoutendijk's theorem

↑ Index



C5 Counterexample / observation *ch2-v2, p. 5*

Let $f(x) = |x|^{4/3}$.

a) Prove that f is C^1 and that f' is not Lipschitz on any neighbourhood of 0.

b) Zoutendijk's theorem states: if f is of class C^1, bounded below, with ∇f Lipschitz with constant L, and $x_{k+1} = x_k + \alpha_k p_k$ with descent directions p_k and steps satisfying the weak Wolfe conditions, then $\sum_k \cos^2\theta_k \|\nabla f(x_k)\|^2 < \infty$, where $\cos\theta_k = -\nabla f(x_k)^\mathsf{T} p_k / (\|\nabla f(x_k)\|\|p_k\|)$. Its proof bounds α_k from below using $(\nabla f(x_{k+1}) - \nabla f(x_k))^\mathsf{T} p_k \le L\alpha_k \|p_k\|^2$. Identify which step is no longer valid for this f.

c) Discuss whether the conclusion of the theorem actually fails for this f (e.g. for steepest descent with Wolfe steps from $x_0 = 1$).

Note. The notes only show that the proof breaks, not that the conclusion is false.

T7 Proof *ch2-v2, Theorem 3.1, pp. 4–5; corollary p. 6*

Setting. $f : \mathbb{R}^n \to \mathbb{R}$ of class C^1, bounded below, with ∇f Lipschitz with constant $L > 0$. Iterates $x_{k+1} = x_k + \alpha_k p_k$, where p_k is a descent direction and α_k satisfies the weak Wolfe conditions, $f(x_k + \alpha_k p_k) \le f(x_k) + c_1 \alpha_k \nabla f(x_k)^\mathsf{T} p_k$ and $\nabla f(x_k + \alpha_k p_k)^\mathsf{T} p_k \ge c_2 \nabla f(x_k)^\mathsf{T} p_k$, with $0 < c_1 < c_2 < 1$. Let $\cos\theta_k = \dfrac{-\nabla f(x_k)^\mathsf{T} p_k}{\|\nabla f(x_k)\|\,\|p_k\|}$.

Problem.

a) Prove Zoutendijk's condition $\displaystyle\sum_{k=0}^{\infty} \cos^2\theta_k \,\|\nabla f(x_k)\|^2 < \infty$.

b) Deduce that, for steepest descent ($p_k = -\nabla f(x_k)$), $\lim_{k\to\infty}\|\nabla f(x_k)\| = 0$.

Q6 Exam problem *added, 4 pts*

Zoutendijk and the angle condition. Let $f : \mathbb{R}^n \to \mathbb{R}$ be of class C^1, bounded below, with ∇f Lipschitz with constant L. Let $x_{k+1} = x_k + \alpha_k p_k$, where p_k is a descent direction, $g_k = \nabla f(x_k)$, $\cos\theta_k = -g_k^\mathsf{T} p_k/(\|g_k\|\|p_k\|)$, and α_k satisfies the weak Wolfe conditions with $0 < c_1 < c_2 < 1$.

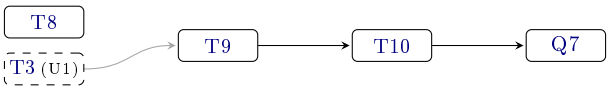
a) From the curvature condition, prove $\alpha_k \geq \dfrac{(c_2-1)\, g_k^\mathsf{T} p_k}{L\|p_k\|^2}$; combine it with Armijo to obtain $f(x_{k+1}) \leq f(x_k) - c\cos^2\theta_k\|g_k\|^2$ with $c = c_1(1-c_2)/L$, and conclude $\sum_k \cos^2\theta_k\|g_k\|^2 < \infty$.

b) Let $p_k = -B_k^{-1} g_k$ with B_k symmetric and $mI \preceq B_k \preceq MI$, $0 < m \leq M$. Prove $\cos\theta_k \geq m/M$ and deduce $\lim_k \|g_k\| = 0$.

c) Apply b) to the modified Newton method $B_k = \nabla^2 f(x_k) + \tau_k I$ (A4) when the iterates stay in a compact set where $\|\nabla^2 f\| \leq \beta$, $\tau_k \leq \bar\tau$ and $\lambda_{\min}(B_k) \geq \delta > 0$: give explicit m and M.

U5. Strict and strong convexity

↑Index



T8 Proof *ch2-v2, Theorem 4.1, p. 6*

Let $f : \mathbb{R}^n \to \mathbb{R}$ be of class C^1 and strictly convex, i.e. $f(y) > f(x) + \nabla f(x)^\mathsf{T}(y-x)$ for all $x \neq y$. Prove that if f has a global minimum, it is unique.

T9 Proof *ch2-v2, Theorem 4.2, pp. 6–7*

Let $f : \mathbb{R}^n \to \mathbb{R}$ be of class C^2 and $m > 0$. Prove that

$$f(y) \geq f(x) + \nabla f(x)^\mathsf{T}(y-x) + \tfrac{m}{2}\|y-x\|^2 \ \ \forall x, y \iff y^\mathsf{T}\nabla^2 f(x)\, y \geq m\|y\|^2 \ \ \forall x, y.$$

T10 Proof *ch2-v2, Theorem 4.3, pp. 7–8*

Definitions.

- *Strong convexity:* f is strongly convex with parameter $m > 0$ if $f(y) \geq f(x) + \nabla f(x)^\mathsf{T}(y-x) + \frac{m}{2}\|y-x\|^2$ for all x, y (for f of class C^2 this is equivalent to $\nabla^2 f(x) \succeq mI$ for all x).

Problem. Let $f : \mathbb{R}^n \to \mathbb{R}$ be of class C^1; Prove that if f is strongly convex with parameter m, then f is coercive ($f(x) \to \infty$ as $\|x\| \to \infty$) and has a unique global minimum.

Q7 Exam problem *added, 4 pts*

Definitions.

- *Strong convexity:* f is strongly convex with parameter $m > 0$ if $f(y) \geq f(x) + \nabla f(x)^\mathsf{T}(y-x) + \frac{m}{2}\|y-x\|^2$ for all x, y (for f of class C^2 this is equivalent to $\nabla^2 f(x) \succeq mI$ for all x).

Problem. Strong convexity and a certified stopping test. Let $X \in \mathbb{R}^{m\times n}$, $y \in \mathbb{R}^m$, $\mu > 0$ and $f(w) = \frac{1}{2}\|Xw - y\|^2 + \frac{\mu}{2}\|w\|^2$ (ridge regression).

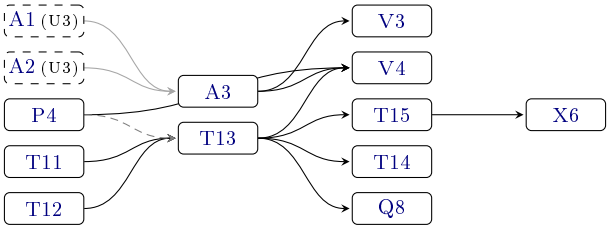
a) Prove that f is strongly convex with parameter $m_f = \mu + \lambda_{\min}(X^\mathsf{T}X)$ and that ∇f is Lipschitz with constant $L_f = \mu + \lambda_{\max}(X^\mathsf{T}X)$.

b) Let h be of class C^1 and strongly convex with parameter $m > 0$, with minimizer x^*. Prove that $h(x) - h(x^*) \leq \frac{1}{2m}\|\nabla h(x)\|^2$ and $\|x - x^*\| \leq \frac{1}{m}\|\nabla h(x)\|$ for all x.

c) Use b) to write a stopping test, computable from $\nabla f(w_k)$ only, that guarantees $f(w_k) - f(w^*) \le \epsilon$ for the ridge problem, and explain why it fails when $\mu = 0$ and X has dependent columns.

U6. Rate of convergence of steepest descent

↑Index



P4 Practice *added*

Let $f(x_1, x_2) = 10x_1^2 + x_2^2$, with Hessian $Q = \mathrm{diag}(20, 2)$ and $\kappa = \lambda_{\max}(Q)/\lambda_{\min}(Q) = 10$, and steepest descent $x_{k+1} = x_k - \alpha_k \nabla f(x_k)$ with exact line search. The minimizer is $x^* = 0$.

a) From $x_0 = (1, 1)^\mathsf{T}$, compute x_1 and x_2, verify that $\nabla f(x_1) \perp \nabla f(x_0)$, and compare the ratio $\big(f(x_{k+1}) - f(x^*)\big)/\big(f(x_k) - f(x^*)\big)$ with the bound $\left(\frac{\kappa-1}{\kappa+1}\right)^2$.

b) Repeat from $x_0 = (1, 10)^\mathsf{T}$ and verify that the bound is attained with equality.

c) Describe the geometric pattern of the iterates in both cases.

A3 Pseudocode *steepest descent with backtracking*

SteepestDescent$(f,\ \nabla f,\ x_0;\ \varepsilon, k_{\max})$

1: $f_0 \leftarrow f(x_0)$; $\quad g_0 \leftarrow \nabla f(x_0)$
2: **for** $k = 0, 1, \ldots, k_{\max}$ **do**
3: $\quad$ **if** $\|g_k\| \le \varepsilon$ **then return** x_k
4: $\quad p_k \leftarrow -g_k$
5: $\quad$ **if** $k = 0$ **then** $\alpha_{\text{init}} \leftarrow \min\{1,\ 1/\|g_0\|\}$ **else** $\alpha_{\text{init}} \leftarrow \min\Big\{1,\ \dfrac{1.01 \cdot 2\,(f_k - f_{k-1})}{g_k^\mathsf{T} p_k}\Big\}$ (N&W, eq. 3.60)
6: $\quad (\alpha_k, f_{k+1}) \leftarrow \text{Backtracking}(\phi_k;\ \alpha_{\text{init}}, 10^{-4}, 0.1, 0.5, 30)$; $\quad$ **if** failure **then return** x_k
7: $\quad x_{k+1} \leftarrow x_k + \alpha_k p_k$; $\quad g_{k+1} \leftarrow \nabla f(x_{k+1})$ $\qquad$ (f_{k+1} already computed in A1)
8: **return** $x_{k_{\max}+1}$ with a warning (iteration limit)

Here $\phi_k(\alpha) = f(x_k + \alpha p_k)$. The direction $-g_k$ carries no information about the scale of the step, so $\alpha = 1$ is a poor first trial. The initial trial of line 5 assumes that the first-order decrease of this step will equal that of the previous one; at $k = 0$ it gives a first step of length about 1. SciPy uses the same rule for the first trial of its BFGS and CG line searches. For a quadratic with Hessian Q, line 6 is replaced by the exact step $\alpha_k = g_k^\mathsf{T} g_k / (g_k^\mathsf{T} Q g_k)$ (V4). Steepest descent is rarely the method of choice (U6 explains why), but this skeleton (direction, initial trial, line search, reuse of f) is shared by every line-search method in this document.

V3 Validation table *A3 with Backtracking (A1)*

$f(x) = 10x_1^2 + x_2^2$, from $x_0 = (1,1)^\mathsf{T}$, $\varepsilon = 10^{-6}$; A1 with $c_1 = 10^{-4}$, $\rho_{\text{lo}} = 0.1$, $\rho_{\text{hi}} = 0.5$. Column "trials" counts the evaluations of f inside A1.

k	x_k	$f(x_k)$	$\|g_k\|$	α_{init}	α_k	trials
0	(1, 1)	11	20.0998	0.049752	0.049752	1
1	(0.004963, 0.900496)	0.811140	1.8037	1	0.486735	2
2	(−0.043349, 0.023890)	0.019362	0.86829	1	0.1	2
3	(0.043349, 0.019112)	0.019156	0.86782	0.000551	0.000551	1
4	(0.042871, 0.019091)	0.018744	0.85827	0.001132	0.001132	1
5	(0.041900, 0.019048)	0.017919	0.83887	0.002366	0.002366	1
6	(0.039917, 0.018958)	0.016293	0.79925	0.005141	0.005141	1
7	(0.035813, 0.018763)	0.013178	0.71724	0.012235	0.012235	1
8	(0.027050, 0.018304)	0.007652	0.54223	0.037964	0.037964	1
9	(0.006511, 0.016914)	0.000710	0.13455	0.774551	0.077455	2
10	(−0.003575, 0.014294)	0.000332	0.07701	0.128718	0.057079	2

The run stops at $k = 41$ with $\|g\| < 10^{-6}$; from $k = 10$ on the steps alternate between 0.0571 and 0.2232. At $k = 0$ the trial $1/\|g_0\|$ is close to the exact step 0.050450 and is accepted at once. At $k = 2$ the step 0.1 (the lower safeguard) gives almost no decrease ($0.019362 \to 0.019156$), so at $k = 3$ rule 3.60 proposes a tiny step, and the steps only recover by doubling over the next iterations.

What it shows. The initial trial matters as much as the line search: most iterations accept the first trial, and a single poor step propagates through rule 3.60. f always decreases, but $\|g_k\|$ is not monotone.

T11 Proof *ch2-v2, equation (11), p. 8*

Let $f(x) = \frac{1}{2}x^\mathsf{T}Qx - b^\mathsf{T}x$ with $Q \succ 0$, $x^* = Q^{-1}b$ and $\|v\|_Q = \sqrt{v^\mathsf{T}Qv}$. Prove that $\frac{1}{2}\|x - x^*\|_Q^2 = f(x) - f(x^*)$ for all $x \in \mathbb{R}^n$.

T12 Proof *ch2-v2, Lemma 4.1 (Kantorovich), pp. 8–10*

Let $Q \succ 0$ with eigenvalues $0 < m = \lambda_1 \le \cdots \le \lambda_n = M$. Prove that for every $g \neq 0$,

$$\frac{(g^\mathsf{T}g)^2}{(g^\mathsf{T}Qg)(g^\mathsf{T}Q^{-1}g)} \ge \frac{4Mm}{(M+m)^2}.$$

T13 Proof *ch2-v2, Theorem 4.4, pp. 10–11*

Let $f(x) = \frac{1}{2}x^\mathsf{T}Qx - b^\mathsf{T}x$ with $Q \succ 0$ with eigenvalues $0 < \lambda_1 \le \cdots \le \lambda_n$, $x^* = Q^{-1}b$ and $\|v\|_Q = \sqrt{v^\mathsf{T}Qv}$. For $x_{k+1} = x_k - \alpha_k g_k$, $g_k = Qx_k - b$, with exact line search:

a) Prove that $\alpha_k = \dfrac{g_k^\mathsf{T}g_k}{g_k^\mathsf{T}Qg_k}$.

b) Prove that $\|x_{k+1} - x^*\|_Q^2 \le \left(\dfrac{\lambda_n - \lambda_1}{\lambda_n + \lambda_1}\right)^2 \|x_k - x^*\|_Q^2$.

V4 VALIDATION TABLE *A3 with exact step; solves P4*

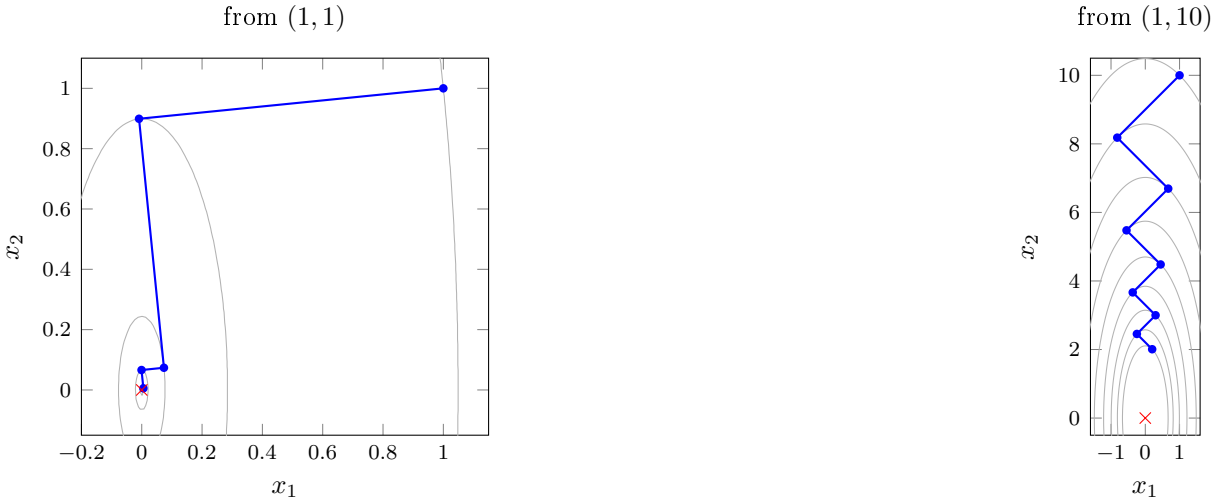
$f(x) = 10x_1^2 + x_2^2 = \frac{1}{2}x^\mathsf{T}Qx$ with $Q = \mathrm{diag}(20, 2)$, $\kappa = 10$, $x^* = 0$; exact step $\alpha_k = g_k^\mathsf{T}g_k/(g_k^\mathsf{T}Qg_k)$, $g_k = Qx_k$. Theoretical bound: $\left(\frac{\kappa-1}{\kappa+1}\right)^2 = \frac{81}{121} \approx 0.6694$.

k	x_k	g_k	α_k	$f(x_k)$	$f(x_{k+1})/f(x_k)$
From $x_0 = (1,1)^\mathsf{T}$					
0	$(1,\ 1)$	$(20,\ 2)$	$\frac{101}{2002} \approx 0.050450$	11	$\frac{810}{11011} \approx 0.07356$
1	$\left(-\frac{9}{1001},\ \frac{900}{1001}\right)$	$\left(-\frac{180}{1001},\ \frac{1800}{1001}\right)$	$\frac{101}{220} \approx 0.459091$	$\frac{810}{1001} \approx 0.809191$	0.07356
2	$\left(\frac{810}{11011},\ \frac{810}{11011}\right)$	$\left(\frac{16200}{11011},\ \frac{1620}{11011}\right)$	$\frac{101}{2002}$	0.059526	0.07356
3	$(-0.000661,\ 0.066140)$	$(-0.013228,\ 0.132281)$	$\frac{101}{220}$	0.004379	0.07356
From $x_0 = (1,10)^\mathsf{T}$ (worst case: the bound is attained)					
0	$(1,\ 10)$	$(20,\ 20)$	$\frac{1}{11}$	110	$\frac{81}{121}$
1	$\left(-\frac{9}{11},\ \frac{90}{11}\right)$	$\left(-\frac{180}{11},\ \frac{180}{11}\right)$	$\frac{1}{11}$	$\frac{810}{11}$	$\frac{81}{121}$
2	$\left(\frac{81}{121},\ \frac{810}{121}\right)$	$\left(\frac{1620}{121},\ \frac{1620}{121}\right)$	$\frac{1}{11}$	$\frac{65610}{1331}$	$\frac{81}{121}$

In both cases $g_{k+1}^\mathsf{T}g_k = 0$ (orthogonal zigzag).

What it shows. The Kantorovich bound is attained in the worst case, and the orthogonal zigzag explains the slowness when $\kappa \gg 1$.

Figure.



Level curves of $f = 10x_1^2 + x_2^2$ through each iterate. Each step is tangent to the level curve at the next point, so $g_{k+1} \perp g_k$. From $(1,10)$ the pattern repeats at scale $\frac{9}{11}$ and the reduction per step is exactly $(\frac{\kappa-1}{\kappa+1})^2$.

T14 OBSERVATION *ch2-v2, pp. 11 and 15*

For $f(x) = \frac{1}{2}x^\mathsf{T}Qx - b^\mathsf{T}x$, $Q \succ 0$ with eigenvalues $\lambda_1 \le \cdots \le \lambda_n$, and steepest descent with exact line search, the contraction factor of T13 is $\frac{\lambda_n-\lambda_1}{\lambda_n+\lambda_1}$ in the norm $\|v\|_Q = \sqrt{v^\mathsf{T}Qv}$.

a) Express the contraction factor in terms of $\kappa(Q) = \lambda_n/\lambda_1$ and explain the cases $\kappa = 1$ (convergence in one iteration) and $\kappa \gg 1$ (zigzag).

b) Explain the difference between the quadratic case (T13: the bound $\|x_{k+1} - x^*\|_Q \le \frac{\lambda_n-\lambda_1}{\lambda_n+\lambda_1}\|x_k - x^*\|_Q$ holds exactly for every $k \ge 0$) and the general C^2 case (T15: $f(x_{k+1}) - f(x^*) \le r^2[f(x_k) - f(x^*)]$ only for large k, and for every r strictly larger than $\frac{\lambda_n-\lambda_1}{\lambda_n+\lambda_1}$, with λ_i the eigenvalues of $\nabla^2 f(x^*)$).

T15 Proof *ch2-v2, Theorem 4.5, pp. 12–15*

Setting. $f:\mathbb{R}^n\to\mathbb{R}$ of class C^2; $\{x_k\}$ generated by steepest descent with exact line search; $x_k\to x^*$ with $\nabla^2 f(x^*)\succ 0$, eigenvalues $\lambda_1\le\cdots\le\lambda_n$.

Problem. Prove that for every $r\in\left(\frac{\lambda_n-\lambda_1}{\lambda_n+\lambda_1},1\right)$ and every sufficiently large k,

$$f(x_{k+1})-f(x^*)\le r^2\,[f(x_k)-f(x^*)].$$

X6 Proposed exercise *ch2-v2, p. 16*

Let $f(x_1,x_2)=100(x_2-x_1^2)^2+(1-x_1)^2$ (Rosenbrock), $\nabla f(x)=\big(-400x_1(x_2-x_1^2)-2(1-x_1),\ 200(x_2-x_1^2)\big)^{\mathsf T}$, $\nabla^2 f(x)=\begin{pmatrix}1200x_1^2-400x_2+2 & -400x_1\\ -400x_1 & 200\end{pmatrix}$.

a) Verify that $x^*=(1,1)^{\mathsf T}$ is the unique global minimum and compute $\nabla^2 f(x^*)$.

b) Find the eigenvalues λ_1,λ_2 of $\nabla^2 f(x^*)$ and $\kappa(\nabla^2 f(x^*))$.

c) Compute the asymptotic rate $r_0=\dfrac{\lambda_2-\lambda_1}{\lambda_2+\lambda_1}$ for steepest descent with exact line search: by T15, eventually $f(x_{k+1})-f(x^*)\le r^2[f(x_k)-f(x^*)]$ for every $r>r_0$.

Q8 Exam problem *added, 4 pts*

Fixed step against exact step on a quadratic. Let $f(x)=\frac12 x^{\mathsf T}Qx-b^{\mathsf T}x$ with $Q\succ 0$, eigenvalues in $[m,M]$, $0<m<M$, $\kappa=M/m$, and $x^*=Q^{-1}b$.

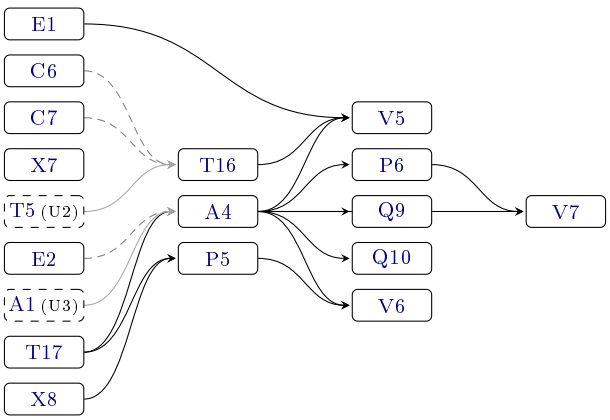
a) For $x_{k+1}=x_k-\alpha\nabla f(x_k)$ with fixed α, show $x_{k+1}-x^*=(I-\alpha Q)(x_k-x^*)$ and that the method converges from every x_0 if and only if $0<\alpha<2/M$.

b) Prove that $\alpha=2/(m+M)$ minimizes $\max_{\lambda\in[m,M]}|1-\alpha\lambda|$ and gives $\|x_k-x^*\|\le\left(\frac{\kappa-1}{\kappa+1}\right)^k\|x_0-x^*\|$.

c) With exact line search the bound (Kantorovich) is $f(x_{k+1})-f(x^*)\le\left(\frac{\kappa-1}{\kappa+1}\right)^2\big(f(x_k)-f(x^*)\big)$. Compare both rates, and compute how many iterations each bound needs to reduce $f(x_k)-f(x^*)$ by a factor 10^{-6} when $\kappa=100$.

U7. Newton's method and modified Hessian

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E1 NUMERICAL EXAMPLE *ch3, p. 4*

Setting. $f(x) = e^x - x - 1$, minimizer $x^* = 0$; Newton $x_{k+1} = x_k - f'(x_k)/f''(x_k)$ from $x_0 = 1$.
Problem.

a) Compute $x_1, \dots, x_4$.

b) Verify numerically that $|x_{k+1}|/|x_k|^2$ settles down, and give its limit.

Note. The values x_2, x_3, x_4 printed in the notes do not match the iteration; see table V5.

C6 COUNTEREXAMPLE *ch3, p. 4*

Setting. $f(x) = x^4$ on $\mathbb{R}$, minimizer $x^* = 0$, Newton iteration $x_{k+1} = x_k - f'(x_k)/f''(x_k)$.
Recall. Local convergence theorem for Newton's method (T16): if $\nabla f(x^*) = 0$, $\nabla^2 f(x^*)$ is invertible and $\nabla^2 f$ is Lipschitz near x^*, then Newton converges quadratically from every x_0 close enough to x^*.
Problem.

a) Prove that the Newton iteration is $x_{k+1} = \frac{2}{3} x_k$, so convergence is only linear.

b) Identify which hypothesis of the theorem fails.

C7 COUNTEREXAMPLE *ch3, p. 4 (marked "verify")*

a) For $f(x) = |x|^{7/3}$, compute the Newton iteration $x_{k+1} = x_k - f'(x_k)/f''(x_k)$ explicitly and its order of convergence. Determine which hypotheses of the local convergence theorem for Newton's method (T16: if $\nabla f(x^*) = 0$, $\nabla^2 f(x^*)$ is invertible and $\nabla^2 f$ is Lipschitz near x^*, then Newton converges quadratically from every x_0 close enough to x^*) fail.

b) For $f(x) = \frac{1}{2} x^2 + |x|^{7/3}$, prove that $f''(0) \neq 0$, that f'' is not Lipschitz near 0, and that Newton converges with order exactly $4/3$.

Note. The notes claim superlinear convergence for $x^{7/3}$; part a) checks that claim and part b) isolates the Lipschitz hypothesis.

T16 PROOF *ch3, Theorem 2.1, p. 3*

Setting. $f : \mathbb{R}^n \to \mathbb{R}$ of class C^2 on an open set D; $x^* \in D$ with

- $\nabla f(x^*) = 0$, $\nabla^2 f(x^*)$ invertible and $\|\nabla^2 f(x^*)^{-1}\| \leq M$;
- $\nabla^2 f$ Lipschitz on D with constant L.

Problem. Prove that there is $r > 0$ such that, for every $x_0 \in B(x^*, r)$, the Newton iteration $x_{k+1} = x_k - \nabla^2 f(x_k)^{-1} \nabla f(x_k)$ is well defined, converges to x^*, and satisfies

$$\|x_{k+1} - x^*\| \leq LM \, \|x_k - x^*\|^2.$$

X7 PROPOSED EXERCISE *ch3, p. 18*

Let $f(x) = \frac{1}{2} x^{\mathsf{T}} A x - b^{\mathsf{T}} x$ with $A \succ 0$.

a) Prove that Newton reaches $x^* = A^{-1} b$ in exactly one iteration from any x_0.

b) For the modified Newton step $x_1 = x_0 - B_0^{-1} \nabla f(x_0)$ with $B_0 = A + \tau I$, $\tau > 0$, express $x_1 - x^*$ in terms of $x_0 - x^*$ and analyse the effect of τ on the speed of convergence.

E2 Numerical example *ch3, p. 6*

Let $f(x_1, x_2) = -2x_1 + 4x_2 - x_1^2 + x_1x_2 + \frac{3}{2}x_2^2$ and $x_k = (0, 0)^\mathsf{T}$.

a) Compute $\nabla f(x_k)$, $\nabla^2 f(x_k)$ and its eigenvalues.

b) Compute the Newton direction and verify that it is not a descent direction.

c) With $B_k = \nabla^2 f(x_k) + 3.5\, I$, compute $p_k = -B_k^{-1}\nabla f(x_k)$ and verify that it is a descent direction.

T17 Proof *ch3, Proposition 3.1, p. 5*

Let B_k be symmetric positive definite and $p_k = -B_k^{-1}\nabla f(x_k)$. Prove that

$$\nabla f(x_k)^\mathsf{T} p_k \le -\frac{1}{\|B_k\|}\|\nabla f(x_k)\|^2.$$

A4 Pseudocode *modified Newton via shifted Cholesky (N&W, Alg. 3.3)*

ModifiedNewton(f, ∇f, $\nabla^2 f$, x_0; $\delta, \varepsilon, k_{\max}$)

1: **for** $k = 0, 1, \ldots, k_{\max}$ **do**
2: $\quad g_k \leftarrow \nabla f(x_k)$; **if** $\|g_k\| < \varepsilon$ **then return** x_k
3: $\quad (L_k, \tau_k) \leftarrow$ ShiftedCholesky($\nabla^2 f(x_k)$; δ) $\qquad (L_k L_k^\mathsf{T} = \nabla^2 f(x_k) + \tau_k I =: B_k \succ 0)$
4: $\quad$ solve $L_k y = -g_k$ (forward substitution); solve $L_k^\mathsf{T} p_k = y$ (back substitution)
5: $\quad \alpha_k \leftarrow$ Backtracking(ϕ_k; $1, 10^{-4}, 0.1, 0.5, 30$) $\qquad$ (A1, $\phi_k(\alpha) = f(x_k + \alpha p_k)$)
6: $\quad x_{k+1} \leftarrow x_k + \alpha_k p_k$

ShiftedCholesky(H; δ) $\quad H$ symmetric $n \times n$, $\delta > 0$

1: $\mu \leftarrow \min_i h_{ii}$; **if** $\mu > 0$ **then** $\tau \leftarrow 0$ **else** $\tau \leftarrow \delta - \mu$
2: **repeat**
3: $\quad (L, \text{ok}) \leftarrow$ Cholesky($H + \tau I$)
4: $\quad$ **if** ok **then return** (L, τ) **else** $\tau \leftarrow \max\{2\tau,\ \delta\}$

Cholesky(M) $\quad M$ symmetric; returns L lower triangular with $LL^\mathsf{T} = M$, or failure

---

1: **for** $j = 1, \ldots, n$ **do**
2: $\quad d \leftarrow m_{jj} - \sum_{s<j} l_{js}^2$; **if** $d \le 0$ **then return** (—, false) $\qquad (M \not\succ 0)$
3: $\quad l_{jj} \leftarrow \sqrt{d}$
4: $\quad$ **for** $i = j + 1, \ldots, n$ **do** $l_{ij} \leftarrow \big(m_{ij} - \sum_{s<j} l_{is} l_{js}\big)/l_{jj}$
5: **return** (L, true)

Why this form is implementable:

- No eigenvalue is computed. Cholesky succeeds (every pivot $d > 0$) if and only if $M \succ 0$, so the factorization *is* the positive-definiteness test. The initial shift uses $\min_i h_{ii}$, which costs n comparisons; since $\lambda_{\min}(H) \le e_i^\mathsf{T} H e_i = h_{ii}$, a nonpositive diagonal entry proves $H \not\succ 0$, but a positive diagonal does not prove $H \succ 0$, which is why the loop may have to increase τ.

- The factor that certifies $B_k \succ 0$ is reused to solve $B_k p_k = -g_k$: two triangular solves of n^2 flops each. B_k^{-1} is never formed.
- Cost per Newton iteration: about $n^3/3$ flops per Cholesky trial, times the number of trials (usually one or two), plus $2n^2$ for the solves. Computing $\lambda_{\min}$ with a symmetric eigensolver costs several times more and still leaves the linear system to be solved.
- Pure Newton is the case $\tau_k = 0$, $\alpha_k = 1$. If $\nabla^2 f(x_k) \succ 0$ and $\min_i h_{ii} > 0$, ShiftedCholesky returns $\tau_k = 0$ on the first trial and A4 takes the Newton direction.

Values used in V6 and V7: $\delta = 1$; A1 with its defaults. N&W use a small δ (e.g. 10^{-3}); a small δ can leave B_k nearly singular and produce a very long p_k that the backtracking must then shorten, and it costs more Cholesky trials. Production Newton codes use the same idea with a modified Cholesky factorization that fixes the pivots in a single pass (Gill–Murray; Julia's `Optim` Newton uses the package `PositiveFactorizations`), or use a trust region instead of a line search (A11).

V5 Validation table *pure Newton on $f(x) = e^x - x - 1$; solves E1*

Iteration $x_{k+1} = x_k - (e^{x_k} - 1)/e^{x_k}$ from $x_0 = 1$; $x^* = 0$ and the ratio tends to $\frac{1}{2} f'''(0)/f''(0) = \frac{1}{2}$.

k	x_k	$f'(x_k)$	$f''(x_k)$	x_{k+1}/x_k^2
0	1	1.71828183	2.71828183	0.3679
1	0.36787944	0.44466786	1.44466786	0.4439
2	0.06008007	0.06192157	1.06192157	0.4901
3	0.00176920	0.00177077	1.00177077	0.4997
4	1.5641×10^{-6}	1.5641×10^{-6}	1.00000156	0.5000
5	1.2233×10^{-12}			

Note. The notes (ch3, p. 4) give $x_2 \approx 0.05807$, $x_3 \approx 0.00159$, $x_4 \approx 1.26 \times 10^{-6}$; these values do not match the iteration. The correct ones are in this table.

X8 Proposed exercise *ch3, p. 18*

Let $f(x_1, x_2) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$ (Rosenbrock).

a) Show that the unique global minimum is $x^* = (1, 1)^\mathsf{T}$.

b) Compute $\nabla^2 f(x)$. Prove that Newton is applicable at $x_0 = (0, 0)^\mathsf{T}$ and find x_1.

c) Decide whether the Hessian is positive definite at $x_0 = (0, 0)^\mathsf{T}$ and at $\tilde{x} = (0, 1)^\mathsf{T}$. If it is not at $\tilde{x}$, find the smallest $\tau > 0$ such that $\nabla^2 f(\tilde{x}) + \tau I$ is positive semidefinite (boundary of positive definiteness).

P5 Practice *added*

Let $f(x_1, x_2) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$ (Rosenbrock), $\nabla f(x) = \big(-400x_1(x_2 - x_1^2) - 2(1 - x_1),\ 200(x_2 - x_1^2)\big)^\mathsf{T}$, $\nabla^2 f(x) = \begin{pmatrix} 1200x_1^2 - 400x_2 + 2 & -400x_1 \\ -400x_1 & 200 \end{pmatrix}$, at the point $\tilde{x} = (0, 1)^\mathsf{T}$, and let $m(p) = f(\tilde{x}) + \nabla f(\tilde{x})^\mathsf{T} p + \frac{1}{2} p^\mathsf{T} \nabla^2 f(\tilde{x}) p$ be the quadratic model.

a) Compute the pure Newton direction and decide whether it is a descent direction.

b) Show that p^N is a saddle point of the model m, not its minimum.

c) With $\tau = 400$ and $B = \nabla^2 f(\tilde{x}) + \tau I$, compute $p = -B^{-1} \nabla f(\tilde{x})$, compare both directions and verify the bound $\nabla f(\tilde{x})^\mathsf{T} p \le -\|\nabla f(\tilde{x})\|^2 / \|B\|$.

V6 VALIDATION TABLE *A4 on Rosenbrock from* $(0,1)^\mathsf{T}$*; solves P5*

$f(x) = 100(x_2 - x_1^2)^2 + (1-x_1)^2$, $x_0 = (0,1)^\mathsf{T}$, $\delta = 1$, A1 with $\alpha_{\text{init}} = 1$ and its defaults; $x^* = (1,1)^\mathsf{T}$.

k	x_k	$f(x_k)$	$\|g_k\|$	$\lambda_{\min}$	τ_k	p_k	α_k
0	(0, 1)	101	200.010	−398	399	$(2,\ -\frac{200}{599})$	0.1
1	(0.2, 0.966611)	86.500798	200.198	−348.316	675.289	(0.177437, −0.195509)	1
2	(0.377437, 0.771102)	39.906751	158.282	−193.426	272.980	(0.627353, −0.065572)	1
3	(1.004790, 0.705530)	9.246064	136.516	22.295	0	(−0.000077, 0.303917)	1
4	(1.004713, 1.009447)	2.2209×10^{-5}	9.43×10^{-3}	0.396	0	(−0.004713, −0.009470)	1
5	(1.000000, 0.999978)	4.93×10^{-8}	9.93×10^{-3}	0.401	0	$(\approx 0,\ 2.22 \times 10^{-5})$	1
6	(1, 1)	6.3×10^{-22}	5.0×10^{-11}				

Detail of $k = 0$: $g_0 = (-2, 200)^\mathsf{T}$, $\nabla^2 f(x_0) = \text{diag}(-398, 200)$, $\min_i h_{ii} = -398$, so $\tau_0 = 1 + 398 = 399$; Cholesky of $B_0 = \text{diag}(1, 599)$ succeeds at the first trial with $L_0 = \text{diag}(1, \sqrt{599}) \approx \text{diag}(1,\ 24.474477)$. Forward substitution gives $y = (2, -8.171778)$ and back substitution $p_0 = (2, -\frac{200}{599})$; $g_0^\mathsf{T} p_0 \approx -70.778$. In A1, $\alpha = 1$ gives $f = 1112.48 > 100.993$ (reject); the quadratic interpolant gives $\alpha_t = \frac{70.778}{2(1112.48 - 101 + 70.778)} \approx 0.0327$, which the safeguard raises to 0.1, and $f(0.2,\ 0.966611) = 86.5008$ is accepted. The pure Newton direction would be $p^N = \left(-\frac{1}{199}, -1\right)^\mathsf{T}$, with $g_0^\mathsf{T} p^N \approx -199.99 < 0$ but pointing to the saddle point of the model. At $k = 1, 2$ the Hessian is indefinite with positive diagonal, so ShiftedCholesky starts at $\tau = 0$, fails, and increases τ until the factorization succeeds. From $k = 3$ the Hessian is positive definite, $\tau_k = 0$ and the steps are Newton steps with $\alpha_k = 1$.

Retry loop of ShiftedCholesky (here on the Hessian of E2). For $H = \left(\begin{smallmatrix} -2 & 1 \\ 1 & 3 \end{smallmatrix}\right)$, $\lambda_{\min}(H) \approx -2.1926$:

δ	trial	τ	Cholesky of $H + \tau I$	result
1	1	3	pivots 1, 5	$L = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 2.236068 \end{smallmatrix}\right)$
0.01	1	2.01	second pivot $d = 5.01 - 1/0.01 = -94.99 \le 0$	fails
0.01	2	4.02	pivots 2.02, 6.525	$L \approx \left(\begin{smallmatrix} 1.421267 & 0 \\ 0.703598 & 2.554398 \end{smallmatrix}\right)$

With $\delta = 0.01$ the first trial has a positive first pivot 0.01 but $H + \tau I$ is still indefinite; Cholesky detects it at the second pivot and the shift is doubled.

P6 PRACTICE *added*

A Hessian that must be modified. Let $f(x_1, x_2) = x_1^2 + x_2^2 + 4x_1x_2 + x_1^4 + x_2^4$, with $\nabla f(x) = (2x_1 + 4x_2 + 4x_1^3,\ 2x_2 + 4x_1 + 4x_2^3)^\mathsf{T}$, $\nabla^2 f(x) = \begin{pmatrix} 2 + 12x_1^2 & 4 \\ 4 & 2 + 12x_2^2 \end{pmatrix}$ and $x_0 = (\frac{1}{4}, 0)^\mathsf{T}$.

a) Show that the origin is a stationary point and a saddle, and that $\pm(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})^\mathsf{T}$ are global minimizers with $f = -\frac{1}{2}$.

b) Show that $\nabla^2 f(x_0)$ is indefinite although both diagonal entries are positive. Describe the region of the plane where $\nabla^2 f \not\succ 0$.

c) Run ShiftedCholesky (A4) with $\delta = 1$ on $\nabla^2 f(x_0)$: for each τ tried, compute the pivots of the Cholesky factorization of $\nabla^2 f(x_0) + \tau I$ and say where it fails, until it succeeds. Then compute p_0 with the two triangular solves and check that $\nabla f(x_0)^\mathsf{T} p_0 < 0$.

d) Continue A4 ($\delta = 1$, A1 with $\alpha_{\text{init}} = 1$ and its defaults) until $\|\nabla f(x_k)\| < 10^{-6}$ (table V7).

e) Run pure Newton, $x_{k+1} = x_k - \nabla^2 f(x_k)^{-1} \nabla f(x_k)$, from the same x_0. Where does it converge? Is the pure Newton direction at x_0 a descent direction? Explain why descent alone does not protect pure Newton here.

V7 VALIDATION TABLE *A4 on a nonconvex function; solves P6*

$f(x) = x_1^2 + x_2^2 + 4x_1x_2 + x_1^4 + x_2^4$, $x_0 = (\frac{1}{4}, 0)^\mathsf{T}$, $x^* = (\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})^\mathsf{T}$, $\delta = 1$, A1 with $\alpha_{\text{init}} = 1$; $e_k = \|x_k - x^*\|$.

a) ShiftedCholesky at $k = 0$. $g_0 = (\frac{9}{16}, 1)^\mathsf{T}$, $\nabla^2 f(x_0) = \left(\begin{smallmatrix} 11/4 & 4 \\ 4 & 2 \end{smallmatrix}\right)$, $\lambda_{\min} \approx -1.6425$, $\min_i h_{ii} = 2 > 0$, so the first trial is $\tau = 0$.

trial	τ	pivots of $\nabla^2 f(x_0) + \tau I$	result
1	0	$d_1 = \frac{11}{4}$, $d_2 = 2 - \frac{16}{11/4} = -\frac{42}{11}$	fails at $j = 2$
2	1	$d_1 = \frac{15}{4}$, $d_2 = 3 - \frac{16}{15/4} = -\frac{19}{15}$	fails at $j = 2$
3	2	$d_1 = \frac{19}{4}$, $d_2 = 4 - \frac{16}{19/4} = \frac{12}{19}$	$L_0 = \begin{pmatrix} \frac{\sqrt{19}}{2} & 0 \\ \frac{8}{\sqrt{19}} & \sqrt{\frac{12}{19}} \end{pmatrix} \approx \begin{pmatrix} 2.179449 & 0 \\ 1.835326 & 0.794719 \end{pmatrix}$

Solves: $L_0 y = -g_0$ gives $y \approx (-0.258093, -0.662266)$; $L_0^\mathsf{T} p_0 = y$ gives $p_0 = (\frac{7}{12}, -\frac{5}{6})^\mathsf{T}$; $g_0^\mathsf{T} p_0 = -\frac{97}{192} \approx -0.505208$; A1 accepts $\alpha_0 = 1$, so $x_1 = (\frac{5}{6}, -\frac{5}{6})^\mathsf{T}$.

b) Iterations of A4.

k	x_k	$f(x_k)$	$\|g_k\|$	$\lambda_{\min}(\nabla^2 f)$	trials	τ_k	e_k
0	(0.25, 0)	0.066406	1.1473	−1.6425	3	2	0.84203
1	(0.833333, −0.833333)	−0.424383	0.91662	6.3333	1	0	0.17851
2	(0.730994, −0.730994)	−0.497640	0.14205	4.4122	1	0	3.378×10^{-2}
3	(0.708229, −0.708229)	−0.499995	6.3625×10^{-3}	4.0191	1	0	1.587×10^{-3}
4	(0.707109, −0.707109)	−0.500000	1.5053×10^{-5}	4.0000	1	0	3.763×10^{-6}
5	(0.707107, −0.707107)	−0.5	8.50×10^{-11}				

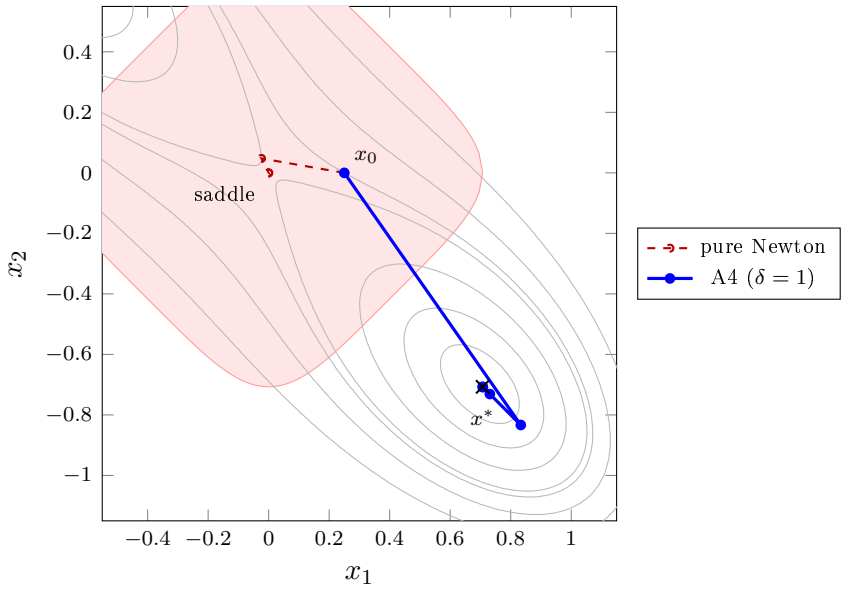
$\alpha_k = 1$ at every iteration.

c) Pure Newton from the same x_0. $p_0^N = (-\frac{23}{84}, \frac{1}{21})^\mathsf{T}$ with $g_0^\mathsf{T} p_0^N = -\frac{143}{1344} \approx -0.1064 < 0$.

k	x_k	$f(x_k)$	$\|g_k\|$	$\lambda_{\min}(\nabla^2 f)$
0	(0.25, 0)	0.066406	1.147	−1.6425
1	$(-\frac{1}{42}, \frac{1}{21}) \approx (-0.023810, 0.047619)$	−0.001695	0.1428	−1.9830
2	(0.000308, −0.000181)	≈ 0	8.757×10^{-4}	−2.0000
3	$\approx (0, 0)$	≈ 0	2.384×10^{-10}	−2.0000

What it shows. The diagonal test cannot see this indefiniteness ($\min_i h_{ii} = 2 > 0$), so Cholesky is what detects it, and the loop doubles τ until $\nabla^2 f(x_0) + \tau I \succ 0$. The pure Newton direction is a descent direction, yet pure Newton converges quadratically to the saddle point, because it jumps to the stationary point of an indefinite model. With the shift, the model has a minimum, the first step leaves the indefinite region, and from $k = 1$ on A4 is pure Newton with quadratic convergence to the minimizer.

Figure.



Level curves of $f = x_1^2+x_2^2+4x_1x_2+x_1^4+x_2^4$. The shaded region is where $\nabla^2 f \not\succ 0$, i.e. $(1+6x_1^2)(1+6x_2^2) < 4$; it contains the saddle point at the origin. Pure Newton models f by a quadratic whose stationary point is a saddle, and it jumps to that saddle. A4 replaces $\nabla^2 f(x_0)$ by $\nabla^2 f(x_0) + 2I \succ 0$, whose model has a minimum, and its first step leaves the shaded region toward x^*; from then on $\tau_k = 0$.

Q9 EXAM PROBLEM *added, 4 pts*

Newton's method on $\sqrt{1+x^2}$. Let $f(x) = \sqrt{1+x^2}$, $x \in \mathbb{R}$, with minimizer $x^* = 0$.

a) Show that the pure Newton iteration is $x_{k+1} = -x_k^3$. Study its behaviour for $|x_0| < 1$, $|x_0| = 1$ and $|x_0| > 1$, and give the order of convergence when it converges.

b) Show that the Newton direction is a descent direction at every $x \neq 0$, and that for $|x_0| > 1$ the unit step fails the Armijo condition. Find the largest $|x_0|$ for which $\alpha = 1$ satisfies Armijo with $c_1 = 10^{-4}$ (an equation to be solved numerically is enough).

c) Prove that Newton's method with Armijo backtracking (A1) converges to 0 from every x_0. (Use that the iterates stay in the bounded sublevel set $\{f \le f(x_0)\}$, where f'' is bounded below by a positive constant and above by 1; bound the accepted steps from below and sum the Armijo decreases.)

Q10 EXAM PROBLEM *added, 4 pts*

Two ways to modify an indefinite Hessian. Let $H = Q\Lambda Q^{\mathsf{T}}$ be symmetric with $\lambda_{\min}(H) < 0$, $g \neq 0$ and $\delta > 0$.

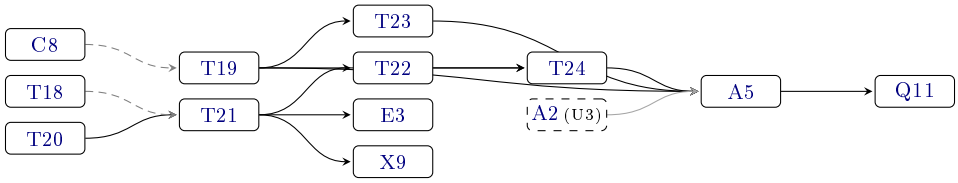
a) Prove that the correction ΔH of smallest Frobenius norm such that $\lambda_{\min}(H + \Delta H) \ge \delta$ is $\Delta H = Q\,\mathrm{diag}(\tau_i)Q^{\mathsf{T}}$ with $\tau_i = \max\{0, \delta - \lambda_i\}$ (eigenvalue modification), and that the smallest shift τI with the same property is $\tau = \delta - \lambda_{\min}(H)$.

b) For $H = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$, $g = (1,0)^{\mathsf{T}}$ and $\delta = 1$, compute the direction $p = -(H+\Delta H)^{-1}g$ for both corrections, check that both are descent directions, and compute cos of the angle of each with $-g$.

c) Explain why practical codes prefer the shift found by trial Cholesky factorizations (A4) to the eigenvalue modification, although the latter changes H less.

U8. Quasi-Newton: DFP and BFGS formulas

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T18 Observation *ch3, p. 7*

Setting. f of class C^2; $s_k = x_{k+1} - x_k$, $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$.
Problem.

a) Prove that $y_k = \big(\int_0^1 \nabla^2 f(x_k + t s_k)\, dt\big) s_k$.

b) Give an example where $\nabla^2 f(x_{k+1}) s_k \neq y_k$.

c) Prove that for quadratic f the exact Hessian satisfies the secant equation $\nabla^2 f\, s_k = y_k$.

C8 Counterexample *ch3, p. 12*

Let $f(x) = -x^2$, $x_0 = 1$, $s_0 = 1$. Compute y_0 and $s_0^{\mathsf T} y_0$, apply the inverse BFGS update $H_{k+1}^{\mathrm{BFGS}} = (I - \rho_k s_k y_k^{\mathsf T}) H_k (I - \rho_k y_k s_k^{\mathsf T}) + \rho_k s_k s_k^{\mathsf T}$, $\rho_k = 1/(y_k^{\mathsf T} s_k)$ (here $k = 0$) with $H_0 = 1$, and show that $H_1 < 0$, so that $p_1 = -H_1 \nabla f(x_1)$ is an ascent direction.

T19 Proof *ch3, Proposition 4.1, pp. 7–8*

Let p_k be a descent direction, $\alpha_k > 0$ such that $\nabla f(x_k + \alpha_k p_k)^{\mathsf T} p_k \geq c_2 \nabla f(x_k)^{\mathsf T} p_k$ with $c_2 < 1$, $s_k = \alpha_k p_k$ and $y_k = \nabla f(x_k + s_k) - \nabla f(x_k)$. Prove that $s_k^{\mathsf T} y_k > 0$.

T20 Proof *ch3, Lemma 4.1, pp. 8–9*

Setting. f of class C^1; $H_k \succ 0$; $s_k = -\alpha_k H_k \nabla f(x_k)$ with $\alpha_k > 0$ from an exact line search; $x_{k+1} = x_k + s_k$, $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$; assume $s_k^{\mathsf T} y_k > 0$ and $\nabla f(x_{k+1}) \neq 0$.
Problem. Prove that s_k and $H_k y_k$ are linearly independent.

T21 Derivation *ch3, p. 9 (DFP) and p. 11 (BFGS, "verify the details")*

Let $s_k = x_{k+1} - x_k$, $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$, with $s_k^{\mathsf T} y_k > 0$.

a) Starting from $H_{k+1} = H_k + \alpha u u^{\mathsf T} + \beta v v^{\mathsf T}$ and $H_{k+1} y_k = s_k$, with $u = s_k$, $v = H_k y_k$, derive the DFP formula $H_{k+1}^{\mathrm{DFP}} = H_k + \dfrac{s_k s_k^{\mathsf T}}{s_k^{\mathsf T} y_k} - \dfrac{H_k y_k y_k^{\mathsf T} H_k}{y_k^{\mathsf T} H_k y_k}$.

b) Starting from $B_{k+1} = B_k + \alpha u u^{\mathsf T} + \beta v v^{\mathsf T}$ and $B_{k+1} s_k = y_k$, with $u = y_k$, $v = B_k s_k$, derive the BFGS formula $B_{k+1}^{\mathrm{BFGS}} = B_k - \dfrac{B_k s_k s_k^{\mathsf T} B_k}{s_k^{\mathsf T} B_k s_k} + \dfrac{y_k y_k^{\mathsf T}}{y_k^{\mathsf T} s_k}$.

c) Justify the "equate coefficients" step (it requires s_k and $H_k y_k$ to be linearly independent; see T20).

T22 Proof *ch3, formula (15), p. 10 ("verify")*

Setting. $s_k = x_{k+1} - x_k$, $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$, $s_k^{\mathsf{T}} y_k > 0$, $\rho_k = 1/(y_k^{\mathsf{T}} s_k)$, $B_k \succ 0$ and $H_k = B_k^{-1}$. Direct BFGS update: $B_{k+1}^{\mathrm{BFGS}} = B_k - \dfrac{B_k s_k s_k^{\mathsf{T}} B_k}{s_k^{\mathsf{T}} B_k s_k} + \dfrac{y_k y_k^{\mathsf{T}}}{y_k^{\mathsf{T}} s_k}$.

Problem. Prove that the inverse of B_{k+1}^{BFGS} is

$$\begin{aligned} H_{k+1}^{\mathrm{BFGS}} &= (I - \rho_k s_k y_k^{\mathsf{T}}) H_k (I - \rho_k y_k s_k^{\mathsf{T}}) + \rho_k s_k s_k^{\mathsf{T}} \\ &= H_k - \rho_k (s_k y_k^{\mathsf{T}} H_k + H_k y_k s_k^{\mathsf{T}}) + \rho_k (1 + \rho_k y_k^{\mathsf{T}} H_k y_k) s_k s_k^{\mathsf{T}}. \end{aligned}$$

E3 Numerical example *ch3, p. 10 (DFP) and p. 12 (BFGS)*

Let $f(x_1, x_2) = \frac{5}{8}x_1^2 + \frac{3}{2}x_2^2 - 2x_1 - x_2$, $x_0 = (0,0)^{\mathsf{T}}$, $H_0 = I_2$ and step $x_1 = x_0 - \alpha_0 H_0 \nabla f(x_0)$ with $\alpha_0 = 1$. Notation: $s_k = x_{k+1} - x_k$, $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$, and the formulas $H_{k+1}^{\mathrm{DFP}} = H_k + \dfrac{s_k s_k^{\mathsf{T}}}{s_k^{\mathsf{T}} y_k} - \dfrac{H_k y_k y_k^{\mathsf{T}} H_k}{y_k^{\mathsf{T}} H_k y_k}$ and $H_{k+1}^{\mathrm{BFGS}} = (I - \rho_k s_k y_k^{\mathsf{T}}) H_k (I - \rho_k y_k s_k^{\mathsf{T}}) + \rho_k s_k s_k^{\mathsf{T}}$, $\rho_k = 1/(y_k^{\mathsf{T}} s_k)$.

a) Compute x_1, s_0, y_0 and verify $s_0^{\mathsf{T}} y_0 > 0$.

b) Compute H_1^{DFP} and H_1^{BFGS}.

c) Verify in both cases the secant equation $H_1 y_0 = s_0$ and positive definiteness.

T23 Proof *ch3, Theorem 4.1, pp. 9–10*

Let H_k be symmetric positive definite, $s_k, y_k \in \mathbb{R}^n$ with $s_k^{\mathsf{T}} y_k > 0$, and $H_{k+1}^{\mathrm{DFP}} = H_k + \dfrac{s_k s_k^{\mathsf{T}}}{s_k^{\mathsf{T}} y_k} - \dfrac{H_k y_k y_k^{\mathsf{T}} H_k}{y_k^{\mathsf{T}} H_k y_k}$. Prove that H_{k+1}^{DFP} is symmetric positive definite.

Note. The notes close with "they cannot vanish simultaneously" without justification; the equality case of Cauchy–Schwarz must be completed.

T24 Proof *ch3, Theorem 4.2, p. 11*

Let $H_k \succ 0$ be symmetric, $s_k, y_k \in \mathbb{R}^n$ with $s_k^{\mathsf{T}} y_k > 0$, and $H_{k+1}^{\mathrm{BFGS}} = (I - \rho_k s_k y_k^{\mathsf{T}}) H_k (I - \rho_k y_k s_k^{\mathsf{T}}) + \rho_k s_k s_k^{\mathsf{T}}$, $\rho_k = 1/(y_k^{\mathsf{T}} s_k)$. Prove that H_{k+1}^{BFGS} is symmetric positive definite.

X9 Proposed exercise *ch3, p. 19*

With $H_0 = I_2$, $x_0 = (1,2)^{\mathsf{T}}$, $\nabla f(x_0) = (3,-1)^{\mathsf{T}}$, $x_1 = (2,0)^{\mathsf{T}}$, $\nabla f(x_1) = (1,1)^{\mathsf{T}}$, $s_0 = x_1 - x_0$, $y_0 = \nabla f(x_1) - \nabla f(x_0)$, and the formulas $H_{k+1}^{\mathrm{DFP}} = H_k + \dfrac{s_k s_k^{\mathsf{T}}}{s_k^{\mathsf{T}} y_k} - \dfrac{H_k y_k y_k^{\mathsf{T}} H_k}{y_k^{\mathsf{T}} H_k y_k}$ and $H_{k+1}^{\mathrm{BFGS}} = (I - \rho_k s_k y_k^{\mathsf{T}}) H_k (I - \rho_k y_k s_k^{\mathsf{T}}) + \rho_k s_k s_k^{\mathsf{T}}$, $\rho_k = 1/(y_k^{\mathsf{T}} s_k)$ (with $k = 0$):

a) Check whether $s_0^{\mathsf{T}} y_0 > 0$ holds.

b) Obtain H_1^{BFGS} with the inverse formula.

c) Prove that, for any $s, y \in \mathbb{R}^n$ with $s^{\mathsf{T}} y > 0$, both H_1^{DFP} and H_1^{BFGS} satisfy $H_1 y = s$.

BFGS$(f,\ \nabla f,\ x_0;\ \varepsilon, k_{\max})$

1: $H \leftarrow I; \quad f_0 \leftarrow f(x_0); \quad g_0 \leftarrow \nabla f(x_0)$
2: **for** $k = 0, 1, \ldots, k_{\max}$ **do**
3: **if** $\|g_k\| \le \varepsilon$ **then return** x_k
4: $p_k \leftarrow -H g_k$
5: $\alpha_k \leftarrow$ StrongWolfe$(\phi_k, \phi_k';\ \alpha_1, 10^{-4}, 0.9)$ with $\alpha_1 = \min\{1, 1/\|g_0\|\}$ if $k = 0$, $\alpha_1 = 1$ otherwise; **if** failure **then return** x_k
6: $s \leftarrow \alpha_k p_k; \quad x_{k+1} \leftarrow x_k + s; \quad g_{k+1} \leftarrow \nabla f(x_{k+1}); \quad y \leftarrow g_{k+1} - g_k$
7: **if** $s^{\mathsf{T}} y \le 10^{-10} \|s\| \|y\|$ **then continue** (skip the update)
8: **if** $k = 0$ **then** $H \leftarrow \dfrac{s^{\mathsf{T}} y}{y^{\mathsf{T}} y}\, I$ (rescale H_0, N&W eq. 6.20)
9: $\rho \leftarrow 1/(s^{\mathsf{T}} y); \quad v \leftarrow Hy$
10: $H \leftarrow H - \rho\,(s\,v^{\mathsf{T}} + v\,s^{\mathsf{T}}) + \left(\rho^2\, y^{\mathsf{T}} v + \rho\right) s\, s^{\mathsf{T}}$ $\qquad (= (I - \rho s y^{\mathsf{T}}) H (I - \rho y s^{\mathsf{T}}) + \rho s s^{\mathsf{T}})$

Here $\phi_k(\alpha) = f(x_k + \alpha p_k)$. Line 10 is the BFGS update expanded so that it costs $O(n^2)$: one product Hy and three rank-one corrections, with no matrix–matrix product. Line 7 never skips when the line search satisfies Wolfe, because curvature implies $s^{\mathsf{T}} y > 0$ (T19); it protects against rounding and against a failed search. Line 8 fixes the scale of H: $H_0 = I$ has no units, and $\frac{s^{\mathsf{T}} y}{y^{\mathsf{T}} y}$ is the inverse of a Rayleigh quotient of the averaged Hessian along s (X38). This is BFGS as in Julia's `Optim` (which rescales H_0 by default); SciPy's `minimize(method='BFGS')` is the same loop without line 8 and with Moré–Thuente as first line search. DFP is the same loop with line 10 replaced by $H \leftarrow H + \frac{ss^{\mathsf{T}}}{s^{\mathsf{T}} y} - \frac{vv^{\mathsf{T}}}{y^{\mathsf{T}} v}$; it is not used in production codes, because it corrects a badly scaled H much more slowly than BFGS.

Q11 Exam problem *added, 4 pts*

Deriving and checking the inverse BFGS update. Let $B \succ 0$, $H = B^{-1}$, and $s, y \in \mathbb{R}^n$ with $s^{\mathsf{T}} y > 0$, $\rho = 1/(s^{\mathsf{T}} y)$. The direct BFGS update is $B_+ = B - \dfrac{Bss^{\mathsf{T}} B}{s^{\mathsf{T}} B s} + \dfrac{yy^{\mathsf{T}}}{y^{\mathsf{T}} s}$.

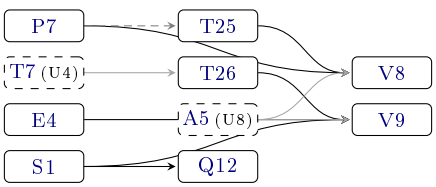
a) Verify that $H_+ = (I - \rho s y^{\mathsf{T}}) H (I - \rho y s^{\mathsf{T}}) + \rho s s^{\mathsf{T}}$ satisfies $B_+ H_+ = I$ (or derive it with the Sherman–Morrison–Woodbury formula).

b) Show that $H_+ y = s$, that $H_+ - H$ has rank at most 2, and that $H_+ \succ 0$ (write $v^{\mathsf{T}} H_+ v$ as a sum of two nonnegative terms and analyse when both vanish).

c) With $H = I$, $s = (1, 0)^{\mathsf{T}}$ and $y = (2, 1)^{\mathsf{T}}$, compute H_+ by hand, check the secant equation and positive definiteness with the leading minors, and compute the expanded $O(n^2)$ form $H_+ = H - \rho(sv^{\mathsf{T}} + vs^{\mathsf{T}}) + (\rho^2 y^{\mathsf{T}} v + \rho) ss^{\mathsf{T}}$, $v = Hy$, to confirm the result.

U9. Convergence of quasi-Newton methods

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P7 PRACTICE *added*

Let $f(x_1,x_2)=\frac{5}{8}x_1^2+\frac{3}{2}x_2^2-2x_1-x_2$, i.e. $A=\operatorname{diag}(\frac{5}{4},3)$, $b=(2,1)^{\mathsf T}$. From $x_0=(0,0)^{\mathsf T}$ and $H_0=I_2$, perform two DFP iterations, $x_{k+1}=x_k+\alpha_k p_k$ with $p_k=-H_k g_k$, $g_k=\nabla f(x_k)$, exact step and $H_{k+1}^{\mathrm{DFP}}=H_k+\dfrac{s_k s_k^{\mathsf T}}{s_k^{\mathsf T}y_k}-\dfrac{H_k y_k y_k^{\mathsf T}H_k}{y_k^{\mathsf T}H_k y_k}$ ($s_k=x_{k+1}-x_k$, $y_k=\nabla f(x_{k+1})-\nabla f(x_k)$), and verify:

a) $g_1^{\mathsf T}s_0=0$ and $s_0^{\mathsf T}As_1=0$.

b) $x_2=x^*=A^{-1}b$.

c) $H_2=A^{-1}$.

T25 PROOF *ch3, Theorem 5.2, pp. 15–17*

Let $f(x)=\frac{1}{2}x^{\mathsf T}Ax-b^{\mathsf T}x$ with $A\succ 0$, $g_k=\nabla f(x_k)$, and $\{x_k\}$ generated by $x_{k+1}=x_k-\alpha_k H_k g_k$ with exact line search and H_{k+1} given by $H_{k+1}^{\mathrm{DFP}}=H_k+\dfrac{s_k s_k^{\mathsf T}}{s_k^{\mathsf T}y_k}-\dfrac{H_k y_k y_k^{\mathsf T}H_k}{y_k^{\mathsf T}H_k y_k}$ or $H_{k+1}^{\mathrm{BFGS}}=(I-\rho_k s_k y_k^{\mathsf T})H_k(I-\rho_k y_k s_k^{\mathsf T})+\rho_k s_k s_k^{\mathsf T}$, $\rho_k=1/(y_k^{\mathsf T}s_k)$ ($s_k=x_{k+1}-x_k$, $y_k=g_{k+1}-g_k$), from arbitrary x_0 and $H_0\succ 0$. Prove by induction that, as long as $g_0,\dots,g_{k-1}\neq 0$:

(P1)$_k$ $s_i^{\mathsf T}As_j=0$ for $0\le i<j\le k-1$;

(P2)$_k$ $g_k^{\mathsf T}s_i=0$ for $0\le i\le k-1$;

(P3)$_k$ $H_k y_i=s_i$ for $0\le i\le k-1$.

Conclude that the method reaches $x^*=A^{-1}b$ in at most n iterations and that, if n nonzero steps are generated, $H_n=A^{-1}$. Complete the BFGS case of (P3), which the notes leave as "analogous".

V8 VALIDATION TABLE *DFP and BFGS updates with exact step; solves P7 and illustrates T25*

$f(x)=\frac{5}{8}x_1^2+\frac{3}{2}x_2^2-2x_1-x_2=\frac{1}{2}x^{\mathsf T}Ax-b^{\mathsf T}x$, $A=\operatorname{diag}(\frac{5}{4},3)$, $b=(2,1)^{\mathsf T}$, $x^*=(\frac{8}{5},\frac{1}{3})^{\mathsf T}$; $x_0=(0,0)^{\mathsf T}$, $H_0=I_2$ (not rescaled, as in T25), exact step $\alpha_k=-g_k^{\mathsf T}p_k/(p_k^{\mathsf T}Ap_k)$. BFGS and DFP generate the same iterates and differ only in p_1, α_1 and H_1.

k	x_k	g_k	p_k	α_k	s_k	y_k
0	$(0,\ 0)$	$(-2,\ -1)$	$(2,\ 1)$	$\frac{5}{8}$	$(\frac{5}{4},\ \frac{5}{8})$	$(\frac{25}{16},\ \frac{15}{8})$
1 (DFP)	$(\frac{5}{4},\ \frac{5}{8})$	$(-\frac{7}{16},\ \frac{7}{8})$	$(\frac{42}{61},\ -\frac{35}{61})$	$\frac{61}{120}$	$(\frac{7}{20},\ -\frac{7}{24})$	$(\frac{7}{16},\ -\frac{7}{8})$
1 (BFGS)	$(\frac{5}{4},\ \frac{5}{8})$	$(-\frac{7}{16},\ \frac{7}{8})$	$(\frac{105}{128},\ -\frac{175}{256})$	$\frac{32}{75}$	$(\frac{7}{20},\ -\frac{7}{24})$	$(\frac{7}{16},\ -\frac{7}{8})$
2	$(\frac{8}{5},\ \frac{1}{3})$	$(0,\ 0)$				

$$H_1^{\mathrm{DFP}}=\begin{pmatrix}\frac{133}{122} & -\frac{59}{244}\\ -\frac{59}{244} & \frac{261}{488}\end{pmatrix},\quad H_1^{\mathrm{BFGS}}=\begin{pmatrix}\frac{77}{64} & -\frac{43}{128}\\ -\frac{43}{128} & \frac{157}{256}\end{pmatrix},\quad H_2^{\mathrm{DFP}}=H_2^{\mathrm{BFGS}}=\begin{pmatrix}\frac{4}{5} & 0\\ 0 & \frac{1}{3}\end{pmatrix}=A^{-1}.$$

Checks: $s_0^{\mathsf T}y_0=\frac{25}{8}$, $s_1^{\mathsf T}y_1=\frac{49}{120}$, $g_1^{\mathsf T}s_0=0$, $s_0^{\mathsf T}As_1=0$, $H_1y_0=s_0$.

What it shows. On a quadratic with exact line search, DFP and BFGS terminate in n steps with A-conjugate directions and $H_n=A^{-1}$.

T26 PROOF *ch3_13-15, Theorem 5.1 (replaces ch3, pp. 13–14)*

Setting.

- $f:\mathbb{R}^n\to\mathbb{R}$ of class C^2 with $m\|d\|^2\le d^{\mathsf T}\nabla^2 f(x)d\le M\|d\|^2$ for all x,d $(0<m\le M)$.
- Iterates $x_{k+1}=x_k+\alpha_k p_k$, $p_k=-B_k^{-1}\nabla f(x_k)$, from any $B_0\succ 0$, with B_{k+1} given by $B_{k+1}^{\mathrm{BFGS}}=B_k-\dfrac{B_k s_k s_k^{\mathsf T}B_k}{s_k^{\mathsf T}B_k s_k}+\dfrac{y_k y_k^{\mathsf T}}{y_k^{\mathsf T}s_k}$ and $s_k=x_{k+1}-x_k$, $y_k=\nabla f(x_{k+1})-\nabla f(x_k)$.
- α_k satisfies the weak Wolfe conditions: $f(x_k+\alpha_k p_k)\le f(x_k)+c_1\alpha_k\nabla f(x_k)^{\mathsf T}p_k$ and $\nabla f(x_k+\alpha_k p_k)^{\mathsf T}p_k\ge c_2\nabla f(x_k)^{\mathsf T}p_k$, with $0<c_1<c_2<1$.

Problem. Prove that, for every x_0, $\lim_{k\to\infty}x_k=x^*$ and $\lim_{k\to\infty}\|\nabla f(x_k)\|=0$, where x^* is the unique global minimizer.

Suggested steps.

(1) Bounds $m\|s_k\|^2\le s_k^{\mathsf T}y_k\le M\|s_k\|^2$.

(2) Lower bound on α_k and sufficient decrease.

(3) Zoutendijk series.

(4) Lower bound on $\cos\theta_k$.

(5) $\|\nabla f(x_k)\|\to 0$ and $\|x_k-x^*\|\le\frac{1}{m}\|\nabla f(x_k)\|$.

Note. Step 4 of the notes bounds $\frac{s_k^{\mathsf T}B_{k+1}s_k}{\|B_{k+1}s_k\|\|s_k\|}$, which is the angle between s_k and y_k; but $\cos\theta_{k+1}$ involves s_{k+1}. That step needs to be revised.

S1 STATEMENT *ch3, Theorem 5.3, p. 17 (no proof in the notes)*

Setting. f of class C^2; x^* a local minimizer with $\nabla^2 f(x^*)\succ 0$ and $\nabla^2 f$ Lipschitz near x^*. BFGS with unit step: $x_{k+1}=x_k-B_k^{-1}\nabla f(x_k)$, B_{k+1} given by $B_{k+1}^{\mathrm{BFGS}}=B_k-\dfrac{B_k s_k s_k^{\mathsf T}B_k}{s_k^{\mathsf T}B_k s_k}+\dfrac{y_k y_k^{\mathsf T}}{y_k^{\mathsf T}s_k}$ ($s_k=x_{k+1}-x_k$, $y_k=\nabla f(x_{k+1})-\nabla f(x_k)$).

Statement. If $x_k\to x^*$, then

$$\lim_{k\to\infty}\frac{\|x_{k+1}-x^*\|}{\|x_k-x^*\|}=0 \qquad \text{(superlinear convergence; the same holds for DFP).}$$

Problem. Be able to state the theorem, and to distinguish linear, superlinear and quadratic convergence.

E4 NUMERICAL EXAMPLE *ch3, p. 18*

Let $f(x_1,x_2)=x_1^4+x_1x_2+(1+x_2)^2$, $x_0=(1,1)^{\mathsf T}$, $B_0=I_2$.

a) Find the unique stationary point x^* of f (solve $\nabla f(x)=0$) and verify that $\nabla^2 f(x^*)\succ 0$.

b) A BFGS run from x_0 (A5; table V9) gives the errors $e_k=\|x_k-x^*\|$, $k=0,\dots,11$: 2.36756, 1.68960, 0.98483, 0.56741, 0.29966, 0.12585, 0.12863, 0.02347, $7.06\cdot10^{-3}$, $3.51\cdot10^{-4}$, $3.88\cdot10^{-5}$, $7.38\cdot10^{-9}$. Compute the ratios $\|x_{k+1}-x^*\|/\|x_k-x^*\|$ and $\|x_{k+1}-x^*\|/\|x_k-x^*\|^2$, and interpret the order of convergence.

Note. The point $x^*\approx(0.160738,-1.080369)^{\mathsf T}$ given in the notes does not make the gradient vanish ($4x_1^3+x_2\approx-1.06$), and the Hessian is indefinite there. The correct stationary point is $x^*\approx(0.695884,-1.347942)^{\mathsf T}$; see table V9.

V9 Validation table *A5 (BFGS + StrongWolfe); complements E4*

$f(x) = x_1^4 + x_1x_2 + (1+x_2)^2$, $\nabla f(x) = (4x_1^3 + x_2,\ x_1 + 2(1+x_2))^{\mathsf T}$, $x^* \approx (0.695884, -1.347942)^{\mathsf T}$; $x_0 = (1,1)^{\mathsf T}$, $c_1 = 10^{-4}$, $c_2 = 0.9$, $\varepsilon = 10^{-6}$. "eval." is the number of steps tried in A2 and $e_k = \|x_k - x^*\|$.

k	x_k	$f(x_k)$	$\|g_k\|$	α_k	eval.	e_k	e_{k+1}/e_k
0	(1, 1)	6	7.07107	0.141421	1	2.36756	0.714
1	(0.292893, 0.292893)	1.764719	2.90544	1	1	1.68960	0.583
2	(0.164425, −0.518820)	0.146958	1.23316	1	1	0.98483	0.576
3	(0.223146, −1.034137)	−0.227119	1.00174	1	1	0.56741	0.528
4	(0.406954, −1.268461)	−0.416707	1.00730	0.228383	2	0.29966	0.420
5	(0.579918, −1.396830)	−0.539472	0.65270	1	1	0.12585	1.022
6	(0.775876, −1.448679)	−0.560298	0.43681	1	1	0.12863	0.182
7	(0.683250, −1.367725)	−0.581346	0.10567	1	1	0.02347	0.301
8	(0.693411, −1.354558)	−0.582367	2.62×10^{-2}	1	1	7.06×10^{-3}	0.050
9	(0.696073, −1.348237)	−0.582445	8.98×10^{-4}	1	1	3.51×10^{-4}	0.111
10	(0.695906, −1.347974)	−0.582445	1.05×10^{-4}	1	1	3.88×10^{-5}	1.9×10^{-4}
11	(0.695884, −1.347942)	−0.582445	3.10×10^{-8}			7.38×10^{-9}	

First step in detail: $g_0 = (5,5)^{\mathsf T}$, $p_0 = (-5,-5)^{\mathsf T}$, first trial $\alpha_1 = 1/\|g_0\| = 0.141421$ (a step of length 1), accepted at once: $\phi' = -16.36$ and $|\phi'| \le 0.9 \cdot 50$. Then $s_0 = (-0.707107, -0.707107)^{\mathsf T}$, $y_0 = (-4.606602, -2.121320)^{\mathsf T}$, $s_0^{\mathsf T} y_0 = 4.757359$, the rescaling $\gamma = s_0^{\mathsf T} y_0 / y_0^{\mathsf T} y_0 = 0.184962$ and

$$H_1 = \begin{pmatrix} 0.141876 & 0.025239 \\ 0.025239 & 0.278525 \end{pmatrix}, \qquad p_1 = -H_1 g_1 = (-0.128469,\ -0.811714)^{\mathsf T}.$$

At $k = 4$ the unit step fails Armijo and Zoom accepts the quadratic interpolant 0.228383.

What it shows. After the rescaling the unit step is accepted in every iteration but one, and $e_{k+1}/e_k \to 0$: the superlinear convergence stated in S1.

Q12 Exam problem *added, 4 pts*

Superlinear convergence: the Dennis–Moré condition. Let f be of class C^2 with $\nabla^2 f$ Lipschitz with constant L near x^*, $\nabla f(x^*) = 0$, $\nabla^2 f(x^*) \succ 0$. Let $x_{k+1} = x_k + p_k$, $p_k = -B_k^{-1} \nabla f(x_k)$, with $x_k \to x^*$ and $x_k \neq x^*$.

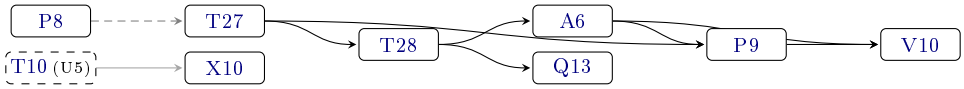
a) Prove that if $\lim_k \dfrac{\|(B_k - \nabla^2 f(x^*))p_k\|}{\|p_k\|} = 0$, then the convergence is superlinear. (Write $\nabla f(x_{k+1}) = \nabla f(x_k) + \nabla^2 f(x^*) p_k + O(\|p_k\| \max\{\|x_k - x^*\|, \|x_{k+1} - x^*\|\})$.)

b) Let $s_k = x_{k+1} - x_k$, $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$. Prove $\|y_k - \nabla^2 f(x^*) s_k\| \le L \max\{\|x_k - x^*\|, \|x_{k+1} - x^*\|\} \|s_k\|$, so that every matrix satisfying the secant equation $B_{k+1} s_k = y_k$ approximates $\nabla^2 f(x^*)$ *along* s_k.

c) Explain why b) does not imply the condition in a): which direction does a) need at iteration $k+1$? What extra property of BFGS closes this gap (state it without proof)?

U10. Optimality over convex sets

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Propositions 1.1 and 1.2 of ch4 (pp. 1–2) repeat T4 and T5; they are not listed again.

P8 Practice *added*

Let $f(x) = \frac{1}{2}\|x - z\|^2$ with $z = (2, -1)^\mathsf{T}$ and $C = [0, 1]^2$.

a) Find the minimizer $\bar{x}$ of f on C and check that $\nabla f(\bar{x}) \neq 0$.

b) Verify the variational inequality $\langle \nabla f(\bar{x}), x - \bar{x} \rangle \geq 0$ for all $x \in C$ and draw $-\nabla f(\bar{x})$ as a normal vector to C at $\bar{x}$.

c) Verify that $\bar{x} = \mathrm{proj}_C(\bar{x} - \alpha \nabla f(\bar{x}))$ for every $\alpha > 0$.

d) Now let $\Omega = \{x \in \mathbb{R}^2 : \ 1 \leq \|x\|_\infty \leq 2\}$ and $f(x) = x_1$, which has a local minimum at $(1, 0)$ on Ω. Show that the variational inequality $\langle \nabla f(1, 0), x - (1, 0) \rangle \geq 0$ fails for some $x \in \Omega$, and explain which hypothesis is violated of the result "if C is convex and f of class C^1 has a local minimum on C at $\bar{x}$, then $\langle \nabla f(\bar{x}), x - \bar{x} \rangle \geq 0$ for all $x \in C$" (T27).

T27 Proof *ch4, Proposition 1.3, p. 2*

Let $C \subset \mathbb{R}^n$ be convex and nonempty and $f : \Omega \to \mathbb{R}$ of class C^1 on an open set $\Omega \supset C$.

a) If f attains a local minimum on C at $\bar{x} \in C$, then $\langle \nabla f(\bar{x}), x - \bar{x} \rangle \geq 0$ for all $x \in C$.

b) If f is convex on C and $\bar{x} \in C$ satisfies the inequality above, then $\bar{x}$ is a global minimum of f on C.

T28 Proof *ch4, Proposition 1.4, pp. 2–3*

Let $C \subset \mathbb{R}^n$ be convex, closed and nonempty.

a) Prove that for every $y \in \mathbb{R}^n$ there is a unique $\mathrm{proj}_C(y) \in C$ minimizing $\|x - y\|$ over C, and that $p = \mathrm{proj}_C(y)$ if and only if $p \in C$ and $\langle y - p, x - p \rangle \leq 0$ for all $x \in C$.

b) Let f be convex of class C^1. Prove that, for any $\alpha > 0$, $\bar{x} \in C$ minimizes f on C if and only if $\bar{x} = \mathrm{proj}_C(\bar{x} - \alpha \nabla f(\bar{x}))$.

c) Prove that proj_C is nonexpansive: $\|\mathrm{proj}_C(y) - \mathrm{proj}_C(z)\| \leq \|y - z\|$.

Note. The notes use characterization a) without proving it; c) is the ingredient for analysing projected gradient (A6).

X10 Proposed exercise *ch4, p. 13*

Setting. $C \subseteq \mathbb{R}^n$ nonempty, closed and convex; $f : \mathbb{R}^n \to \mathbb{R}$ continuous on C and strongly convex with parameter $\alpha > 0$ on C:

$$f(\lambda x + (1 - \lambda) y) \leq \lambda f(x) + (1 - \lambda) f(y) - \tfrac{\alpha}{2} \lambda (1 - \lambda) \|x - y\|^2 \qquad (x, y \in C, \ \lambda \in [0, 1]).$$

Problem. Prove that $\min_{x \in C} f(x)$ has a unique optimal solution.

Hint. (i) $f - \frac{\alpha}{2}\|\cdot\|^2$ is convex on C, hence bounded below on C by an affine function, so minimizing sequences are bounded. (ii) For uniqueness use the inequality with $\lambda = \frac{1}{2}$.

A6 Pseudocode *spectral projected gradient (SPG; Birgin–Martínez–Raydan)*

SPG$(f,\ \nabla f,\ \mathrm{proj}_C,\ x_0;\ M, \varepsilon, \alpha_{\min}, \alpha_{\max}, \gamma, \sigma_1, \sigma_2)$ $\quad C$ closed convex

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1: $x_0 \leftarrow \mathrm{proj}_C(x_0)$; $\quad g_0 \leftarrow \nabla f(x_0)$; $\quad \alpha_0 \leftarrow \min\{\alpha_{\max}, \max\{\alpha_{\min},\ 1/\|\mathrm{proj}_C(x_0 - g_0) - x_0\|_\infty\}\}$
2: **for** $k = 0, 1, \ldots, k_{\max}$ **do**
3: $\quad$ **if** $\|\mathrm{proj}_C(x_k - g_k) - x_k\|_\infty \le \varepsilon$ **then return** x_k $\quad$ (projected-gradient residual, T28 b)
4: $\quad d_k \leftarrow \mathrm{proj}_C(x_k - \alpha_k g_k) - x_k$ $\quad$ (feasible descent direction)
5: $\quad f_{\max} \leftarrow \max\{f(x_{k-j}) :\ 0 \le j \le \min(k, M-1)\}$; $\quad \lambda \leftarrow 1$
6: $\quad$ **while** $f(x_k + \lambda d_k) > f_{\max} + \gamma\lambda\, g_k^{\mathsf T} d_k$ **do** $\quad$ (nonmonotone Armijo)
7: $\quad\quad \lambda_t \leftarrow -\dfrac{\lambda^2\, g_k^{\mathsf T} d_k}{2\,[f(x_k + \lambda d_k) - f(x_k) - \lambda g_k^{\mathsf T} d_k]}$; $\quad$ **if** $\lambda_t \in [\sigma_1\lambda, \sigma_2\lambda]$ **then** $\lambda \leftarrow \lambda_t$ **else** $\lambda \leftarrow \lambda/2$
8: $\quad x_{k+1} \leftarrow x_k + \lambda d_k$; $\quad g_{k+1} \leftarrow \nabla f(x_{k+1})$; $\quad s \leftarrow x_{k+1} - x_k$; $\quad y \leftarrow g_{k+1} - g_k$
9: $\quad$ **if** $s^{\mathsf T} y \le 0$ **then** $\alpha_{k+1} \leftarrow \alpha_{\max}$ **else** $\alpha_{k+1} \leftarrow \min\{\alpha_{\max}, \max\{\alpha_{\min},\ s^{\mathsf T} s / s^{\mathsf T} y\}\}$ $\quad$ (Barzilai–Borwein)

Defaults: $M = 10$, $\gamma = 10^{-4}$, $\sigma_1 = 0.1$, $\sigma_2 = 0.9$, $\alpha_{\min} = 10^{-30}$, $\alpha_{\max} = 10^{30}$. For a box $C = \prod_i [l_i, u_i]$: $[\mathrm{proj}_C(y)]_i = \min\{\max\{y_i, l_i\}, u_i\}$. Since x_k and $\mathrm{proj}_C(x_k - \alpha_k g_k)$ lie in C, every point $x_k + \lambda d_k$, $\lambda \in [0,1]$, is feasible: one projection per iteration. Line 3 is the stopping test of bound-constrained codes (`pgtol` in L-BFGS-B). The spectral step $s^{\mathsf T}s / s^{\mathsf T}y$ is the inverse of a Rayleigh quotient of the averaged Hessian, so it adapts the step to the curvature without knowing L. The nonmonotone test lets f increase for a few iterations, which the spectral step needs to be effective. This is SPG2 (R package `BB`, TANGO project). With $M = 1$ and $\alpha_k \equiv \alpha$ fixed (line 9 removed), A6 is the fixed-step projected gradient of ch4, Prop. 1.4; if ∇f is Lipschitz with constant L and $\alpha < 2/L$, then $\lambda = 1$ is always accepted (for $\gamma \le 1 - L\alpha/2$).

P9 Practice *added*

Projected gradient by hand. Let $f(x_1, x_2) = 2x_1^2 + x_2^2 - 2x_1x_2 + 2x_1 - 4x_2$ (the one in X2) on the box $C = [0,2]^2$, with proj_C computed componentwise.

a) Verify with the variational inequality $\langle \nabla f(\bar x), x - \bar x\rangle \ge 0$ $(\forall x \in C)$ that $\bar x = (\tfrac12, 2)^{\mathsf T}$ is the minimizer of f on C, and find $f(\bar x)$.

b) With $L = \lambda_{\max}(\nabla^2 f)$, check that the fixed step $\alpha = \tfrac15$ satisfies $\alpha < 2/L$.

c) Run A6 (defaults, $\varepsilon = 10^{-6}$) from $x_0 = (0,0)^{\mathsf T}$: for each k record g_k, α_k, $x_k - \alpha_k g_k$, d_k, $f_{\max}$, the accepted λ and the residual of line 3 (table V10 a). At which iteration does the nonmonotone test accept a step that a monotone test ($M = 1$) would reject?

d) Run the fixed-step version $x_{k+1} = \mathrm{proj}_C(x_k - \tfrac15 \nabla f(x_k))$ from the same point (table V10 b). Identify the iteration from which the bound $x_2 \le 2$ stays active forever, explain why from then on the step $\|x_{k+1} - x_k\|$ is multiplied by $\tfrac15$ at each iteration, and compare with a).

V10 Validation table *A6 on $C = [0,2]^2$; solves P9*

$f(x) = 2x_1^2 + x_2^2 - 2x_1x_2 + 2x_1 - 4x_2$, $\nabla f(x) = (4x_1 - 2x_2 + 2,\ -2x_1 + 2x_2 - 4)^{\mathsf T}$, $C = [0,2]^2$, $\bar x = (\tfrac12, 2)^{\mathsf T}$, $f(\bar x) = -\tfrac92$; from $x_0 = (0,0)^{\mathsf T}$.

a) A6 (SPG, $M = 10$). $\alpha_0 = 1/\|\mathrm{proj}_C(x_0 - g_0) - x_0\|_\infty = 1/\|(0,2)\|_\infty = \tfrac12$; $r_k = \|\mathrm{proj}_C(x_k - g_k) - x_k\|_\infty$.

k	x_k	$f(x_k)$	g_k	r_k	α_k	$x_k - \alpha_k g_k$	d_k	$f_{\max}$, λ
0	$(0,\ 0)$	0	$(2,\ -4)$	2	$\frac{1}{2}$	$(-1,\ 2)$	$(0,\ 2)$	0, 1
1	$(0,\ 2)$	-4	$(-2,\ 0)$	2	$\frac{1}{2}$	$(1,\ 2)$	$(1,\ 0)$	0, 1
2	$(1,\ 2)$	-4	$(2,\ -2)$	1	$\frac{1}{4}$	$(\frac{1}{2},\ \frac{5}{2})$	$(-\frac{1}{2},\ 0)$	0, 1
3	$(\frac{1}{2},\ 2)$	$-\frac{9}{2}$	$(0,\ -1)$	0				

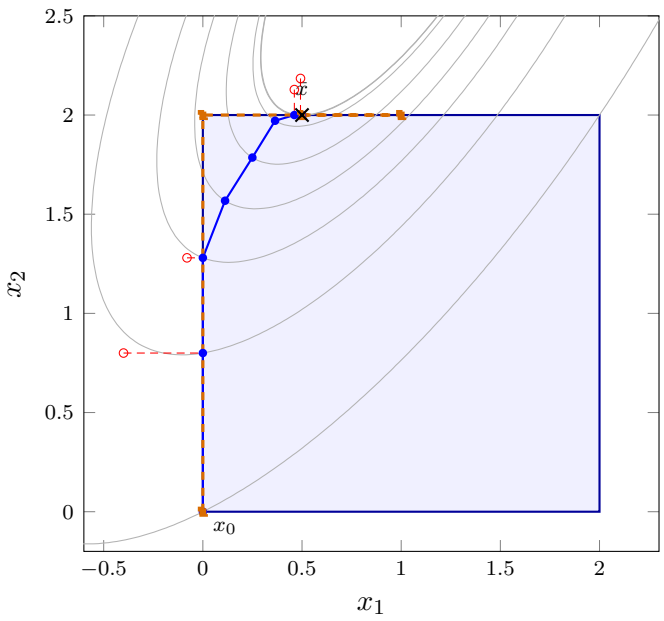
Spectral steps: $s_0 = (0,2)$, $y_0 = (-4,4)$, $\alpha_1 = \frac{4}{8} = \frac{1}{2}$; $s_1 = (1,0)$, $y_1 = (4,-2)$, $\alpha_2 = \frac{1}{4}$. At $k=1$, $g_1^{\mathsf{T}} d_1 = -2$ and $f(x_1 + d_1) = -4 = f(x_1)$: the step does not decrease f, so the monotone test ($M = 1$, bound -4.0002) would reject it and backtrack, while the nonmonotone test (bound $f_{\max} - 0.0002$ with $f_{\max} = 0$) accepts it. A6 stops at $k = 3$ with the exact solution.

b) Fixed step $\alpha = \frac{1}{5}$ (ch4, Prop. 1.4). Rows 0–3 in exact fractions; $y_k = x_k - \alpha g_k$, $x_{k+1} = \mathrm{proj}_C(y_k)$, $r_k = \|x_{k+1} - x_k\|$.

k	x_k	g_k	$y_k = x_k - \alpha g_k$	$f(x_k)$	r_k
0	$(0,\ 0)$	$(2,\ -4)$	$(-\frac{2}{5},\ \frac{4}{5})$	0	0.8
1	$(0,\ \frac{4}{5})$	$(\frac{2}{5},\ -\frac{12}{5})$	$(-\frac{2}{25},\ \frac{32}{25})$	-2.56	0.48
2	$(0,\ \frac{32}{25})$	$(-\frac{14}{25},\ -\frac{36}{25})$	$(\frac{14}{125},\ \frac{196}{125})$	-3.4816	0.3090
3	$(\frac{14}{125},\ \frac{196}{125})$	$(-\frac{86}{125},\ -\frac{136}{125})$	$(\frac{156}{625},\ \frac{1116}{625})$	-3.915520	0.2575
4	$(0.249600,\ 1.785600)$	$(-0.572800,\ -0.928000)$	$(0.364160,\ 1.971200)$	-4.221604	0.2181
5	$(0.364160,\ 1.971200)$	$(-0.485760,\ -0.785920)$	$(0.461312,\ 2.128384)$	-4.441290	0.1013
6	$(0.461312,\ 2)$	$(-0.154752,\ -0.922624)$	$(0.492262,\ 2.184525)$	-4.497006	3.095×10^{-2}
7	$(0.492262,\ 2)$	$(-0.030950,\ -0.984525)$	$(0.498452,\ 2.196905)$	-4.499880	6.190×10^{-3}
8	$(0.498452,\ 2)$	$(-0.006190,\ -0.996905)$	$(0.499690,\ 2.199381)$	-4.499995	1.238×10^{-3}
9	$(0.499690,\ 2)$	$(-0.001238,\ -0.999381)$	$(0.499938,\ 2.199876)$	-4.500000	2.476×10^{-4}

At $k = 0, 1$ the projection clips x_1 to 0; from $k = 5$ it clips x_2 to 2 and the bound $x_2 \le 2$ stays active forever. On that face, $x_1^{k+1} = \frac{1}{5} x_1^k + \frac{2}{5}$, so $r_{k+1}/r_k = \frac{1}{5}$: linear convergence with no end in finite arithmetic, against 3 iterations for a). At $\bar{x} = (\frac{1}{2}, 2)$: $g = (0, -1)$, the outward normal of the face $x_2 = 2$, and the variational inequality holds.

Figure.



Box $C = [0,2]^2$, level curves of f through the iterates and through $f(\bar{x}) = -\frac{9}{2}$. In blue, the fixed-step iterates of V10 b); in red, the points $y_k = x_k - \alpha g_k$ that fall outside C and their projection (dashed line). First the bound $x_1 \geq 0$ becomes active, then the bound $x_2 \leq 2$, and on that face the method becomes a one-variable gradient method with factor $\frac{1}{5}$. In orange (squares), the SPG iterates of V10 a): $(0,0) \to (0,2) \to (1,2) \to \bar{x}$; the second step moves along the face without decreasing f and is accepted by the nonmonotone test.

Q13 EXAM PROBLEM *added, 4 pts*

Optimality and projection on the simplex. Let $\Delta_n = \{x \in \mathbb{R}^n : \ x \geq 0,\ \sum_i x_i = 1\}$ and $f : \mathbb{R}^n \to \mathbb{R}$ convex and of class C^1.

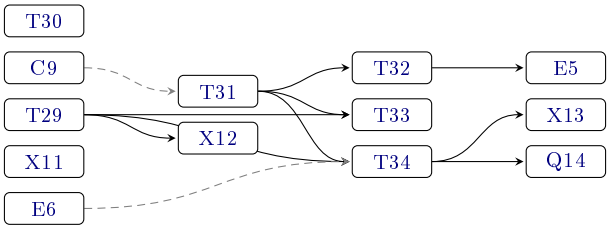
a) From the variational inequality $\nabla f(\bar{x})^{\mathsf{T}}(x - \bar{x}) \geq 0$ for all $x \in \Delta_n$, prove that $\bar{x}$ minimizes f on Δ_n if and only if $\partial_i f(\bar{x}) = \min_j \partial_j f(\bar{x})$ for every i with $\bar{x}_i > 0$.

b) Apply a) to $f(x) = \frac{1}{2}\|x - y\|^2$ to prove that $\mathrm{proj}_{\Delta_n}(y)_i = \max\{y_i - \theta, 0\}$, where θ is the unique number with $\sum_i \max\{y_i - \theta, 0\} = 1$.

c) Compute $\mathrm{proj}_{\Delta_3}(y)$ for $y = (0.8,\ 0.6,\ -0.2)$. Then, for $f(x) = \frac{1}{2}x^{\mathsf{T}}Dx$ with $D = \mathrm{diag}(1,2,3)$, perform one projected-gradient step $x_1 = \mathrm{proj}_{\Delta_3}(x_0 - \alpha \nabla f(x_0))$ from $x_0 = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ with $\alpha = \frac{1}{2}$, and check whether x_1 satisfies the condition of a).

U11. Tangent cone and constraint qualifications

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C9 COUNTEREXAMPLE *ch4, p. 4*

Definitions.

- *Tangent cone:* $T(a,C) = \{d : \ \exists\, d_k \to d,\ t_k \to 0^+ \text{ with } a + t_k d_k \in C\}$.
- *Linearized cone:* $L(x,C) = \{d : \ \langle \nabla g_i(x), d\rangle \leq 0 \ (i \in I(x)),\ \langle \nabla h_j(x), d\rangle = 0 \ (\forall j)\}$.
- *KKT multipliers at $\bar{x}$:* $\lambda_i \geq 0$, $\mu_j \in \mathbb{R}$ with $\nabla f(\bar{x}) + \sum_i \lambda_i \nabla g_i(\bar{x}) + \sum_j \mu_j \nabla h_j(\bar{x}) = 0$ and $\lambda_i g_i(\bar{x}) = 0$.

Problem. Let $C = \{x \in \mathbb{R}^2 : \ g_1(x) = x_1^2 + x_2 \leq 0,\ g_2(x) = x_1^2 - x_2 \leq 0\}$ and $\bar{x} = (0,0)$.

a) Prove that $C = \{(0,0)\}$, that $T(\bar{x}, C) = \{0\}$ and that $L(\bar{x}, C) = \{d : \ d_2 = 0\}$.

b) With $f(x) = x_1$, check that $\bar{x}$ is a global minimum on C and that there are no KKT multipliers. Relate this to the failure of the constraint qualification $T(\bar{x}, C) = L(\bar{x}, C)$, which is the hypothesis under which local minima admit KKT multipliers.

T29 PROOF *ch4, Proposition 2.1, p. 3*

Definitions.

- *Tangent cone:* $T(a,C) = \{d:\ \exists\, d_k \to d,\ t_k \to 0^+ \text{ with } a + t_k d_k \in C\}$.

Problem. Let $C \subset \mathbb{R}^n$ be nonempty. Prove that, for every $a \in C$, $T(a,C)$ is a closed cone containing the origin.

Note. The notes close the closedness part with "selecting a suitable diagonal"; the diagonal subsequence must be constructed explicitly.

T30 PROOF *ch4, Theorem 2.1, p. 3*

Definitions.

- *Tangent cone:* $T(a,C) = \{d:\ \exists\, d_k \to d,\ t_k \to 0^+ \text{ with } a + t_k d_k \in C\}$.

Problem. Let $C \subset \mathbb{R}^n$. If $f : \mathbb{R}^n \to \mathbb{R}$ of class C^1 attains a local minimum on C at $a \in C$, prove that $\langle \nabla f(a), d\rangle \geq 0$ for all $d \in T(a,C)$.

T31 PROOF *ch4, Proposition 2.2, p. 4*

Setting. Let $C = \{x \in \mathbb{R}^n :\ g_i(x) \leq 0\ (i = 1,\dots,p),\ h_j(x) = 0\ (j = 1,\dots,q)\}$ with g_i, h_j of class C^1, and let $I(x) = \{i :\ g_i(x) = 0\}$ be the set of active indices at $x \in C$.

Definitions.

- *Tangent cone:* $T(a,C) = \{d:\ \exists\, d_k \to d,\ t_k \to 0^+ \text{ with } a + t_k d_k \in C\}$.
- *Linearized cone:* $L(x,C) = \{d:\ \langle \nabla g_i(x), d\rangle \leq 0\ (i \in I(x)),\ \langle \nabla h_j(x), d\rangle = 0\ (\forall j)\}$.

Problem. Prove that $T(x,C) \subset L(x,C)$ for all $x \in C$.

T32 PROOF *ch4, Proposition 2.3, p. 5*

Setting. Let $C = \{x \in \mathbb{R}^n :\ g_i(x) \leq 0\ (i = 1,\dots,p),\ h_j(x) = 0\ (j = 1,\dots,q)\}$ with g_i, h_j of class C^1, and let $I(x) = \{i :\ g_i(x) = 0\}$ be the set of active indices at $x \in C$.

Definitions.

- *Tangent cone:* $T(a,C) = \{d:\ \exists\, d_k \to d,\ t_k \to 0^+ \text{ with } a + t_k d_k \in C\}$.
- *Linearized cone:* $L(x,C) = \{d:\ \langle \nabla g_i(x), d\rangle \leq 0\ (i \in I(x)),\ \langle \nabla h_j(x), d\rangle = 0\ (\forall j)\}$.

Problem. Suppose $g_i(x) = a_i^{\mathsf{T}} x - b_i$ and $h_j(x) = a_j'^{\mathsf{T}} x - b_j'$ are affine. Prove that $T(x,C) = L(x,C)$ for all $x \in C$.

E5 NUMERICAL EXAMPLE *ch4, p. 11 (solved exercise)*

Setting. $C = \{x \in \mathbb{R}^2 :\ x_1 + 2x_2 \leq 4,\ x_1 \geq 0,\ x_2 \geq 0\}$, $\bar{x} = (0,2)^{\mathsf{T}}$. Linearized cone: $L(\bar{x},C) = \{d:\ \langle \nabla g_i(\bar{x}), d\rangle \leq 0,\ i \text{ active}\}$; for affine constraints $T = L$ (T32).

Problem.

a) Identify the active constraints at $\bar{x}$.

b) Compute $L(\bar{x},C)$ and conclude $T(\bar{x},C) = \{d:\ d_1 + 2d_2 \leq 0,\ d_1 \geq 0\}$.

c) Draw C and the cone translated to $\bar{x}$.

X11 PROPOSED EXERCISE *ch4, p. 12*

For $C = \{x \in \mathbb{R}^2 :\ x_2 \geq x_1^2,\ x_1^2 + x_2^2 \leq 2\}$, compute the tangent cone $T(\bar{x}, C) = \{d :\ \exists\, d_k \to d,\ t_k \to 0^+,\ \bar{x} + t_k d_k \in C\}$ at the intersection point $\bar{x} = (1,1)^\mathsf{T}$.

X12 PROPOSED EXERCISE *ch4, p. 12*

Definitions.

- *Tangent cone:* $T(a, C) = \{d :\ \exists\, d_k \to d,\ t_k \to 0^+ \text{ with } a + t_k d_k \in C\}$.

Problem. Prove that if $C \subset \mathbb{R}^n$ is closed and convex and $\bar{x} \in C$, then $T(\bar{x}, C) = \mathrm{cl}\big(\bigcup_{\gamma>0} \gamma(C - \bar{x})\big)$.

T33 PROOF *ch4, Proposition 2.4, pp. 5–6*

Setting. $C = \{x :\ g_i(x) \leq 0,\ i = 1, \ldots, p\}$ with g_i convex of class C^1; $I(x) = \{i :\ g_i(x) = 0\}$. Slater's condition holds: there is $\hat{x}$ with $g_i(\hat{x}) < 0$ for all i.

Definitions.

- *Tangent cone:* $T(a, C) = \{d :\ \exists\, d_k \to d,\ t_k \to 0^+ \text{ with } a + t_k d_k \in C\}$.
- *Linearized cone:* $L(x, C) = \{d :\ \langle \nabla g_i(x), d\rangle \leq 0,\ i \in I(x)\}$.

Problem. Prove that $T(x, C) = L(x, C)$ for all $x \in C$.

Note. The step "by Taylor's theorem there is t_ϵ" must treat active constraints (sign of the directional derivative) and inactive constraints (continuity) separately.

E6 NUMERICAL EXAMPLE *ch4, p. 12 (solved exercise)*

Setting. $C = \{x \in \mathbb{R}^2 :\ x_2 \geq (1 - x_1)^3,\ x_2 \geq 0\}$, $\bar{x} = (1,0)^\mathsf{T}$, written with $g_1(x) = (1-x_1)^3 - x_2 \leq 0$ and $g_2(x) = -x_2 \leq 0$. MFCQ (no equalities): there is d with $\langle \nabla g_i(\bar{x}), d\rangle < 0$ for every active i.

Problem.

a) Check that both constraints are active at $\bar{x}$ and compute their gradients.

b) Show that $d = (0,1)^\mathsf{T}$ satisfies MFCQ.

T34 PROOF *ch4, Theorem 2.2, p. 6*

Setting. Let $C = \{x \in \mathbb{R}^n :\ g_i(x) \leq 0\ (i = 1, \ldots, p),\ h_j(x) = 0\ (j = 1, \ldots, q)\}$ with g_i, h_j of class C^1, and let $I(x) = \{i :\ g_i(x) = 0\}$ be the set of active indices at $x \in C$.

Definitions.

- *Tangent cone:* $T(a, C) = \{d :\ \exists\, d_k \to d,\ t_k \to 0^+ \text{ with } a + t_k d_k \in C\}$.
- *Linearized cone:* $L(x, C) = \{d :\ \langle \nabla g_i(x), d\rangle \leq 0\ (i \in I(x)),\ \langle \nabla h_j(x), d\rangle = 0\ (\forall j)\}$.
- *MFCQ at $\bar{x}$:* $\{\nabla h_j(\bar{x})\}$ linearly independent and there is d with $\langle \nabla g_i(\bar{x}), d\rangle < 0\ (i \in I(\bar{x}))$ and $\langle \nabla h_j(\bar{x}), d\rangle = 0\ (\forall j)$.

Problem. Prove that if $\bar{x} \in C$ satisfies MFCQ, then $T(\bar{x}, C) = L(\bar{x}, C)$.

Tool. Implicit function theorem: if $F : \mathbb{R}^q \times \mathbb{R}^s \to \mathbb{R}^q$ is of class C^1, $F(\bar{u}, \bar{v}) = 0$ and $D_u F(\bar{u}, \bar{v})$ is invertible, then there are a neighbourhood $V \ni \bar{v}$ and a C^1 map $u : V \to \mathbb{R}^q$ with $u(\bar{v}) = \bar{u}$ and $F(u(v), v) = 0$ for $v \in V$.

Note. One must also check that the curve $\gamma_k(t)$ respects the inactive constraints and that $(\gamma_k(t) - \bar{x})/t \to d^{(k)}$, which is what gives $d^{(k)} \in T(\bar{x}, C)$.

X13 Proposed exercise *ch4, p. 12*

Definitions.

- *Tangent cone:* $T(a,C)=\{d:\ \exists\, d_k\to d,\ t_k\to 0^+ \text{ with } a+t_kd_k\in C\}$.
- *Linearized cone:* $L(x,C)=\{d:\ \langle\nabla g_i(x),d\rangle\le 0\ (i\in I(x)),\ \langle\nabla h_j(x),d\rangle=0\ (\forall j)\}$.
- *LICQ at $\bar x$:* the gradients $\{\nabla g_i(\bar x)\}_{i\in I(\bar x)}\cup\{\nabla h_j(\bar x)\}_j$ are linearly independent.
- *MFCQ at $\bar x$:* $\{\nabla h_j(\bar x)\}$ linearly independent and there is d with $\langle\nabla g_i(\bar x),d\rangle<0$ $(i\in I(\bar x))$ and $\langle\nabla h_j(\bar x),d\rangle=0$ $(\forall j)$.
- *KKT multipliers at $\bar x$:* $\lambda_i\ge 0$, $\mu_j\in\mathbb{R}$ with $\nabla f(\bar x)+\sum_i\lambda_i\nabla g_i(\bar x)+\sum_j\mu_j\nabla h_j(\bar x)=0$ and $\lambda_i g_i(\bar x)=0$.

Setting. $C=\{x\in\mathbb{R}^2:\ (x_1-1)^3+x_2\le 0,\ x_1\ge 0,\ x_2\ge 0\}$, $\bar x=(1,0)^{\mathsf T}$.

Problem.

a) Decide whether LICQ and MFCQ hold at $\bar x$.

b) Compute $T(\bar x,C)$ and $L(\bar x,C)$.

c) Show that for $f(x)=-x_1$ the point $\bar x$ is a global minimum on C without KKT multipliers.

Q14 Exam problem *added, 4 pts*

A cusp: tangent cone, LICQ and KKT. Let $\Omega=\{x\in\mathbb{R}^2:\ g_1(x)=x_2-x_1^3\le 0,\ g_2(x)=-x_2\le 0\}$ and $x^*=(0,0)$.

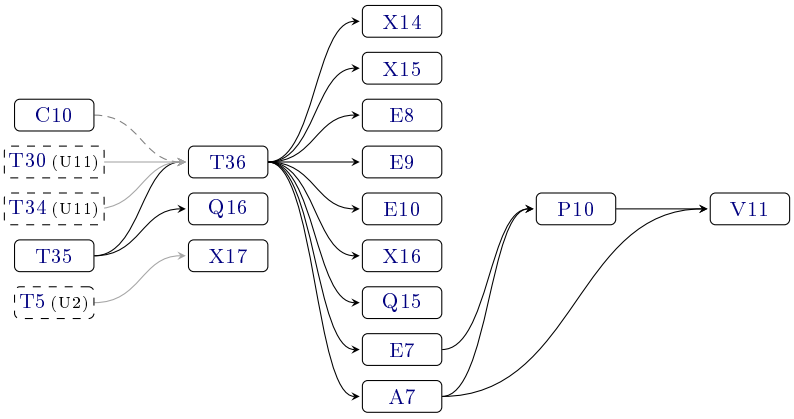
a) Compute the tangent cone $T_\Omega(x^*)$ and the linearized cone $\mathcal{F}(x^*)=\{d:\ \nabla g_i(x^*)^{\mathsf T}d\le 0,\ i=1,2\}$, and show that they differ. Check that LICQ and MFCQ fail at x^*.

b) Show that x^* is the global minimizer of $f(x)=x_1$ on Ω, but that there are no multipliers $\lambda\ge 0$ with $\nabla f(x^*)+\lambda_1\nabla g_1(x^*)+\lambda_2\nabla g_2(x^*)=0$.

c) Verify that the geometric condition $\nabla f(x^*)^{\mathsf T}d\ge 0$ for all $d\in T_\Omega(x^*)$ does hold, and explain why Slater's condition cannot be invoked here.

U12. Farkas' lemma and KKT conditions

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C10 COUNTEREXAMPLE *ch4, p. 7*

Definitions.

- *Tangent cone:* $T(a,C) = \{d:\ \exists\, d_k \to d,\ t_k \to 0^+ \text{ with } a + t_k d_k \in C\}$.
- *Linearized cone:* $L(x,C) = \{d:\ \langle \nabla g_i(x), d\rangle \le 0\ (i \in I(x)),\ \langle \nabla h_j(x), d\rangle = 0\ (\forall j)\}$.
- *LICQ at $\bar{x}$:* the gradients $\{\nabla g_i(\bar{x})\}_{i\in I(\bar{x})} \cup \{\nabla h_j(\bar{x})\}_j$ are linearly independent.
- *MFCQ at $\bar{x}$:* $\{\nabla h_j(\bar{x})\}$ linearly independent and there is d with $\langle \nabla g_i(\bar{x}), d\rangle < 0$ $(i \in I(\bar{x}))$ and $\langle \nabla h_j(\bar{x}), d\rangle = 0$ $(\forall j)$.

Problem. Consider $\min x_1$ subject to $g_1(x) = -x_2 \le 0$ and $g_2(x) = x_2 - x_1^3 \le 0$.

a) Prove that $\bar{x} = (0,0)^{\mathsf{T}}$ is the only optimum.

b) Show that there are no KKT multipliers at $\bar{x}$: $\lambda_1, \lambda_2 \ge 0$ with $\nabla f(\bar{x}) + \lambda_1 \nabla g_1(\bar{x}) + \lambda_2 \nabla g_2(\bar{x}) = 0$.

c) Compute $T(\bar{x}, C)$ and $L(\bar{x}, C)$, and verify that MFCQ and LICQ fail at $\bar{x}$.

T35 PROOF *ch4, Lemma 3.1 (Farkas), p. 6 (no proof in the notes)*

Let $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{q\times n}$ and $c \in \mathbb{R}^n$. Prove that the following are equivalent:

i) $Ad \le 0$, $Bd = 0$ implies $c^{\mathsf{T}} d \ge 0$.

ii) There exist $u \in \mathbb{R}^m$, $v \in \mathbb{R}^q$ with $u \ge 0$ and $c + A^{\mathsf{T}} u + B^{\mathsf{T}} v = 0$.

The separation theorem for a point and a closed convex set may be cited; one must prove that the cone $\{A^{\mathsf{T}} u + B^{\mathsf{T}} v:\ u \ge 0\}$ is closed.

T36 PROOF *ch4, Theorem 3.1 (KKT), p. 7*

Setting. Let $C = \{x \in \mathbb{R}^n:\ g_i(x) \le 0\ (i = 1,\dots,p),\ h_j(x) = 0\ (j = 1,\dots,q)\}$ with g_i, h_j of class C^1, and let $I(x) = \{i:\ g_i(x) = 0\}$ be the set of active indices at $x \in C$. $\bar{x}$ is a local minimum of $f \in C^1$ on C, and satisfies the constraint qualification $T(\bar{x}, C) = L(\bar{x}, C)$ (true, for instance, under MFCQ, LICQ, Slater, or if C is a polyhedron).

Definitions.

- *Tangent cone:* $T(a,C) = \{d:\ \exists\, d_k \to d,\ t_k \to 0^+ \text{ with } a + t_k d_k \in C\}$.
- *Linearized cone:* $L(x,C) = \{d:\ \langle \nabla g_i(x), d\rangle \le 0\ (i \in I(x)),\ \langle \nabla h_j(x), d\rangle = 0\ (\forall j)\}$.

Problem. Prove that there exist $\lambda_1, \dots, \lambda_p \ge 0$ and $\mu_1, \dots, \mu_q \in \mathbb{R}$ such that

$$\nabla f(\bar{x}) + \sum_{i=1}^{p} \lambda_i \nabla g_i(\bar{x}) + \sum_{j=1}^{q} \mu_j \nabla h_j(\bar{x}) = 0, \qquad \lambda_i g_i(\bar{x}) = 0 \ \ (i = 1,\dots,p).$$

Tool. Farkas' lemma (T35): $Ad \le 0,\ Bd = 0 \Rightarrow c^{\mathsf{T}} d \ge 0$ holds if and only if there are $u \ge 0$ and v with $c + A^{\mathsf{T}} u + B^{\mathsf{T}} v = 0$.

E7 NUMERICAL EXAMPLE *ch4, p. 11 (solved exercise)*

Let $f(x) = x_1^2 + x_2^2 - 4x_1 - 6x_2$, $g_1(x) = 2 - x_1 - x_2 \le 0$, $g_2(x) = x_1 + 2x_2 - 5 \le 0$. Write the KKT conditions, solve the case g_2 active, g_1 inactive, and justify that $(1.4,\ 1.8)^{\mathsf{T}}$ with $\lambda = (0, \frac{6}{5})^{\mathsf{T}}$ is the unique global minimum.

A7 Pseudocode *primal active-set method for convex QP (N&W, Alg. 16.3)*

Solves $\min \tfrac{1}{2}x^{\mathsf T}Gx + c^{\mathsf T}x$ s.t. $a_i^{\mathsf T}x \le b_i$ $(i = 1, \dots, m)$, with $G \succ 0$, from a feasible x_0.

ActiveSetQP$(G,\ c,\ \{a_i, b_i\},\ x_0,\ \mathcal{W}_0)$ $\quad x_0$ feasible, $\mathcal{W}_0 \subseteq \{i :\ a_i^{\mathsf T}x_0 = b_i\}$

1: **for** $k = 0, 1, 2, \dots$ **do**

2: solve $\begin{pmatrix} G & A_{\mathcal{W}}^{\mathsf T} \\ A_{\mathcal{W}} & 0 \end{pmatrix}\begin{pmatrix} p \\ \lambda_{\mathcal{W}} \end{pmatrix} = \begin{pmatrix} -(Gx_k + c) \\ 0 \end{pmatrix}$ $(A_{\mathcal{W}}$: rows $a_i^{\mathsf T},\ i \in \mathcal{W}_k)$

3: **if** $\|p\| \le$ tol **then**

4: **if** $\lambda_i \ge -$tol for all $i \in \mathcal{W}_k$ **then return** (x_k, λ) (KKT point = global minimum)

5: $j \leftarrow \arg\min_{i \in \mathcal{W}_k} \lambda_i$; $\quad \mathcal{W}_{k+1} \leftarrow \mathcal{W}_k \setminus \{j\}$; $\quad x_{k+1} \leftarrow x_k$ (drop a constraint)

6: **else**

7: $\alpha_k \leftarrow \min\Big\{1,\ \min_{i \notin \mathcal{W}_k,\ a_i^{\mathsf T}p > 0} \dfrac{b_i - a_i^{\mathsf T}x_k}{a_i^{\mathsf T}p}\Big\}$; $\quad x_{k+1} \leftarrow x_k + \alpha_k p$

8: **if** $\alpha_k < 1$ **then** $\mathcal{W}_{k+1} \leftarrow \mathcal{W}_k \cup \{\text{blocking } i\}$ **else** $\mathcal{W}_{k+1} \leftarrow \mathcal{W}_k$

Each iteration solves one equality-constrained QP on the working set, and the signs of the KKT multipliers (T36) decide when to stop and which constraint to drop. Only the working sets on the path are visited, not all 2^m. In an implementation: tol $\approx 10^{-10}$ relative to the data; the system of line 2 is solved by the null-space method with a QR factorization of $A_{\mathcal{W}}^{\mathsf T}$ that is *updated* when one constraint enters or leaves ($O(n^2)$ instead of $O(n^3)$ per iteration); a feasible x_0 comes from a phase-I linear program; the rows a_i, $i \in \mathcal{W}_k$, are kept linearly independent, and ties in the ratio test of line 7 are broken by a fixed rule to avoid cycling. This is the method of MATLAB's `quadprog` (`'active-set'`) and of qpOASES (which adds warm starts for sequences of QPs). The dual method of Goldfarb–Idnani (R and Python package `quadprog`) works on the same equality-constrained subproblems but needs no feasible x_0.

P10 Practice *added*

Let $\min f(x) = x_1^2 + x_2^2 - 4x_1 - 6x_2$ s.t. $g_1(x) = 2 - x_1 - x_2 \le 0$, $g_2(x) = x_1 + 2x_2 - 5 \le 0$.

a) For each of the four sets $\mathcal{A} \subseteq \{1, 2\}$, solve the stationarity system with $g_i = 0$ for $i \in \mathcal{A}$, and decide whether it gives a KKT point ($\lambda_i \ge 0$ for $i \in \mathcal{A}$, $g_i \le 0$ for $i \notin \mathcal{A}$).

b) Run A7 from $x_0 = (2, 0)^{\mathsf T}$ with $\mathcal{W}_0 = \{1\}$ (write $g_i \le 0$ as $a_i^{\mathsf T}x \le b_i$). Record x_k, $\mathcal{W}_k$, p, $\lambda_{\mathcal{W}}$, α_k and the blocking constraint.

c) Compare the sets visited in b) with the four in a), and explain the role of a negative multiplier.

V11 Validation table *A7; solves P10 a) and b)*

$f = x_1^2 + x_2^2 - 4x_1 - 6x_2$ $(G = 2I,\ c = (-4, -6))$, $a_1 = (-1, -1)$, $b_1 = -2$, $a_2 = (1, 2)$, $b_2 = 5$.

a) The four candidate active sets.

$\mathcal{A}$	x	$\lambda_{\mathcal{A}}$	$(g_1(x),\ g_2(x))$	$f(x)$	decision
$\emptyset$	$(2,\ 3)$	—	$(-3,\ 3)$	-13	$g_2 > 0$: infeasible
$\{1\}$	$(0.5,\ 1.5)$	$\lambda_1 = -3$	$(0,\ -1.5)$	-8.5	$\lambda_1 < 0$
$\{2\}$	$(1.4,\ 1.8)$	$\lambda_2 = 1.2$	$(-1.2,\ 0)$	-11.2	**KKT**
$\{1, 2\}$	$(-1,\ 3)$	$(\lambda_1, \lambda_2) = (-12,\ -6)$	$(0,\ 0)$	-4	$\lambda < 0$

b) A7 from $x_0 = (2, 0)$, $\mathcal{W}_0 = \{1\}$.

k	x_k	$\mathcal{W}_k$	p	$\lambda_{\mathcal{W}}$	α_k (blocking)	$f(x_k)$
0	$(2,\ 0)$	$\{1\}$	$\left(-\frac{3}{2},\ \frac{3}{2}\right)$		1	-4
1	$\left(\frac{1}{2},\ \frac{3}{2}\right)$	$\{1\}$	0	$\lambda_1 = -3$	drop 1	$-\frac{17}{2}$
2	$\left(\frac{1}{2},\ \frac{3}{2}\right)$	$\emptyset$	$\left(\frac{3}{2},\ \frac{3}{2}\right)$		$\frac{1}{3}$ (constraint 2)	$-\frac{17}{2}$
3	$(1,\ 2)$	$\{2\}$	$\left(\frac{2}{5},\ -\frac{1}{5}\right)$		1	-11
4	$\left(\frac{7}{5},\ \frac{9}{5}\right)$	$\{2\}$	0	$\lambda_2 = \frac{6}{5}$	stop	$-\frac{56}{5}$

What it shows. The negative multiplier $\lambda_1 = -3$ tells A7 that constraint 1 is holding x in the wrong direction; dropping it lets f decrease until constraint 2 blocks the step. The method reaches the KKT point of a) while visiting three working sets, without enumerating all of them.

X14 Proposed exercise *ch4, p. 12*

Minimize $f(x_1, x_2) = 2x_1^2 + x_2^2 - 12x_1 - 8x_2$ subject to $x_1 + x_2 \leq 4$ and $x_1, x_2 \geq 0$. Find the global minimum with KKT and justify uniqueness using the Hessian.

X15 Proposed exercise *ch4, p. 13*

Let $f(x) = \frac{1}{2}(x_1^2 + x_2^2 + x_3^2)$ subject to $x_1 + x_2 + x_3 = 10$, $x_1 \geq 1$, $x_2 \geq 2$, $x_3 \geq 0$. Compute the optimal Lagrange multipliers analytically.

E8 Example *ch4, p. 7*

Setting. $A \in \mathbb{R}^{m \times n}$ of rank $m < n$, $b \in \mathbb{R}^m$, $y \in \mathbb{R}^n$; the projection of y onto $\{x :\ Ax = b\}$ solves $\min \frac{1}{2}\|x - y\|^2$ s.t. $Ax = b$.
Problem.

a) Derive with the KKT conditions that the projection is $x = y - A^{\mathsf{T}}(AA^{\mathsf{T}})^{-1}(Ay - b)$.

b) Justify that the qualification $T = L$ holds (affine constraints) and that the KKT point is the global minimum.

E9 Example *ch4, p. 8*

Setting. $M \in \mathbb{R}^{n \times n}$ symmetric; problems $\min x^{\mathsf{T}}Mx$ and $\max x^{\mathsf{T}}Mx$ subject to $\|x\|^2 = 1$.
Problem.

a) Show that the KKT points are the unit eigenvectors of M, with multiplier equal to the eigenvalue.

b) Conclude that the optimal values are $\lambda_{\min}(M)$ and $\lambda_{\max}(M)$.

Note. The notes justify neither the existence of optima (compactness of the sphere) nor the qualification (LICQ on the sphere); both are needed for a complete argument.

E10 Example *ch4, p. 8*

Setting. Entropy $H(p) = -\sum_{i=1}^{n} p_i \ln p_i$ on the simplex $\{p \geq 0,\ \sum_i p_i = 1\}$ (with $0 \ln 0 = 0$).
Problem.

a) Show that the maximum of H is attained at $p_i = 1/n$.

b) Deduce $H(p) \leq \ln n$.

Note. The notes omit the constraints $p_i \geq 0$ from the Lagrangian, and ln is not differentiable at 0. One must argue with the concavity of H (or with T27 on the simplex) that the interior stationary point is the global maximum.

X16 Proposed exercise *ch4, p. 13*

Definitions.

- *KKT multipliers at $\bar{x}$:* $\lambda_i \geq 0$, $\mu_j \in \mathbb{R}$ with $\nabla f(\bar{x}) + \sum_i \lambda_i \nabla g_i(\bar{x}) + \sum_j \mu_j \nabla h_j(\bar{x}) = 0$ and $\lambda_i g_i(\bar{x}) = 0$.

Setting. Problems of the form $\min f(x)$ s.t. $g_i(x) \leq 0$, $h_j(x) = 0$.

Problem.

a) Give a nonconvex problem of this form in which KKT multipliers exist at a point $\bar{x}$ that is a local *maximum* of f on the feasible set.

b) Explain why the KKT conditions are only necessary.

X17 Proposed exercise *ch4, p. 13*

Let $f(x, y) = x^2 - y^2$. Prove that $(0, 0)^\mathsf{T}$ satisfies $\nabla f(0, 0) = 0$ but is not a local extremum. Identify the directions of strict descent from the spectral analysis of $\nabla^2 f(0, 0)$.

Q15 Exam problem *added, 4 pts*

A parametric QP: KKT and sensitivity. For $t \in \mathbb{R}$ consider $v(t) = \min \tfrac{1}{2}(x_1^2 + x_2^2)$ s.t. $g_1(x) = t - x_1 - x_2 \leq 0$, $g_2(x) = x_1 - 1 \leq 0$.

a) Write the KKT conditions and solve them for every t: find the regimes of t, the solution $x^*(t)$ and the multipliers $\lambda^*(t)$.

b) Compute $v(t)$ and verify that $v'(t) = \lambda_1^*(t)$ in each regime. Interpret the sign and the size of λ_1^*.

c) Justify that the KKT points found are the global minimizers, and identify the values of t at which strict complementarity fails.

Q16 Exam problem *added, 4 pts*

Farkas' lemma and certificates of infeasibility. Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

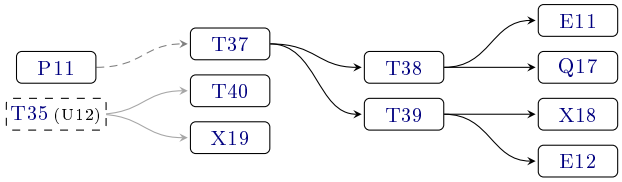
a) From Farkas' lemma, prove that exactly one of the systems (I) $Ax = b$, $x \geq 0$ and (II) $A^\mathsf{T} y \geq 0$, $b^\mathsf{T} y < 0$ has a solution.

b) For $A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ and $b = (1, 3)^\mathsf{T}$, decide which system is solvable and give an explicit solution of it.

c) Interpret the solution of (II) geometrically, as a hyperplane that separates b from the cone generated by the columns of A, and draw it.

U13. Lagrangian, saddle points and duality

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P11 PRACTICE *added*

Inf-sup gap. Let $A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ and $\Phi(x,y) = x^{\mathsf{T}} A y$, minimized in x and maximized in y.

a) With $x, y \in \{e_1, e_2\}$ (vertices), compute $\max_y \min_x \Phi$ and $\min_x \max_y \Phi$, and verify that the inf-sup inequality is strict.

b) With $x, y \in \Delta_2 = \{z \in \mathbb{R}^2 : \ z \geq 0, \ z_1 + z_2 = 1\}$, compute both values and verify equality, with saddle point $x = y = (\frac{1}{2}, \frac{1}{2})^{\mathsf{T}}$.

T37 PROOF *ch4, Proposition 4.1, p. 8*

For any $L : X \times Y \to \mathbb{R}$, prove that $\sup_{y \in Y} \inf_{x \in X} L(x,y) \leq \inf_{x \in X} \sup_{y \in Y} L(x,y)$.

T38 PROOF *ch4, Theorem 4.1, p. 9*

Setting. For $\min f(x)$ s.t. $g_i(x) \leq 0$ $(i = 1, \ldots, p)$, $h_j(x) = 0$ $(j = 1, \ldots, q)$, let $L(x, \lambda, \mu) = f(x) + \sum_i \lambda_i g_i(x) + \sum_j \mu_j h_j(x)$ with $\lambda \geq 0$, the dual function $q(\lambda, \mu) = \inf_{x \in \mathbb{R}^n} L(x, \lambda, \mu)$, p^* the primal optimal value and $d^* = \sup_{\lambda \geq 0, \mu} q(\lambda, \mu)$.

Definition. $(\bar{x}, \bar{\lambda}, \bar{\mu})$ with $\bar{\lambda} \geq 0$ is a saddle point of L if

$$L(\bar{x}, \lambda, \mu) \leq L(\bar{x}, \bar{\lambda}, \bar{\mu}) \leq L(x, \bar{\lambda}, \bar{\mu}) \qquad \text{for all } x, \ \lambda \geq 0, \ \mu.$$

Problem. Prove that $(\bar{x}, \bar{\lambda}, \bar{\mu})$ is a saddle point if and only if $\bar{x}$ is primal optimal, $(\bar{\lambda}, \bar{\mu})$ is dual optimal and $p^* = d^*$.

Note. In ($\Leftarrow$) complementary slackness must be deduced from $f(\bar{x}) = q(\bar{\lambda}, \bar{\mu}) \leq L(\bar{x}, \bar{\lambda}, \bar{\mu}) \leq f(\bar{x})$ before asserting the two saddle inequalities.

E11 NUMERICAL EXAMPLE *ch4, p. 11 (solved exercise)*

Setting. $L(x, \lambda) = x^2 + 4x + \lambda(1 - x)$, $x \in \mathbb{R}$, $\lambda \geq 0$.

Problem.

a) Identify the primal problem ($\min x^2 + 4x$ s.t. $1 - x \leq 0$).

b) Verify that $(\bar{x}, \bar{\lambda}) = (1, 6)$ is a saddle point: $L(\bar{x}, \lambda) \leq L(\bar{x}, \bar{\lambda}) \leq L(x, \bar{\lambda})$ for all $x \in \mathbb{R}$, $\lambda \geq 0$.

c) Compute the dual function $q(\lambda) = \inf_x L(x, \lambda)$ and check $p^* = d^* = 5$.

T39 PROOF *ch4, Theorem 5.1, p. 10*

Setting. For $\min f(x)$ s.t. $g_i(x) \leq 0$ $(i = 1, \ldots, p)$, $h_j(x) = 0$ $(j = 1, \ldots, q)$, let $L(x, \lambda, \mu) = f(x) + \sum_i \lambda_i g_i(x) + \sum_j \mu_j h_j(x)$ with $\lambda \geq 0$, the dual function $q(\lambda, \mu) = \inf_{x \in \mathbb{R}^n} L(x, \lambda, \mu)$, p^* the primal optimal value and $d^* = \sup_{\lambda \geq 0, \mu} q(\lambda, \mu)$.

Problem.

a) (Weak duality) Prove that $q(\lambda, \mu) \leq f(x)$ for every feasible x and $\lambda \geq 0$; in particular $d^* \leq p^*$.

b) (Strong duality) If f, g_i are convex, h_j affine, p^* finite and Slater holds (there is $\hat{x}$ with $g_i(\hat{x}) < 0$ for all i and $h_j(\hat{x}) = 0$ for all j), prove that there is (λ^*, μ^*) with $\lambda^* \geq 0$ and $q(\lambda^*, \mu^*) = p^*$.

Note. The notes apply *strict* separation between the set $\mathcal{A}$ and the point $(0, 0, p^*)$, but that point lies in the closure of $\mathcal{A}$. The correct argument separates $\mathcal{A}$ from $\mathcal{B} = \{(0, 0, t) : \ t < p^*\}$ (disjoint convex sets) and uses Slater to show the hyperplane is not vertical. With equality constraints a relative-interior hypothesis is also required.

X18 Proposed exercise *ch4, p. 12*

For $\min (x_1-3)^2+(x_2-4)^2$ subject to $x_1+x_2=5$, build the dual function $q(\mu)=\inf_x\big[(x_1-3)^2+(x_2-4)^2+\mu(x_1+x_2-5)\big]$, solve $\max_\mu q(\mu)$ and check that there is no duality gap.

E12 Numerical example *ch4, p. 12 (solved exercise)*

Setting. Primal: $\min 3x_1+5x_2$ s.t. $x_1+2x_2\geq 4$, $2x_1+x_2\geq 3$, $x\geq 0$. Lagrangian (minimized over $x\geq 0$): $L(x,\lambda)=3x_1+5x_2+\lambda_1(4-x_1-2x_2)+\lambda_2(3-2x_1-x_2)$, $\lambda\geq 0$.

Problem.

a) Formulate the dual problem.

b) Solve both problems by vertices and verify $x^*=(\frac{2}{3},\frac{5}{3})^\mathsf{T}$, $\lambda^*=(\frac{7}{3},\frac{1}{3})^\mathsf{T}$, common value $\frac{31}{3}$.

c) Verify complementary slackness.

T40 Proof *ch4, Theorem 5.3, p. 10 (no proof in the notes)*

Let $A\in\mathbb{R}^{m\times n}$. Prove that exactly one of the following holds: (1) there is $x>0$ with $Ax=0$; (2) there is $y\in\mathbb{R}^m$ with $A^\mathsf{T}y\geq 0$ and $A^\mathsf{T}y\neq 0$.

Note. The notes call it Gordan's theorem; with this statement it is Stiemke's theorem. Gordan's is: there is $x\geq 0$, $x\neq 0$ with $Ax=0$, or there is y with $A^\mathsf{T}y>0$. It can be derived from Farkas' lemma: $Ad\leq 0$, $Bd=0\Rightarrow c^\mathsf{T}d\geq 0$ if and only if there are $u\geq 0$, v with $c+A^\mathsf{T}u+B^\mathsf{T}v=0$ (T35).

X19 Proposed exercise *ch4, p. 13*

Setting. Primal LP $\min c^\mathsf{T}x$ s.t. $Ax=b$, $x\geq 0$; dual LP $\max b^\mathsf{T}y$ s.t. $A^\mathsf{T}y\leq c$.

Problem. Prove the strong duality theorem of linear programming: if the primal or the dual has a finite optimal solution, then both do and $c^\mathsf{T}x^*=b^\mathsf{T}y^*$. Do not use differentiability.

Tool. Farkas' lemma (T35): $Ad\leq 0,\ Bd=0\Rightarrow c^\mathsf{T}d\geq 0$ holds if and only if there are $u\geq 0$ and v with $c+A^\mathsf{T}u+B^\mathsf{T}v=0$.

Q17 Exam problem *added, 4 pts*

Duals of least-norm problems.

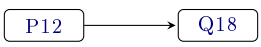
a) Let $A\in\mathbb{R}^{m\times n}$ have full row rank and $b\in\mathbb{R}^m$. For $\min\frac{1}{2}\|x\|^2$ s.t. $Ax=b$, compute the dual function $q(\lambda)=\inf_x L(x,\lambda)$ with $L(x,\lambda)=\frac{1}{2}\|x\|^2+\lambda^\mathsf{T}(b-Ax)$, and the dual problem.

b) Solve the dual, recover $x^*=A^\mathsf{T}(AA^\mathsf{T})^{-1}b$, and check that there is no duality gap.

c) For $a\neq 0$, solve $\min\frac{1}{2}\|x\|^2$ s.t. $a^\mathsf{T}x\geq 1$ through its dual ($\mu\geq 0$): compute $q(\mu)$, μ^*, x^* and both optimal values, and verify that Slater's condition holds.

U14. Trust region: model, ratio ρ_k and radius

↑Index



P12 Practice *merges ch5, p. 24 (solved ex. 4) and p. 26 (ex. 8)*

Definitions.

- *Acceptance and radius rule of the notes:* with $\rho_k = \dfrac{f(x_k) - f(x_k + p_k)}{m_k(0) - m_k(p_k)}$ and $0 < \eta_1 \le \eta_2 < 1$: if $\rho_k < \eta_1$ the step is rejected ($x_{k+1} = x_k$) and $\Delta_{k+1} = \frac{1}{2}\Delta_k$; if $\rho_k \ge \eta_1$ it is accepted ($x_{k+1} = x_k + p_k$), with $\Delta_{k+1} = \min(2\Delta_k, \Delta_{\max})$ if moreover $\rho_k > \eta_2$ and $\|p_k\| = \Delta_k$, and $\Delta_{k+1} = \Delta_k$ otherwise.
- *Predicted and actual reduction:* $\text{pred}_k = m_k(0) - m_k(p_k) = -g_k^{\mathsf{T}} p_k - \frac{1}{2} p_k^{\mathsf{T}} B_k p_k$ and $\text{ared}_k = f(x_k) - f(x_k + p_k)$, so $\rho_k = \text{ared}_k / \text{pred}_k$.

Data. $\eta_1 = 0.1$, $\eta_2 = 0.75$, $\Delta_{\max} = 4$ in every case.

Problem.

a) With $\Delta_k = 1$, decide x_{k+1} and Δ_{k+1} in each case: (i) $\rho_k = -0.5$; (ii) $\rho_k = 0.05$; (iii) $\rho_k = 0.5$, $\|p_k\| = 1$; (iv) $\rho_k = 0.9$, $\|p_k\| = 0.6$; (v) $\rho_k = 0.9$, $\|p_k\| = 1$.

b) (ch5, p. 24) $f(x_k) = 100$, $x_k = (0,0)^{\mathsf{T}}$, $g_k = (-2,0)^{\mathsf{T}}$, $B_k = 2I$, $p_k = (1,0)^{\mathsf{T}}$, $\Delta_k = 1$, $f(x_k + p_k) = 98.8$. Compute pred_k, ared_k and ρ_k, decide the step and Δ_{k+1}, and interpret $\rho_k > 1$.

c) (ch5, p. 26) $f(x_k) = 25.0$, $\text{pred}_k = 4.0$, $f(x_k + p_k) = 21.8$, $\Delta_k = 0.5$, $\|p_k\| = 0.5$. Compute ρ_k, decide the step and Δ_{k+1}. Which of the cases in a) is this?

d) Explain why in case (iv) enlarging the radius would not change the next iteration if the model were the same.

e) This rule does not shrink the radius when a step is accepted with small ρ_k. The rule of A11 (used in Q18) instead accepts if $\rho_k > \eta = 0.15$, sets $\Delta \leftarrow \frac{1}{4}\Delta$ whenever $\rho_k < \frac{1}{4}$, and doubles Δ only if $\rho_k > \frac{3}{4}$ and the step is on the boundary. Redo cases (i)–(v) with the A11 rule and compare.

Q18 Exam problem *added, 4 pts*

The agreement ratio and the radius update. Let $f(x) = e^x - 2x$ on $\mathbb{R}$, $x_0 = 0$, and the quadratic model $m(p) = f(0) + f'(0)p + \frac{1}{2}f''(0)p^2$.

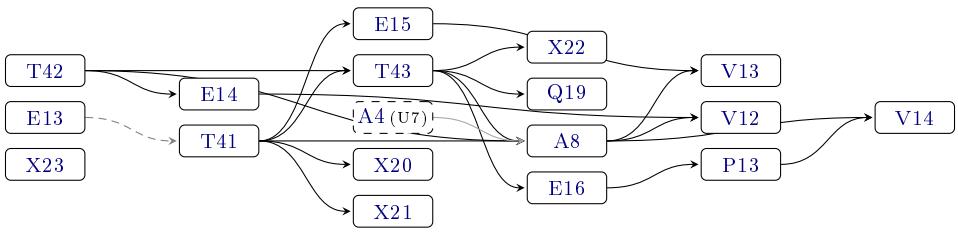
a) Write $m(p)$, find its unconstrained minimizer, and compute the step $p(\Delta)$ that minimizes m on $|p| \le \Delta$.

b) Compute $\rho(\Delta) = \dfrac{f(0) - f(p(\Delta))}{m(0) - m(p(\Delta))}$ for $\Delta = 0.5$, 1 and 2, and prove that $\rho \to 1$ as $\Delta \to 0$ using Taylor's theorem ($\text{ared} - \text{pred} = O(\Delta^3)$).

c) Apply the update of A11 ($\Delta \leftarrow \Delta/4$ if $\rho < \frac{1}{4}$; $\Delta \leftarrow \min\{2\Delta, \Delta_{\max}\}$ if $\rho > \frac{3}{4}$ and the step is on the boundary; accept if $\rho > \eta = 0.15$) for $\Delta_0 = 0.5$ and for $\Delta_0 = 2$, and give x_1 and Δ_1 in each case.

U15. Trust-region subproblem: exact solution

↑Index



E13 NUMERICAL EXAMPLE *ch5, p. 7*

For the subproblem $\min\{g^\mathsf{T}p+\frac{1}{2}p^\mathsf{T}Bp:\ \|p\|_2\le\Delta\}$ with $g=(-4,-6)^\mathsf{T}$, $B=\begin{pmatrix}2&1\\1&4\end{pmatrix}$ and $\Delta=1$:

a) Compute $p^N=-B^{-1}g=(\frac{10}{7},\frac{8}{7})^\mathsf{T}$ and verify $\|p^N\|>\Delta$, so the solution lies on the boundary.

b) Set up $\|p(\lambda)\|=1$ with $p(\lambda)=-(B+\lambda I)^{-1}g$ and find λ^* (table V12).

c) Compare $g^\mathsf{T}p^*+\frac{1}{2}p^{*\mathsf{T}}Bp^*$ with the value at the Cauchy point $p^C=-\frac{\Delta}{\|g\|}g$ (truncated here; see V15).

Note. The notes leave the example at "the minimum must be on the boundary", without computing λ^*.

T41 PROOF *ch5, Theorem 3.1, pp. 7–11*

Setting. Let $m_k(p)=f(x_k)+g_k^\mathsf{T}p+\frac{1}{2}p^\mathsf{T}B_kp$ with $g_k=\nabla f(x_k)$ and B_k symmetric, and the subproblem $\min\{m_k(p):\ \|p\|_2\le\Delta_k\}$.

Problem.

a) Prove that p^* is a global solution of the subproblem if and only if $\|p^*\|\le\Delta_k$ and there is $\lambda^*\ge 0$ with

$$(B_k+\lambda^*I)p^*=-g_k,\qquad \lambda^*(\|p^*\|-\Delta_k)=0,\qquad B_k+\lambda^*I\succeq 0.$$

b) Derive the identity $m_k(p)-m_k(p^*)=\frac{1}{2}(p-p^*)^\mathsf{T}(B_k+\lambda^*I)(p-p^*)+\frac{\lambda^*}{2}\big(\|p^*\|^2-\|p\|^2\big)$, used in a) and obtained in the notes by "rearranging".

Note. In ($\Rightarrow$), the case $v^\mathsf{T}p^*=0$ assumes $\|p^*\|=\Delta_k$; the case $\|p^*\|<\Delta_k$ ("follows directly") requires saying that then $\lambda^*=0$, p^* is an unconstrained local minimum of m_k, and SONC (T5) gives $B_k\succeq 0$.

X20 PROPOSED EXERCISE *ch5, p. 25 (exercise 1)*

Setting. Let $m_k(p)=f(x_k)+g_k^\mathsf{T}p+\frac{1}{2}p^\mathsf{T}B_kp$ with $g_k=\nabla f(x_k)$ and B_k symmetric, and the subproblem $\min\{m_k(p):\ \|p\|_2\le\Delta_k\}$.

Problem. Prove that if $B_k\succ 0$ and $\|B_k^{-1}g_k\|\le\Delta_k$, then $p_k^N=-B_k^{-1}g_k$ is the unique global solution of the subproblem.

T42 PROOF *ch5, Proposition 3.1, pp. 12–13*

Setting. $B_k=\sum_i\lambda_iq_iq_i^\mathsf{T}$ symmetric (spectral decomposition), $g_k\neq 0$, $\gamma_i=q_i^\mathsf{T}g_k$, $p(\lambda)=-(B_k+\lambda I)^{-1}g_k$, so that $\|p(\lambda)\|^2=\sum_i\gamma_i^2/(\lambda_i+\lambda)^2$; $p_k^N=-B_k^{-1}g_k$ when B_k is invertible.

Problem. On every interval where $B_k+\lambda I\succ 0$:

a) Prove $p'(\lambda)=-(B_k+\lambda I)^{-1}p(\lambda)$.

b) Prove that $\lambda\mapsto\|p(\lambda)\|$ is strictly decreasing and that $p(\lambda)\to 0$ as $\lambda\to+\infty$; note $p(0)=p_k^N$ if $B_k\succ 0$.

c) Deduce that if $B_k\succ 0$ and $\|p_k^N\|>\Delta_k$, there is a unique $\lambda^*>0$ with $\|p(\lambda^*)\|=\Delta_k$.

E14 NUMERICAL EXAMPLE *ch5, p. 23 (solved exercise 1)*

Setting. the subproblem $\min\{g^\mathsf{T}p+\frac{1}{2}p^\mathsf{T}Bp:\ \|p\|_2\le\Delta\}$ with $g=(-2,-4)^\mathsf{T}$, $B=\mathrm{diag}(4,2)$, $\Delta=1$; the solution has the form $p^*=p(\lambda^*)=-(B+\lambda^*I)^{-1}g$.

Problem.

a) Verify $\|p^N\|=\sqrt{4.25}>1$, so the solution is on the boundary.

b) Set up $\frac{4}{(4+\lambda)^2}+\frac{16}{(2+\lambda)^2}=1$ and find λ^* and p^*.

Note. The notes give $\lambda^*\approx 2.21$ and $p^*\approx(0.322,0.950)^\mathsf{T}$, but with $\lambda=2.21$ the equation gives 1.0064, and $\|(0.322,0.950)\|\approx 1.003$. The correct values are $\lambda^*\approx 2.224019$, $p^*\approx(0.321336,0.946965)^\mathsf{T}$; see V12.

E15 Numerical example *ch5, p. 11*

For the subproblem $\min\{g^{\mathsf T}p + \tfrac12 p^{\mathsf T}Bp :\ \|p\|_2 \le \Delta\}$ with $g = (2,-4)^{\mathsf T}$, $B = \operatorname{diag}(-1,3)$ and $\Delta = 2$, and $p(\lambda) = -(B+\lambda I)^{-1}g$:

a) Justify that one must look for $\lambda^* \ge 1$ (so that $B + \lambda^* I \succeq 0$) and that this is the easy case: the component of g in the eigenspace of $\lambda_{\min}(B) = -1$ is nonzero.

b) Solve $\frac{4}{(\lambda-1)^2} + \frac{16}{(\lambda+3)^2} = 4$ with Newton on $\phi(\lambda) = 1/\|p(\lambda)\| - 1/\Delta$ from $\lambda^{(0)} = 2$ (table V13) and verify $(B+\lambda^* I)p^* = -g$, $\lambda^*(\|p^*\| - \Delta) = 0$, $B + \lambda^* I \succeq 0$, $\lambda^* \ge 0$.

Note. The notes give $\lambda^* \approx 2.083$, $p^* \approx (-1.846, 0.786)^{\mathsf T}$; that p^* has norm ≈ 2.006. The correct values are $\lambda^* \approx 2.087561$, $p^* \approx (-1.838978, 0.786231)^{\mathsf T}$.

X21 Proposed exercise *ch5, p. 26 (exercise 6)*

With $g = (5,-12)^{\mathsf T}$, $B = \begin{pmatrix} 6 & 2 \\ 2 & 3 \end{pmatrix}$ and $\Delta = 1$, find the exact solution of the subproblem $\min\{g^{\mathsf T}p + \tfrac12 p^{\mathsf T}Bp :\ \|p\|_2 \le \Delta\}$ through the multiplier λ^*, with $(B+\lambda^* I)p^* = -g$, $\lambda^*(\|p^*\| - \Delta) = 0$, $B + \lambda^* I \succeq 0$, $\lambda^* \ge 0$.

T43 Proof *ch5, pp. 13–15 (easy case, hard case and Proposition 3.2)*

Setting. Let $m_k(p) = f(x_k) + g_k^{\mathsf T}p + \tfrac12 p^{\mathsf T}B_k p$ with $g_k = \nabla f(x_k)$ and B_k symmetric, and the subproblem $\min\{m_k(p) :\ \|p\|_2 \le \Delta_k\}$.

Problem. Let $B_k = \sum_i \lambda_i q_i q_i^{\mathsf T}$ with $\lambda_1 \le \cdots \le \lambda_n$, $\lambda_1 < 0$, $\gamma_i = q_i^{\mathsf T} g_k$, $E_1 = \{i : \lambda_i = \lambda_1\}$, $p(\lambda) = -(B_k + \lambda I)^{-1} g_k$ and $\psi(\lambda) = \|p(\lambda)\|^2 = \sum_i \gamma_i^2/(\lambda_i + \lambda)^2$. A solution is characterized (T41) by $(B_k + \lambda^* I)p^* = -g_k$, $\lambda^*(\|p^*\| - \Delta_k) = 0$, $B_k + \lambda^* I \succeq 0$, $\lambda^* \ge 0$.

a) (Easy case) If $\gamma_i \neq 0$ for some $i \in E_1$, prove that there is a unique $\lambda^* \in (-\lambda_1, \infty)$ with $\|p(\lambda^*)\| = \Delta_k$, and that $p(\lambda^*)$ is the solution.

b) (Hard case) If $\gamma_i = 0$ for all $i \in E_1$ and $\psi(-\lambda_1) < \Delta_k^2$, prove that $p^* = p(-\lambda_1) + \tau z$, with $z \in \operatorname{Ker}(B_k - \lambda_1 I)$ of unit norm and $\tau = \sqrt{\Delta_k^2 - \psi(-\lambda_1)}$, satisfies those three conditions.

c) In the hard case with $\psi(-\lambda_1) \ge \Delta_k^2$, show that the solution is obtained as in the easy case.

A8 Pseudocode *exact subproblem: Moré–Sorensen (as in SciPy `trust-exact`; N&W, Alg. 4.3)*

Solves $\min\{g^{\mathsf T}p + \tfrac12 p^{\mathsf T}Bp :\ \|p\| \le \Delta\}$ for any symmetric B, with $p(\lambda) = -(B+\lambda I)^{-1}g$, using only Cholesky factorizations.

ExactSubproblem$(g,\ B,\ \Delta;\ \kappa_{\text{easy}}, \kappa_{\text{hard}})$

$\lambda_L \leftarrow \max\{0,\ -\min_i b_{ii},\ \|g\|/\Delta - \min(\bar\beta, \|B\|_F, \|B\|_\infty)\}$
  $\lambda_U \leftarrow \max\{0,\ \|g\|/\Delta + \min(-\underline\beta, \|B\|_F, \|B\|_\infty)\}$ (bracket for λ^*)
if $\lambda_L = 0$ **then** $\lambda \leftarrow 0$ **else** $\lambda \leftarrow$ Safeguard(Safeguard $= \max\{\sqrt{\lambda_L \lambda_U},\ \lambda_L + 0.01(\lambda_U - \lambda_L)\}$)
loop
  $(L, \text{ok}, j, d) \leftarrow \text{Cholesky}(B + \lambda I)$ (on failure: index j and pivot $d \le 0$)
  if not ok **then**
    $v \leftarrow e_j - \sum_{i<j} w_i e_i$, where $L_{1:j-1}^{\mathsf T} w = \ell_j$ $(v^{\mathsf T}(B+\lambda I)v = d)$
    $\lambda_L \leftarrow \max\{\lambda_L,\ \lambda - d/\|v\|^2\}$; $\lambda \leftarrow$ Safeguard; **continue**
  solve $LL^{\mathsf T}p = -g$; **if** $\lambda = 0$ **and** $\|p\| \le \Delta$ **then return** p (interior: Newton step)
  solve $Lq = p$; $\lambda_N \leftarrow \lambda + \big(\|p\|/\|q\|\big)^2(\|p\| - \Delta)/\Delta$ (Newton on $1/\|p(\lambda)\| - 1/\Delta$)
  if $\|p\| > \Delta$ **then**
    if $\|p\| - \Delta \le \kappa_{\text{easy}}\Delta$ **then return** p; $\lambda_L \leftarrow \lambda$; $\lambda \leftarrow \lambda_N$

12: **else** ($\|p\| < \Delta$ with $\lambda > 0$: possible hard case)
13: $(\sigma, z) \leftarrow$ estimate of the smallest singular value of L^{T} and its unit vector
14: $\tau \leftarrow$ root of $\|p + \tau z\| = \Delta$ of smaller $|\tau|$
15: **if** $\tau^2\sigma^2 \le \kappa_{\text{hard}}\big(p^{\mathsf{T}}(B+\lambda I)p + \lambda\Delta^2\big)$ **then return** $p + \tau z$
16: $\lambda_U \leftarrow \lambda$; $\lambda_L \leftarrow \max\{\lambda_L,\ \lambda - \sigma^2\}$
17: **if** Cholesky$(B+\lambda_N I)$ succeeds **then** $\lambda \leftarrow \lambda_N$ **else** $\lambda_L \leftarrow \max\{\lambda_L, \lambda_N\}$; $\lambda \leftarrow$ Safeguard

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$\underline{\beta}$, $\bar{\beta}$ are the Gershgorin bounds $\min_i(b_{ii} - \sum_{k\neq i}|b_{ik}|)$ and $\max_i(b_{ii} + \sum_{k\neq i}|b_{ik}|)$, so $\lambda^* \in [\lambda_L, \lambda_U]$ before any factorization; ℓ_j (line 6) is the computed part of row j of the partial factor L, and $\sigma^2 \approx \lambda_{\min}(B+\lambda I)$ (line 13). Defaults (Moré–Sorensen; SciPy): $\kappa_{\text{easy}} = 0.1$, $\kappa_{\text{hard}} = 0.2$. These tolerances are loose on purpose: they guarantee $m(p) \le (1-\kappa)^2\, m(p^*)$, which is all the outer method needs, and they stop the hard case after one or two factorizations. Line 6: when Cholesky fails at pivot j, the partial factor gives v with $v^{\mathsf{T}}(B+\lambda I)v = d \le 0$, hence $\lambda_{\min}(B) \le -\lambda + d/\|v\|^2$, a free lower bound. Line 13 uses the LINPACK condition estimator (Cline–Moler–Stewart–Wilkinson), which costs $O(n^2)$ with the factor already available; one or two steps of inverse iteration with LL^{T} serve the same purpose by hand. Line 15 is the hard-case test: if moving along z to the boundary changes the model by a small fraction, $p + \tau z$ is accepted. Since $\|q\|^2 = p^{\mathsf{T}}(B+\lambda I)^{-1}p$, line 9 is the update of the notes (ch5, p. 11).

Is it used in practice? Yes, for small and medium dense problems, where a few Cholesky factorizations per iteration are affordable: Moré–Sorensen (GQTPAR in MINPACK-2; SciPy `trust-exact`) typically needs 2–3 factorizations. For large n the subproblem is solved approximately: Dogleg (A9) when $B \succ 0$, and Steihaug–CG (A10) in general, which only needs products Bv. The Cauchy point by itself is the reference step for the convergence theory (T45, T48), not a step used on its own.

V12 Validation table *A8 on E14 and E13*

Subproblem $\min\{g^{\mathsf{T}}p + \frac{1}{2}p^{\mathsf{T}}Bp : \|p\| \le \Delta\}$, $\Delta = 1$, $B \succ 0$ in both cases. The runs use $\kappa_{\text{easy}} = \kappa_{\text{hard}} = 10^{-6}$ to show the convergence of λ; with the defaults (0.1, 0.2) A8 stops at the row marked †. "rel." is $(\|p\| - \Delta)/\Delta$.

l	λ	$p(\lambda)$	$\|p\|$	rel.	action
E14: $g = (-2,-4)$, $B = \text{diag}(4,2)$; Gershgorin $\underline{\beta} = 2$, $\bar{\beta} = 4$; $\lambda_L = 0.472136$, $\lambda_U = 2.472136$					
0	1.0803630	(0.393673, 1.298548)	1.3569104	3.6×10^{-1}	outside: $\lambda_L \leftarrow \lambda$, Newton
1†	2.2174555	(0.321675, 0.948439)	1.0015047	1.5×10^{-3}	outside: Newton
2	2.2240193	(0.321336, 0.946965)	1.0000000	3.5×10^{-8}	boundary: return
E13: $g = (-4,-6)$, $B = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 4 \end{smallmatrix}\right)$; $\underline{\beta} = 1$, $\bar{\beta} = 5$; $\lambda_L = 2.520687$, $\lambda_U = 6.211103$					
0	3.9567972	(0.556656, 0.684112)	0.8819727	-1.2×10^{-1}	inside: $\lambda_U \leftarrow \lambda$, Newton
1†	3.0084501	(0.646125, 0.763917)	1.0005231	5.2×10^{-4}	outside: Newton
2	3.0121232	(0.645719, 0.763575)	1.0000000	1.2×10^{-8}	boundary: return

In E13 the first λ is above λ^* ($\|p\| < \Delta$), so line 15 is evaluated: $\sigma = 2.395923$, $z = (0.789762, -0.613413)$, $\tau = 0.451741$, $\tau^2\sigma^2/(p^{\mathsf{T}}(B+\lambda I)p + \lambda\Delta^2) = 0.114$, which is not small, so the Newton step (now downward) is taken. With the default $\kappa_{\text{hard}} = 0.2$ the test would pass at $l = 0$ and A8 would return $p + \tau z = (0.913424, 0.407008)$, with model value -4.558321 instead of -5.088225. In E13: $g^{\mathsf{T}}p^* + \frac{1}{2}p^{*\mathsf{T}}Bp^* \approx -5.088225$, against -5.057256 with the Cauchy point and with Dogleg (V15), which coincide here because the minimizer along the ray $-g$ lies outside the region.

V13 VALIDATION TABLE *A8 on E15*

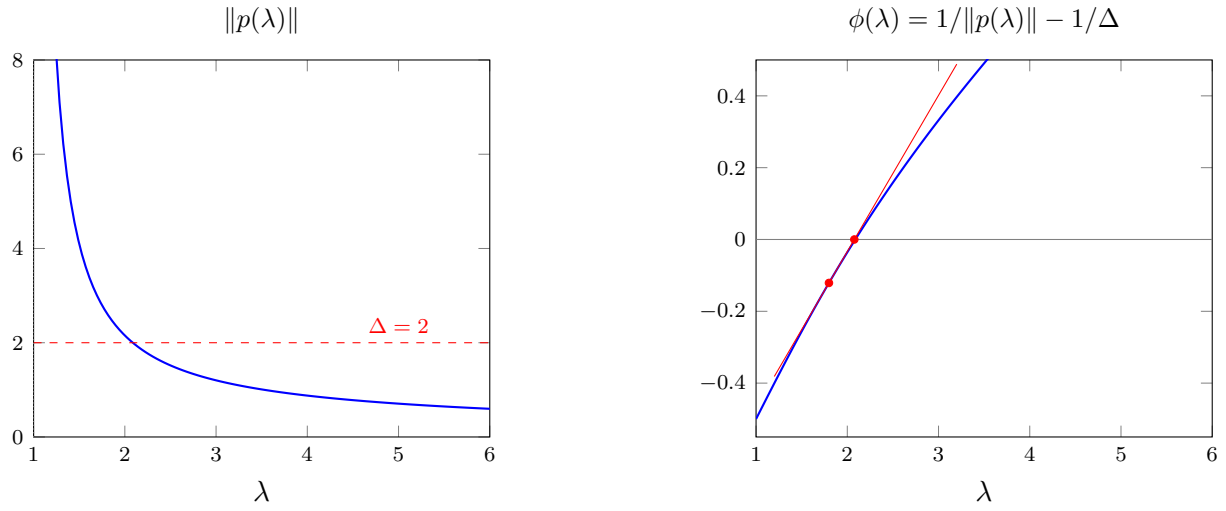
Subproblem $\min\{g^{\mathsf T}p + \tfrac12 p^{\mathsf T}Bp :\ \|p\| \le \Delta\}$ with $g = (2,-4)$, $B = \operatorname{diag}(-1,3)$, $\Delta = 2$; easy case ($g_1 = 2 \neq 0$). Here $p(\lambda) = \left(-\frac{2}{\lambda-1},\ \frac{4}{\lambda+3}\right)$. Line 1: $\lambda_L = \max\{0,\ 1,\ \sqrt{5}-3\} = 1$, $\lambda_U = \sqrt{5}+1 = 3.236068$; line 2: $\lambda^{(0)} = \sqrt{\lambda_L \lambda_U} = 1.798907$. Run with $\kappa_{\text{easy}} = 10^{-6}$; the default 0.1 stops at †.

l	$\lambda^{(l)}$	$p(\lambda^{(l)})$	$\|p\|$	$(\|p\| - \Delta)/\Delta$
0	1.7989074	$(-2.503419,\ 0.833523)$	2.6385350	3.2×10^{-1}
1†	2.0771143	$(-1.856813,\ 0.787849)$	2.0170427	8.5×10^{-3}
2	2.0875467	$(-1.839001,\ 0.786234)$	2.0000224	1.1×10^{-5}
3	2.0875606	$(-1.838978,\ 0.786231)$	2.0000000	2.0×10^{-11}

$\lambda^* \approx 2.087561$, $B + \lambda^* I = \operatorname{diag}(1.087561,\ 5.087561) \succ 0$, $g^{\mathsf T}p^* + \frac12 p^{*\mathsf T}Bp^* \approx -7.586562$. Cholesky succeeds at every λ because the bracket starts at $\lambda_L = -\lambda_{\min}(B) = 1$.

What it shows. The Gershgorin bracket gives a good start without any eigenvalue computation; from $\|p(\lambda)\| > \Delta$, Newton on ϕ increases λ monotonically and converges quadratically, even with indefinite B.

Figure.



$g = (2,-4)$, $B = \operatorname{diag}(-1,3)$. Left: $\|p(\lambda)\|$ has a pole at $\lambda = -\lambda_{\min}(B) = 1$ and is strongly nonlinear near λ^*, so Newton on $\|p\| - \Delta$ would take poor steps. Right: ϕ is almost linear and concave; the tangent at $\lambda^{(0)} = 1.7989$ (red) lands close to the root, which explains the fast convergence in the table.

X22 PROPOSED EXERCISE *ch5, p. 26 (exercise 7)*

With $g = (1,1)^{\mathsf T}$, $B = \operatorname{diag}(-2,4)$ and $\Delta = 0.8$, find the exact solution of the subproblem $\min\{g^{\mathsf T}p + \frac12 p^{\mathsf T}Bp :\ \|p\|_2 \le \Delta\}$. Decide first whether it is the easy or the hard case (hard: g orthogonal to the eigenspace of $\lambda_{\min}(B)$).

E16 NUMERICAL EXAMPLE *ch5, p. 25 (solved exercise 5)*

Setting. the subproblem $\min\{g^{\mathsf T}p + \frac12 p^{\mathsf T}Bp :\ \|p\|_2 \le \Delta\}$ with $g = (0,-3)^{\mathsf T}$, $B = \operatorname{diag}(-4,2)$, $\Delta = 1$ and $p(\lambda) = -(B+\lambda I)^{-1}g$. Optimality conditions: $(B+\lambda^* I)p^* = -g$, $\lambda^*(\|p^*\| - \Delta) = 0$, $B + \lambda^* I \succeq 0$, $\lambda^* \ge 0$.

Problem.

a) Show that $\|p(\lambda)\| = 1$ has no solution with $\lambda > 4$, and identify the hard case.

b) Obtain $p^* = (\pm\frac{\sqrt{3}}{2}, \frac12)^{\mathsf T}$ with $\lambda^* = 4$ and verify the optimality conditions.

c) Check that both solutions give $g^{\mathsf T}p^* + \frac12 p^{*\mathsf T}Bp^* = -\frac{11}{4}$.

P13 Practice *added*

Near the hard case. For the subproblem $\min\{g^{\mathsf{T}}p + \frac{1}{2}p^{\mathsf{T}}Bp : \ \|p\|_2 \le \Delta\}$ with $B = \mathrm{diag}(-4, 2)$, $\Delta = 1$ and $g = (\varepsilon, -3)^{\mathsf{T}}$, $\varepsilon > 0$; $p(\lambda) = -(B + \lambda I)^{-1}g$. With $\varepsilon = 0$ the solutions are $(\pm\frac{\sqrt{3}}{2}, \frac{1}{2})^{\mathsf{T}}$ with $\lambda^* = 4$ (E16).

a) Show that for $\varepsilon > 0$ it is the easy case and compute $\lambda^*(\varepsilon)$ and $p^*(\varepsilon)$ for $\varepsilon = 1$, 0.1, 0.01, 0.001 (table V14).

b) Verify that $p^*(\varepsilon) \to (-\frac{\sqrt{3}}{2}, \frac{1}{2})^{\mathsf{T}}$ and $\lambda^*(\varepsilon) \to 4^+$ as $\varepsilon \to 0^+$: the sign of ε selects one of the two solutions of the case $\varepsilon = 0$.

c) Explain why Newton on $\phi(\lambda) = 1/\|p(\lambda)\| - 1/\Delta$ becomes difficult as $\varepsilon \to 0$ (pole of $\|p(\lambda)\|$ at $\lambda = 4$ with weight ε^2).

V14 Validation table *near-hard case; solves P13*

Subproblem $\min\{g^{\mathsf{T}}p + \frac{1}{2}p^{\mathsf{T}}Bp : \ \|p\| \le \Delta\}$ with $B = \mathrm{diag}(-4, 2)$, $\Delta = 1$, $g = (\varepsilon, -3)$; for $\varepsilon > 0$ it is the easy case and $\lambda^* > 4$.

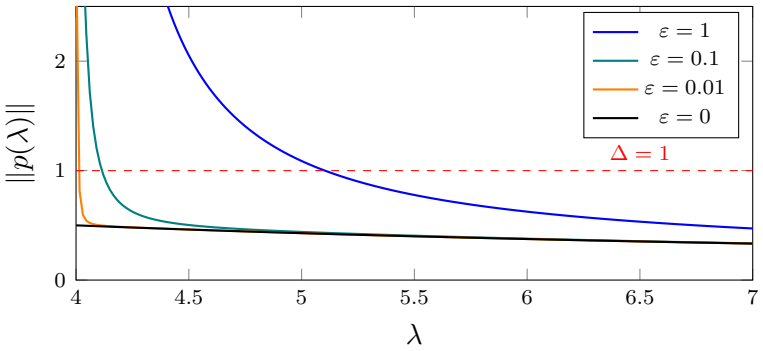
ε	$\lambda^*(\varepsilon)$	$p^*(\varepsilon)$	$g^{\mathsf{T}}p^* + \frac{1}{2}p^{*\mathsf{T}}Bp^*$
1	5.103222	(−0.906436, 0.422344)	−3.638344
0.1	4.114761	(−0.871376, 0.490616)	−2.836873
0.01	4.011540	(−0.866579, 0.499040)	−2.758663
0.001	4.001155	(−0.866081, 0.499904)	−2.750866
0 (E16)	4	(±0.866025, 0.5)	−2.75

A8 (defaults) on the exact hard case $\varepsilon = 0$ (E16): Gershgorin gives $\lambda_L = 4$, $\lambda_U = \frac{3}{1} + 4 = 7$, so $\lambda^{(0)} = \sqrt{28} = 5.291503$. Cholesky of $B + \lambda I = \mathrm{diag}(1.291503, 7.291503)$ succeeds, $p = (0, 0.411438)$ and $\|p\| < \Delta$, so line 13 runs: the estimator gives $\sigma = 1.214090$, $z = (0.984673, 0.174410)$, $\tau = 0.842500$, and the test value $\tau^2\sigma^2/(p^{\mathsf{T}}(B+\lambda I)p+\lambda) = 0.160 \le 0.2$. A8 returns $p + \tau z = (0.829587, 0.558378)$ after *one* factorization, with model value −2.739776, against −2.75 at the solutions of E16 (0.4% off). By hand, with the exact eigenvector $z = (1, 0)$ and $\sigma^2 = \lambda_{\min}(B + \lambda I) = 1.291503$: $\tau = 0.911438$, test value 0.164, step (0.911438, 0.411438) with model value −2.726470.

With $\kappa_{\mathrm{hard}} = 10^{-6}$ the test only passes after 19 factorizations: at each one the Newton value $\lambda_N = 1$ lies below $\lambda_L = 4$ (Cholesky fails), so the safeguard moves λ, and $\lambda - 4$ is roughly halved per step (5.2915, 4.6007, 4.2898, 4.1424, ..., 4.000004), until $p + \tau z \approx (0.866025, 0.5)$.

What it shows. As $\varepsilon \to 0$ the solution tends to one of the hard-case solutions and $\lambda^* \to -\lambda_1$, where the secular equation becomes ill-conditioned. In the exact hard case Newton on ϕ points below λ_L at every step and only the safeguard moves λ; the loose test of line 15 is what makes A8 practical, because it accepts a near-optimal boundary step built from the factor already available.

Figure.



$B = \mathrm{diag}(-4, 2)$, $g = (\varepsilon, -3)$, for $\lambda > -\lambda_{\min}(B) = 4$. The pole at $\lambda = 4$ has weight ε^2: as ε decreases the curve drops almost vertically and the crossing with $\Delta = 1$ approaches $\lambda = 4$. With $\varepsilon = 0$ the pole disappears, $\|p(\lambda)\| \le \frac{1}{2} < \Delta$ and there is no crossing: this is the hard case, and the missing component is added along the eigenvector of $\lambda_{\min}$.

X23 PROPOSED EXERCISE *ch5, p. 25 (exercise 3)*

Consider the trust-region subproblem in the ℓ_∞ norm: $\min g_k^\mathsf{T} p + \frac{1}{2} p^\mathsf{T} B_k p$ s.t. $\|p\|_\infty \le \Delta_k$. If $B_k = \mathrm{diag}(d_1, \ldots, d_n)$ with $d_i > 0$, find the exact analytic expression of the optimal step p^*.

Q19 EXAM PROBLEM *added, 4 pts*

An exact subproblem with an indefinite model. Let $B = \mathrm{diag}(-1, 2)$, $g = (1, 2)^\mathsf{T}$ and $p(\lambda) = -(B + \lambda I)^{-1} g$.

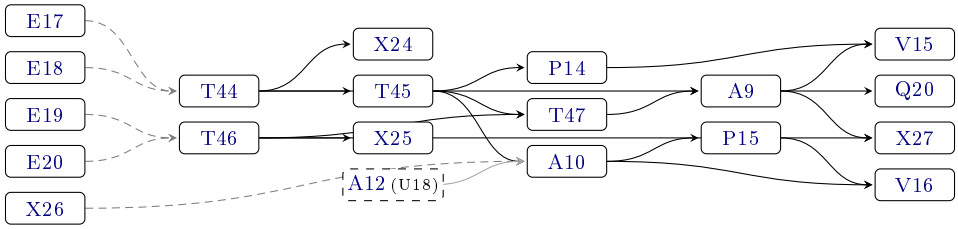
a) Explain why the solution of $\min\{g^\mathsf{T} p + \frac{1}{2} p^\mathsf{T} B p : \ \|p\| \le \Delta\}$ has $\lambda^* > 1$ for every $\Delta > 0$, and write the secular equation $\|p(\lambda)\| = \Delta$ explicitly.

b) For $\Delta = 1$, perform two Newton steps on $\phi(\lambda) = 1/\|p(\lambda)\| - 1/\Delta$ from $\lambda^{(0)} = 2$, and check the optimality conditions at the result: $(B + \lambda I)p = -g$, $B + \lambda I \succeq 0$, $\lambda(\Delta - \|p\|) = 0$.

c) Keep $g_2 = 2$ and $\Delta = 1$ and let $g_1 \to 0$. Show that $g_1 = 0$ is the hard case, and find the solution(s) in that case.

U16. Approximate methods: Cauchy point and Dogleg

↑Index



E17 NUMERICAL EXAMPLE *ch5, p. 18*

Definitions.

- *Cauchy point:* $p^C = -\alpha^C g$, where α^C minimizes $g^{\mathsf{T}}(-\alpha g) + \frac{1}{2}\alpha^2 g^{\mathsf{T}} B g$ over $0 \le \alpha \le \Delta/\|g\|$; and $p^U = -\frac{g^{\mathsf{T}} g}{g^{\mathsf{T}} B g}\, g$ (if $g^{\mathsf{T}} B g > 0$) is the minimizer along the ray with no radius constraint.

Data. $g = (3, 4)^{\mathsf{T}}$, $B = 5I$, $\Delta = 0.5$.

Problem.

a) Compute $\alpha^* = \|g\|^2 / g^{\mathsf{T}} B g$ and check that the unconstrained step p^U leaves the region.

b) Obtain $p^C = (-0.3, -0.4)^{\mathsf{T}}$.

c) Explain why here p^C is also the exact solution of the subproblem $\min\{g^{\mathsf{T}} p + \frac{1}{2} p^{\mathsf{T}} B p : \ \|p\|_2 \le \Delta\}$.

E18 NUMERICAL EXAMPLE *ch5, p. 23 (solved exercise 2)*

Definitions.

- *Cauchy point:* $p^C = -\alpha^C g$, where α^C minimizes $g^{\mathsf{T}}(-\alpha g) + \frac{1}{2}\alpha^2 g^{\mathsf{T}} B g$ over $0 \le \alpha \le \Delta/\|g\|$; and $p^U = -\frac{g^{\mathsf{T}} g}{g^{\mathsf{T}} B g}\, g$ (if $g^{\mathsf{T}} B g > 0$) is the minimizer along the ray with no radius constraint.

Data. $g = (3, -4)^{\mathsf{T}}$, $B = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, $\Delta = 0.5$.

Problem. Compute $g^{\mathsf{T}} B g = 26$, the unconstrained step length $\alpha^* = \frac{25}{26}$, and the Cauchy point $p^C = (-0.3, 0.4)^{\mathsf{T}}$.

T44 DERIVATION *ch5, Definition 4.1 and eq. (12), p. 16*

Setting. Let $m_k(p) = f(x_k) + g_k^{\mathsf{T}} p + \frac{1}{2} p^{\mathsf{T}} B_k p$ with $g_k = \nabla f(x_k)$ and B_k symmetric, and the subproblem $\min\{m_k(p) : \ \|p\|_2 \le \Delta_k\}$. The Cauchy point $p_k^C = -\alpha_k^C g_k$ minimizes m_k over the segment $\{-\alpha g_k : \ 0 \le \alpha \le \Delta_k / \|g_k\|\}$.

Problem. By minimizing $\alpha \mapsto m_k(-\alpha g_k)$ on $[0, \Delta_k/\|g_k\|]$, derive

$$\alpha_k^C = \begin{cases} \dfrac{\|g_k\|^2}{g_k^{\mathsf{T}} B_k g_k}, & g_k^{\mathsf{T}} B_k g_k > 0 \text{ and } \dfrac{\|g_k\|^3}{g_k^{\mathsf{T}} B_k g_k} \le \Delta_k, \\ \dfrac{\Delta_k}{\|g_k\|}, & \text{otherwise.} \end{cases}$$

T45 PROOF *ch5, Proposition 4.1, p. 17*

Setting. Let $m_k(p) = f(x_k) + g_k^{\mathsf{T}} p + \frac{1}{2} p^{\mathsf{T}} B_k p$ with $g_k = \nabla f(x_k)$ and B_k symmetric, and the subproblem $\min\{m_k(p) : \ \|p\|_2 \le \Delta_k\}$. p_k^C is the Cauchy point: the minimizer of m_k over $\{-\alpha g_k : \ 0 \le \alpha \le \Delta_k/\|g_k\|\}$.

Problem. Prove that $m_k(0) - m_k(p_k^C) \ge \frac{1}{2}\|g_k\| \min\big(\Delta_k, \|g_k\|/\|B_k\|\big)$.

Note. In subcase 2b the notes appeal to "the slope at the origin dominates". It must be made precise: if $\alpha^C \le \alpha^*$ then $\alpha^C g^{\mathsf{T}} B g \le \|g\|^2$, hence $m_k(0) - m_k(p^C) \ge \frac{1}{2}\alpha^C \|g\|^2 = \frac{1}{2}\Delta_k \|g\|$.

X24 Proposed exercise *ch5, p. 26 (exercise 4)*

Definitions.

- *Cauchy point:* $p^C = -\alpha^C g$, where α^C minimizes $g^{\mathsf T}(-\alpha g) + \tfrac{1}{2}\alpha^2 g^{\mathsf T} B g$ over $0 \le \alpha \le \Delta/\|g\|$; and $p^U = -\frac{g^{\mathsf T} g}{g^{\mathsf T} B g}\, g$ (if $g^{\mathsf T} B g > 0$) is the minimizer along the ray with no radius constraint.

Setting. $f(x_1, x_2) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$ (Rosenbrock), $\nabla f(x) = \big(-400x_1(x_2 - x_1^2) - 2(1 - x_1),\ 200(x_2 - x_1^2)\big)^{\mathsf T}$, $\nabla^2 f(x) = \begin{pmatrix} 1200x_1^2 - 400x_2 + 2 & -400x_1 \\ -400x_1 & 200 \end{pmatrix}$; $x_0 = (-1.2, 1.0)^{\mathsf T}$, $\Delta_0 = 0.25$.

Problem. Compute $g_0 = \nabla f(x_0)$, $B_0 = \nabla^2 f(x_0)$ and the Cauchy step p_0^C.

E19 Numerical example *ch5, p. 20*

Definitions.

- *Dogleg step* (for $B \succ 0$): with $p^N = -B^{-1}g$ and $p^U = -\frac{g^{\mathsf T} g}{g^{\mathsf T} B g} g$, the Dogleg path is $\tilde p(\tau) = \tau p^U$ on $[0, 1]$ and $\tilde p(\tau) = p^U + (\tau - 1)(p^N - p^U)$ on $[1, 2]$; the Dogleg step is $\tilde p(\tau^*)$ with $\tau^* = \max\{\tau \in [0, 2] :\ \|\tilde p(\tau)\| \le \Delta\}$.

Data. $g = (2, 2)^{\mathsf T}$, $B = \operatorname{diag}(2, 8)$, $\Delta = 0.5$.

Problem.

a) Compute $p^N = (-1, -0.25)^{\mathsf T}$ and $p^U = (-0.4, -0.4)^{\mathsf T}$.

b) Conclude that the Dogleg step is on the first segment: $p^{DL} \approx (-0.3536, -0.3536)^{\mathsf T}$.

E20 Numerical example *ch5, p. 24 (solved exercise 3)*

Definitions.

- *Dogleg step* (for $B \succ 0$): with $p^N = -B^{-1}g$ and $p^U = -\frac{g^{\mathsf T} g}{g^{\mathsf T} B g} g$, the Dogleg path is $\tilde p(\tau) = \tau p^U$ on $[0, 1]$ and $\tilde p(\tau) = p^U + (\tau - 1)(p^N - p^U)$ on $[1, 2]$; the Dogleg step is $\tilde p(\tau^*)$ with $\tau^* = \max\{\tau \in [0, 2] :\ \|\tilde p(\tau)\| \le \Delta\}$.

Data. $g = (1, 2)^{\mathsf T}$, $B = \operatorname{diag}(3, 1)$, $\Delta = 1.5$.

Problem.

a) Compute $p^N = (-\tfrac{1}{3}, -2)^{\mathsf T}$ and $p^U = -\tfrac{5}{7}(1, 2)^{\mathsf T}$, with $\|p^U\| = \tfrac{5}{7}\sqrt{5} \approx 1.597$.

b) Compute the Dogleg step $p^{DL} \approx (-0.6708, -1.3416)^{\mathsf T}$ and say on which segment it lies.

T46 Proof *ch5, Lemma 4.1, p. 19*

Definitions.

- *Dogleg step* (for $B \succ 0$): with $p^N = -B^{-1}g$ and $p^U = -\frac{g^{\mathsf T} g}{g^{\mathsf T} B g} g$, the Dogleg path is $\tilde p(\tau) = \tau p^U$ on $[0, 1]$ and $\tilde p(\tau) = p^U + (\tau - 1)(p^N - p^U)$ on $[1, 2]$; the Dogleg step is $\tilde p(\tau^*)$ with $\tau^* = \max\{\tau \in [0, 2] :\ \|\tilde p(\tau)\| \le \Delta\}$.

Problem. Let $g \neq 0$. Prove that $\|\tilde p(\tau)\|$ is strictly increasing and that $m(\tilde p(\tau))$, with $m(p) = g^{\mathsf T} p + \tfrac{1}{2} p^{\mathsf T} B p$, is strictly decreasing on $[0, 2]$.

Note. The notes state "one shows that $p_k^{C\,\mathsf T}(p_k^N - p_k^C) \ge 0$" without doing it, and part 2 only invokes convexity. Both need a computation: write $p^C = -\frac{g^{\mathsf T} g}{g^{\mathsf T} B g} g$ and use Cauchy–Schwarz in the B and B^{-1} inner products.

X25 Proposed exercise *ch5, p. 25 (exercise 2)*

Definitions.

- *Dogleg step* (for $B \succ 0$): with $p^N = -B^{-1}g$ and $p^U = -\frac{g^\mathsf{T}g}{g^\mathsf{T}Bg}g$, the Dogleg path is $\tilde{p}(\tau) = \tau p^U$ on $[0,1]$ and $\tilde{p}(\tau) = p^U + (\tau-1)(p^N - p^U)$ on $[1,2]$; the Dogleg step is $\tilde{p}(\tau^*)$ with $\tau^* = \max\{\tau \in [0,2] : \ \|\tilde{p}(\tau)\| \le \Delta\}$.

Problem. Prove formally that $h(\tau) = \|\tilde{p}(\tau)\|^2$ is strictly increasing on $[1,2]$.

P14 Practice *added*

Definitions.

- *Cauchy point:* $p^C = -\alpha^C g$, where α^C minimizes $g^\mathsf{T}(-\alpha g) + \frac{1}{2}\alpha^2 g^\mathsf{T}Bg$ over $0 \le \alpha \le \Delta/\|g\|$; and $p^U = -\frac{g^\mathsf{T}g}{g^\mathsf{T}Bg}\,g$ (if $g^\mathsf{T}Bg > 0$) is the minimizer along the ray with no radius constraint.
- *Dogleg step* (for $B \succ 0$): with $p^N = -B^{-1}g$ and $p^U = -\frac{g^\mathsf{T}g}{g^\mathsf{T}Bg}g$, the Dogleg path is $\tilde{p}(\tau) = \tau p^U$ on $[0,1]$ and $\tilde{p}(\tau) = p^U + (\tau-1)(p^N - p^U)$ on $[1,2]$; the Dogleg step is $\tilde{p}(\tau^*)$ with $\tau^* = \max\{\tau \in [0,2] : \ \|\tilde{p}(\tau)\| \le \Delta\}$.

Problem. Dogleg on the second segment. With $g = (2,2)^\mathsf{T}$, $B = \mathrm{diag}(2,8)$ (the data of E19).

a) For $\Delta = 0.8$, verify $\|p^U\| < \Delta < \|p^N\|$, set up the quadratic $\|p^U + s(p^N - p^U)\|^2 = \Delta^2$ in $s = \tau - 1 \in [0,1]$ and find τ^* and p^{DL}.

b) For $\Delta = 1.2$, verify that $p^{DL} = p^N$.

c) For the three radii 0.5, 0.8, 1.2, compare the reduction $m(0) - m(p) = -g^\mathsf{T}p - \frac{1}{2}p^\mathsf{T}Bp$ of Dogleg with that of the Cauchy point and with the bound $m(0) - m(p) \ge \frac{1}{2}\|g\| \min\big(\Delta, \|g\|/\|B\|\big)$ (table V15).

T47 Proof *ch5, Theorem 4.1, p. 20*

Definitions.

- *Cauchy point:* $p^C = -\alpha^C g$, where α^C minimizes $g^\mathsf{T}(-\alpha g) + \frac{1}{2}\alpha^2 g^\mathsf{T}Bg$ over $0 \le \alpha \le \Delta/\|g\|$; and $p^U = -\frac{g^\mathsf{T}g}{g^\mathsf{T}Bg}\,g$ (if $g^\mathsf{T}Bg > 0$) is the minimizer along the ray with no radius constraint.
- *Dogleg step* (for $B \succ 0$): with $p^N = -B^{-1}g$ and $p^U = -\frac{g^\mathsf{T}g}{g^\mathsf{T}Bg}g$, the Dogleg path is $\tilde{p}(\tau) = \tau p^U$ on $[0,1]$ and $\tilde{p}(\tau) = p^U + (\tau-1)(p^N - p^U)$ on $[1,2]$; the Dogleg step is $\tilde{p}(\tau^*)$ with $\tau^* = \max\{\tau \in [0,2] : \ \|\tilde{p}(\tau)\| \le \Delta\}$.

Problem. Let $m(p) = g^\mathsf{T}p + \frac{1}{2}p^\mathsf{T}Bp$ with $B \succ 0$. Prove that $m(0) - m(p^{DL}) \ge m(0) - m(p^C)$ and hence that the Dogleg step satisfies the bound $m(0) - m(p) \ge \frac{1}{2}\|g\| \min\big(\Delta, \|g\|/\|B\|\big)$ (T45).

Note. The chain "$m_k(p^{DL}) \le m_k(p^C) \le m_k(p^C)$" in the notes has a typo: in the case $\|p^U\| < \Delta_k$ one must observe that the constrained Cauchy point coincides with p^U (minimum along the ray inside the region) and then apply T46.

A9 Pseudocode *Cauchy point and Dogleg (ch5, eq. 12 and Def. 4.2)*

CauchyPoint(g, B, Δ)

```
1: γ ← g^T B g
2: if γ > 0 and ‖g‖^3 ≤ Δ γ then α ← ‖g‖^2/γ else α ← Δ/‖g‖
3: return −α g
```

Dogleg$(g,\ B,\ \Delta)$

1: $(L, \text{ok}) \leftarrow \text{Cholesky}(B)$; **if not** ok **then return** CauchyPoint(g, B, Δ)(Dogleg needs $B \succ 0$)
2: solve $LL^{\mathsf{T}} p^N = -g$; **if** $\|p^N\| \le \Delta$ **then return** p^N ($\tau^* = 2$)
3: $p^U \leftarrow -\dfrac{g^{\mathsf{T}} g}{g^{\mathsf{T}} B g}\, g$; **if** $\|p^U\| \ge \Delta$ **then return** $(\Delta/\|p^U\|)\, p^U$ (first segment)
4: $d \leftarrow p^N - p^U$; $a \leftarrow \|d\|^2$; $b \leftarrow 2\,(p^U)^{\mathsf{T}} d$; $c \leftarrow \|p^U\|^2 - \Delta^2 < 0$
5: $s \leftarrow \dfrac{-2c}{b + \sqrt{b^2 - 4ac}}$ **if** $b > 0$ **else** $s \leftarrow \dfrac{-b + \sqrt{b^2 - 4ac}}{2a}$ (positive root, no cancellation)
6: **return** $p^U + s\, d$ (second segment, $\tau^* = 1 + s$)

Here $m(p) = g^{\mathsf{T}} p + \frac{1}{2} p^{\mathsf{T}} B p$, p^U is the minimizer of m along the ray $-g$ with no radius constraint (the notation p^C in Definition 4.2 of the notes), and the Cauchy point of line 2 of CauchyPoint is its version truncated to the ball $\|p\| \le \Delta$. The Cholesky factor of line 1 is computed once per outer iteration and serves both as the test $B \succ 0$ and to solve for p^N. SciPy's `trust-dogleg` stops with an error if B is not positive definite; Dogleg is also the step of Powell's hybrid method for nonlinear equations (MINPACK `hybrd`, behind SciPy's `fsolve`), where $B = J^{\mathsf{T}} J$.

V15 Validation table *A9; solves P14 and checks E17, E19, E18, E20, E13*

Model $m(p) = g^{\mathsf{T}} p + \frac{1}{2} p^{\mathsf{T}} B p$; $p^U = -\frac{g^{\mathsf{T}} g}{g^{\mathsf{T}} B g} g$, $p^N = -B^{-1} g$, p^C the Cauchy point and p^{DL} the Dogleg step. Data: E17: $g = (3, 4)$, $B = 5I$; E18: $g = (3, -4)$, $B = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$; E20: $g = (1, 2)$, $B = \text{diag}(3, 1)$; E19 and P14: $g = (2, 2)$, $B = \text{diag}(2, 8)$; E13: $g = (-4, -6)$, $B = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 4 \end{smallmatrix}\right)$. Model reduction $\text{red}(p) = m(0) - m(p) = -g^{\mathsf{T}} p - \frac{1}{2} p^{\mathsf{T}} B p$ and Cauchy bound $\kappa = \frac{1}{2} \|g\| \min(\Delta, \|g\|/\|B\|_2)$.

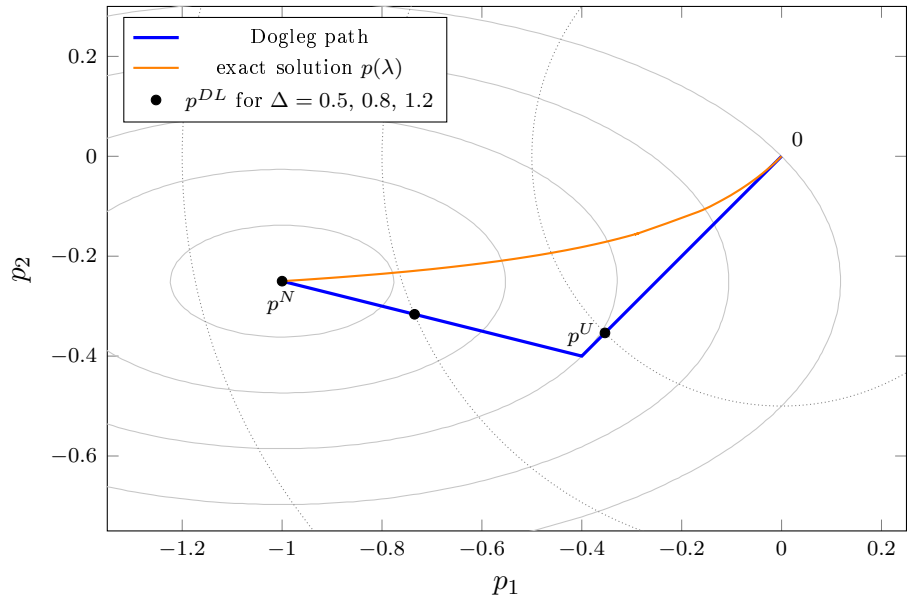
case	Δ	$\|p^U\|$	$\|p^N\|$	p^C	$\text{red}(p^C)$	κ
E17	0.5	1	1	(−0.3, −0.4)	1.875	1.25
E18	0.5	4.807692	4.955356	(−0.3, 0.4)	2.37	1.25
E20	1.5	1.597191	2.027588	(−0.670820, −1.341641)	1.779102	0.833333
E19	0.5	0.565685	1.030776	(−0.353553, −0.353553)	0.789214	0.5
P14 a)	0.8	0.565685	1.030776	(−0.4, −0.4)	0.8	0.5
P14 b)	1.2	0.565685	1.030776	(−0.4, −0.4)	0.8	0.5
E13	1	1.674006	1.829464	(0.554700, 0.832050)	5.057256	3.605551

case	Δ	segment	p^{DL}	$\text{red}(p^{DL})$
E17	0.5	1	(−0.3, −0.4)	1.875
E18	0.5	1	(−0.3, 0.4)	2.37
E20	1.5	1	(−0.670820, −1.341641)	1.779102
E19	0.5	1	(−0.353553, −0.353553)	0.789214
P14 a)	0.8	2, $s = 0.558030$, $\tau^* = 1.558030$	(−0.734818, −0.316296)	1.162098
P14 b)	1.2	Newton, $\tau^* = 2$	(−1, −0.25)	1.25
E13	1	1	(0.554700, 0.832050)	5.057256

Detail of P14 a): $p^U = (-0.4, -0.4)$, $d = p^N - p^U = (-0.6,\ 0.15)$, the equation is $0.3825 s^2 + 0.36 s - 0.32 = 0$. In E17, $\|p^U\| = \|p^N\| = 1$ because $B = 5I$: the gradient ray passes through p^N.

What it shows. In every case $\text{red}(p^{DL}) \ge \text{red}(p^C) \ge \kappa$, which is the inequality used by global convergence.

Figure.



$g = (2, 2)$, $B = \mathrm{diag}(2, 8)$. Level curves of the model (centred at p^N), dotted circles $\|p\| = \Delta$. The Dogleg path goes from 0 to p^U along the ray $-g$ and then straight to p^N; the exact solution follows the curve $p(\lambda)$, tangent to $-g$ at the origin ($\lambda \to \infty$) and ending at p^N ($\lambda = 0$). Dogleg approximates that curve with two segments; the norm increases and the model decreases along both (T46).

X26 PROPOSED EXERCISE *ch5, p. 26 (exercise 10)*

Setting. Let $m_k(p) = f(x_k) + g_k^\mathsf{T} p + \frac{1}{2} p^\mathsf{T} B_k p$ with $g_k = \nabla f(x_k)$ and B_k symmetric, and the subproblem $\min\{m_k(p) : \|p\|_2 \le \Delta_k\}$.

Method. Steihaug–CG applies linear CG to $B_k p = -g_k$ from $p = 0$ (iterates z_j, directions d_j, $d_0 = -g_k$) and stops at the first of three events:

(i) the residual $B_k z_j + g_k$ is small: return z_j;

(ii) the next iterate would leave the region $\|p\| \le \Delta_k$: return the point where the segment from z_j along d_j crosses the boundary;

(iii) a direction with $d_j^\mathsf{T} B_k d_j \le 0$ appears: move along d_j to the boundary.

Problem.

a) Explain why the method only needs products $B_k d$ and never B_k^{-1}, and why it suits large n.

b) Show that the first CG iterate z_1 is the minimizer of m_k along $-g_k$, so the returned step reduces m_k at least as much as the Cauchy point.

c) Explain why event (iii) cannot occur when $B_k \succ 0$, and why going to the boundary is the right decision when $d_j^\mathsf{T} B_k d_j \le 0$.

Note. ch5 says Steihaug–CG is covered in the next chapter, but ch6 only develops linear CG (U18–U19), on which Steihaug–CG is built. A10 is one answer.

A10 Pseudocode *Steihaug–CG: truncated CG for the trust-region subproblem (N&W, Alg. 7.2)*

Approximately solves $\min\{g^{\mathsf T}p + \tfrac12 p^{\mathsf T}Bp :\ \|p\| \le \Delta\}$ for any symmetric B (also indefinite), using only products Bd.

SteihaugCG$(g,\ B,\ \Delta;\ \varepsilon, j_{\max})$ default $\varepsilon = \min\{\tfrac12, \sqrt{\|g\|}\}\|g\|$

1: $z \leftarrow 0$; $r \leftarrow g$; $d \leftarrow -r$; **if** $\|r\| < \varepsilon$ **then return** 0
2: **for** $j = 0, 1, \dots, j_{\max} - 1$ **do**
3: $w \leftarrow Bd$; $\kappa \leftarrow d^{\mathsf T}w$ (the only use of B)
4: **if** $\kappa \le 0$ **then return** the one of $z + \tau_{\pm}d$ with smaller model value, $\|z + \tau_{\pm}d\| = \Delta$ (nonpositive curvature)
5: $\alpha \leftarrow r^{\mathsf T}r/\kappa$; $z_{\text{new}} \leftarrow z + \alpha d$
6: **if** $\|z_{\text{new}}\| \ge \Delta$ **then return** $z + \tau d$, $\tau \ge 0$ with $\|z + \tau d\| = \Delta$ (leaves the region)
7: $r_{\text{new}} \leftarrow r + \alpha w$; **if** $\|r_{\text{new}}\| < \varepsilon$ **then return** z_{new} (interior: CG converged)
8: $\beta \leftarrow r_{\text{new}}^{\mathsf T}r_{\text{new}}/r^{\mathsf T}r$; $d \leftarrow -r_{\text{new}} + \beta d$; $z \leftarrow z_{\text{new}}$; $r \leftarrow r_{\text{new}}$
9: **return** z (iteration cap, e.g. $j_{\max} = n$)

In line 6, τ is the positive root of $\|d\|^2\tau^2 + 2z^{\mathsf T}d\,\tau + \|z\|^2 - \Delta^2 = 0$; in line 4 both roots $\tau_- < 0 < \tau_+$ are tried, because along a direction of negative curvature the model may decrease in either sense. This is linear CG (A12) applied to $Bp = -g$ from $p = 0$, stopped at the boundary or at nonpositive curvature. Its first iterate is the minimizer of the model along $-g$, so the step is at least as good as the Cauchy point and satisfies the bound of T45; when $B \succ 0$ and CG converges inside the region it returns the Newton step. The default ε is relative to $\|g\|$: far from the solution a rough step suffices, and near it the steps approach the Newton step, which gives superlinear convergence of the outer method. A preconditioner M is added as in A12 (the trust region is then measured in the M-norm). This is the subproblem solver of SciPy's `trust-ncg` and of the Newton-CG trust-region codes for large problems (e.g. in TRON, and in Ipopt's and Knitro's CG options).

P15 Practice *added*

Steihaug–CG by hand. For the subproblem $\min\{g^{\mathsf T}p + \tfrac12 p^{\mathsf T}Bp :\ \|p\| \le \Delta\}$, run A10 with $\varepsilon = 10^{-12}$ (instead of the default) in the cases below, record each CG step and the reason it stops, and compare the model value with that of the exact solution (A8).

a) $g = (-4, -6)^{\mathsf T}$, $B = \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}$, $\Delta = 1$ and $\Delta = 2$.

b) $g = (2, -4)^{\mathsf T}$, $B = \operatorname{diag}(-1, 3)$, $\Delta = 2$.

c) $g = (2, -0.5)^{\mathsf T}$, $B = \operatorname{diag}(-1, 3)$, $\Delta = 1$.

d) $g = (1, -4)^{\mathsf T}$, $B = \operatorname{diag}(-1, 3)$, $\Delta = 10$.

Explain why, in a), the step for $\Delta = 1$ is exactly the Cauchy point, and why in b) A10 works although B is indefinite.

V16 Validation table *A10; solves P15*

Model $m(p) = g^{\mathsf T}p + \tfrac12 p^{\mathsf T}Bp$. "Stop": B = leaves the region, NC = nonpositive curvature, I = interior (CG converged).

case	j	z_j	d_j	$d_j^{\mathsf{T}} B d_j$	α_j	stop
a) $\Delta = 1$	0	$(0, 0)$	$(4, 6)$	224	0.232143	B, $\tau = 0.138675$
a) $\Delta = 2$	0	$(0, 0)$	$(4, 6)$	224	0.232143	
	1	$(0.928571,\ 1.392857)$	$(0.8125,\ -0.40625)$	1.320313	0.615385	I
b)	0	$(0, 0)$	$(-2, 4)$	44	0.454545	B, $\tau = 0.447214$
c)	0	$(0, 0)$	$(-2, 0.5)$	-3.25		NC, $\tau = 0.485071$
d)	0	$(0, 0)$	$(-1, 4)$	47	0.361702	
	1	$(-0.361702,\ 1.446809)$	$(-1.477592,\ 0.123133)$	-2.137792		NC, $\tau = 6.352705$

case	returned p	$m(p)$, A10	m, exact (A8)
a) $\Delta = 1$	$(0.554700,\ 0.832050)$ (Cauchy point, V15)	-5.057256	-5.088225
a) $\Delta = 2$	$(\frac{10}{7},\ \frac{8}{7}) = p^N$	-6.285714	-6.285714
b)	$(-0.894427,\ 1.788854)$	-4.544272	-7.586562
c)	$(-0.970143,\ 0.242536)$	-2.443906	-2.520821
d)	$(-9.748405,\ 2.229034)$	-58.727357	-61.951106

What it shows. Steihaug–CG never needs $B \succ 0$ or a factorization: it stops at the boundary or as soon as it detects nonpositive curvature, and otherwise it is plain CG and returns the Newton step. Its model value is between the Cauchy point and the exact solution, which is all the global convergence theory needs; the gap to the exact value (case b) is the price for using only products Bd. With the default $\varepsilon = \min\{\frac{1}{2}, \sqrt{\|g\|}\}\|g\|$ the cases that stop at the boundary or by curvature are unchanged, but in a) with $\Delta = 2$ and in d) CG stops after one step, because $\|r_1\|$ is already below $\frac{1}{2}\|g\|$: it returns $z_1 = (0.928571,\ 1.392857)$ (model -6.035714) and $z_1 = (-0.361702,\ 1.446809)$ (model -3.074468). Inside a trust-region method this is enough: the step only has to beat the Cauchy point.

X27 Proposed exercise

ch5, p. 26 (exercise 5)

Definitions.

- *Dogleg step* (for $B \succ 0$): with $p^N = -B^{-1}g$ and $p^U = -\frac{g^{\mathsf{T}}g}{g^{\mathsf{T}}Bg}g$, the Dogleg path is $\tilde{p}(\tau) = \tau p^U$ on $[0, 1]$ and $\tilde{p}(\tau) = p^U + (\tau - 1)(p^N - p^U)$ on $[1, 2]$; the Dogleg step is $\tilde{p}(\tau^*)$ with $\tau^* = \max\{\tau \in [0, 2] :\ \|\tilde{p}(\tau)\| \le \Delta\}$.

Setting. $f(x_1, x_2) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$ (Rosenbrock), $\nabla f(x) = \big(-400x_1(x_2 - x_1^2) - 2(1 - x_1),\ 200(x_2 - x_1^2)\big)^{\mathsf{T}}$, $\nabla^2 f(x) = \begin{pmatrix} 1200x_1^2 - 400x_2 + 2 & -400x_1 \\ -400x_1 & 200 \end{pmatrix}$; $x_0 = (-1.2, 1.0)^{\mathsf{T}}$, $\Delta_0 = 0.25$.

Problem.

a) Check whether $\nabla^2 f(x_0)$ is positive definite.

b) Build the Dogleg step p_0^{DL}.

Q20 Exam problem *added, 4 pts*

Cauchy point, Dogleg and the exact step. Let $m(p) = g^{\mathsf{T}}p + \frac{1}{2}p^{\mathsf{T}}Bp$ with $g = (1,1)^{\mathsf{T}}$ and $B = \mathrm{diag}(1,10)$.

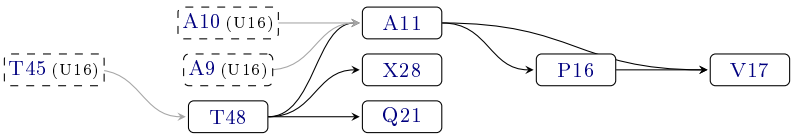
a) Compute the minimizer p^U of m along $-g$, the Newton step p^N, and their norms.

b) Compute the Dogleg step for $\Delta = 0.2$, 0.5 and 1.2, saying on which segment of the path each one lies, and compare each with the Cauchy point.

c) For $\Delta = 0.5$, find the exact solution through the secular equation (a few Newton steps on $\phi(\lambda)$ are enough) and compare $m(p^{DL})$ with $m(p^*)$. Explain why the Dogleg path bends away from the exact solution curve $\lambda \mapsto p(\lambda)$ when $\kappa(B)$ is large.

U17. Global convergence of trust region

↑Index



T48 Proof *ch5, Theorem 5.1, pp. 21–22*

Setting. Let $m_k(p) = f(x_k) + g_k^{\mathsf{T}}p + \frac{1}{2}p^{\mathsf{T}}B_kp$ with $g_k = \nabla f(x_k)$ and B_k symmetric, and the subproblem $\min\{m_k(p) : \|p\|_2 \le \Delta_k\}$.

Definitions.

- *Acceptance and radius rule of the notes:* with $\rho_k = \dfrac{f(x_k) - f(x_k + p_k)}{m_k(0) - m_k(p_k)}$ and $0 < \eta_1 \le \eta_2 < 1$: if $\rho_k < \eta_1$ the step is rejected ($x_{k+1} = x_k$) and $\Delta_{k+1} = \frac{1}{2}\Delta_k$; if $\rho_k \ge \eta_1$ it is accepted ($x_{k+1} = x_k + p_k$), with $\Delta_{k+1} = \min(2\Delta_k, \Delta_{\max})$ if moreover $\rho_k > \eta_2$ and $\|p_k\| = \Delta_k$, and $\Delta_{k+1} = \Delta_k$ otherwise.

Hypotheses.

- $f : \mathbb{R}^n \to \mathbb{R}$ of class C^2, bounded below on $S = \{x : f(x) \le f(x_0)\}$;
- $\|B_k\| \le M$ for all k;
- iterates generated with the acceptance and radius rule above;
- steps satisfy the Cauchy bound $m_k(0) - m_k(p_k) \ge \frac{1}{2}\|g_k\| \min(\Delta_k, \|g_k\|/\|B_k\|)$.

Problem. Prove that $\lim_{k\to\infty} \|\nabla f(x_k)\| = 0$.

Note. (1) Step 1 uses a bound on $\nabla^2 f$ on a compact set, but S is not assumed bounded; one must assume $\nabla^2 f$ bounded on a neighbourhood of S (or ∇f Lipschitz). (2) The proof uses $\eta_1 < \frac{1}{4}$, which is not in the statement. (3) Assuming $\|g_k\| \ge \epsilon$ for infinitely many indices and concluding $\Delta_k \ge \bar{\Delta}$ for all k is not valid; the argument of the notes proves $\liminf_k \|g_k\| = 0$ (N&W, Theorem 4.5). For $\lim$ the additional argument of N&W, Theorem 4.6, with ∇f Lipschitz and $\eta_1 > 0$, is needed.

A11 Pseudocode *trust region with Steihaug–CG (N&W, Alg. 4.1; SciPy `trust-ncg`)*

TrustRegion$(f,\ \nabla f,\ \nabla^2 f,\ x_0;\ \Delta_0, \Delta_{\max}, \eta, \varepsilon, k_{\max})$

1: $\Delta \leftarrow \Delta_0$; $f_0 \leftarrow f(x_0)$; $g_0 \leftarrow \nabla f(x_0)$
2: **for** $k = 0, 1, \ldots, k_{\max}$ **do**
3: **if** $\|g_k\| < \varepsilon$ **then return** x_k
4: $(p_k, \text{stop}) \leftarrow \text{SteihaugCG}(g_k, \nabla^2 f(x_k), \Delta)$ (A10; only products $\nabla^2 f(x_k)\, d$ are needed)
5: $\text{pred} \leftarrow -g_k^{\mathsf{T}} p_k - \frac{1}{2} p_k^{\mathsf{T}} \nabla^2 f(x_k) p_k$; **if** pred ≤ 0 **then return** x_k with a warning
6: $f_{\text{new}} \leftarrow f(x_k + p_k)$; $\rho_k \leftarrow (f_k - f_{\text{new}})/\text{pred}$
7: **if** $\rho_k < \frac{1}{4}$ **then** $\Delta \leftarrow \frac{1}{4}\Delta$
8: **else if** $\rho_k > \frac{3}{4}$ **and** stop $\in$ {B, NC} **then** $\Delta \leftarrow \min\{2\Delta, \Delta_{\max}\}$ (step reached the boundary)
9: **if** $\rho_k > \eta$ **then**
10: $x_{k+1} \leftarrow x_k + p_k$; $f_{k+1} \leftarrow f_{\text{new}}$; $g_{k+1} \leftarrow \nabla f(x_{k+1})$ (accept)
11: **else**
12: $x_{k+1} \leftarrow x_k$; $f_{k+1} \leftarrow f_k$; $g_{k+1} \leftarrow g_k$ (reject: no new gradient or Hessian)

Defaults (SciPy `trust-ncg`, `trust-exact`, `trust-krylov`): $\Delta_0 = 1$, $\Delta_{\max} = 1000$, $\eta = 0.15$. Line 4 can use Dogleg (A9) when $\nabla^2 f(x_k)$ or its approximation is positive definite, or ExactSubproblem (A8) for small n (`trust-exact`). Any of the three keeps the Cauchy bound $m_k(0) - m_k(p_k) \ge \frac{1}{2}\|g_k\| \min(\Delta_k,\ \|g_k\|/\|B_k\|)$, and hence the hypotheses of T48 (whose acceptance rule is the same with other constants: accept if $\rho_k \ge \eta_1$, halve or quarter Δ if ρ_k is small). Line 8 asks for "the step reached the boundary" through the flag returned by A10 instead of testing $\|p_k\| = \Delta$ in floating point. A rejected step costs one evaluation of f only.

P16 Practice *added*

Setting. $f(x) = x_1^4 + x_1 x_2 + (1 + x_2)^2$, $\nabla f(x) = (4x_1^3 + x_2,\ x_1 + 2(1 + x_2))^{\mathsf{T}}$, $\nabla^2 f(x) = \begin{pmatrix} 12x_1^2 & 1 \\ 1 & 2 \end{pmatrix}$ (indefinite if $|x_1| < 1/\sqrt{24} \approx 0.204$).

Data. A11 from $x_0 = (0, 0)^{\mathsf{T}}$ with $\Delta_0 = 1$, $\Delta_{\max} = 4$, $\eta = 0.15$; p_k from SteihaugCG (A10) with its default tolerance.

Problem.

a) For $k = 0, \ldots, 4$: decide whether $B_k \succ 0$; compute p_k and the reason CG stops; compute the predicted and actual reductions, ρ_k, and the update applied (table V17).

b) Point out the iteration in which the model predicts a reduction but f increases, and explain it.

V17 Validation table *A11 from $(0, 0)^{\mathsf{T}}$; solves P16*

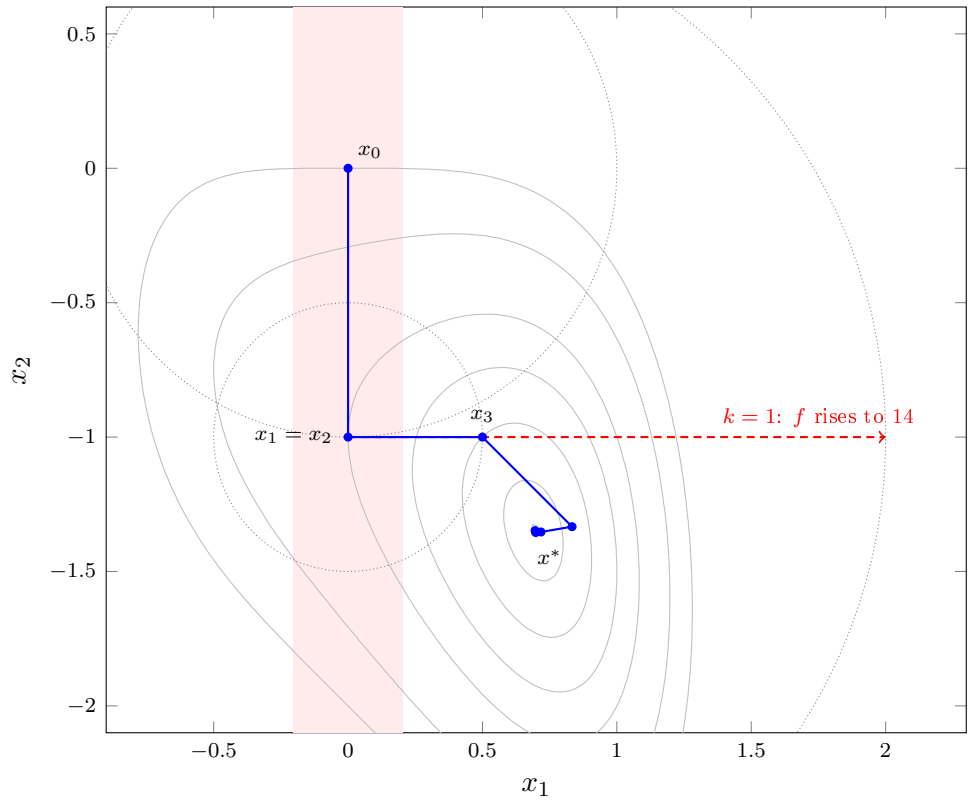
$f(x) = x_1^4 + x_1 x_2 + (1 + x_2)^2$, $B_k = \nabla^2 f(x_k)$, $x^* \approx (0.695884, -1.347942)^{\mathsf{T}}$; $\Delta_0 = 1$, $\Delta_{\max} = 4$, $\eta = 0.15$, $\varepsilon = 10^{-6}$; SteihaugCG with its default tolerance. Column "stop": B = leaves the region, NC = nonpositive curvature ($d^{\mathsf{T}} B_k d \le 0$), I = interior, with the number of CG steps. Reproduced exactly by SciPy's `minimize(method='trust-ncg')` with the same options.

k	x_k	$f(x_k)$	$\|g_k\|$	$\lambda_{\min}(B_k)$	Δ
0	$(0,\ 0)$	1	2	-0.4142	1
1	$(0,\ -1)$	0	1	-0.4142	2
2	$(0,\ -1)$	0	1	-0.4142	0.5
3	$(0.5,\ -1)$	-0.4375	0.707107	1.3820	1
4	$(0.833333,\ -1.333333)$	-0.517747	0.995532	1.8459	1
5	$(0.717673,\ -1.352974)$	-0.581121	0.126138	1.7731	1
6	$(0.697832,\ -1.354826)$	-0.582400	1.2635×10^{-2}	1.7554	1
7	$(0.695890,\ -1.347945)$	-0.582445	3.1528×10^{-5}	1.7535	1
8	$(0.695884,\ -1.347942)$	-0.582445	2.94×10^{-10}		

k	stop	p_k	pred	ared	ρ_k	decision
0	B (1)	$(0,\ -1)$	1	1	1.0000	accept, $\Delta \leftarrow 2$
1	NC (1)	$(2,\ 0)$	2	-14	-7.0000	reject, $\Delta \leftarrow 0.5$
2	NC (1)	$(0.5,\ 0)$	0.5	0.4375	0.8750	accept, $\Delta \leftarrow 1$
3	I (1)	$(\frac{1}{3},\ -\frac{1}{3})$	$\frac{1}{6}$	0.080247	0.4815	accept
4	I (1)	$(-0.115660,\ -0.019640)$	0.058396	0.063374	1.0853	accept
5	I (2)	$(-0.019842,\ -0.001853)$	1.2569×10^{-3}	1.2791×10^{-3}	1.0177	accept
6	I (2)	$(-0.001941,\ 0.006881)$	4.5002×10^{-5}	4.5023×10^{-5}	1.0005	accept
7	I (2)	$(-5.94 \times 10^{-6},\ 2.97 \times 10^{-6})$	9.358×10^{-11}	9.358×10^{-11}	1.0000	accept

Detail by hand. $k = 0$: $g_0 = (0, 2)$, $B_0 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, $d = (0, -2)$, $d^{\mathsf{T}} B_0 d = 8$, $\alpha = \frac{4}{8} = \frac{1}{2}$ and $z_1 = (0, -1)$ lands exactly on the boundary, so $p_0 = (0, -1)$ (the Cauchy point). $k = 1$: $g_1 = (-1, 0)$, $d = (1, 0)$, $d^{\mathsf{T}} B_1 d = 0$: nonpositive curvature; the two boundary points $(\pm 2, -1)$ have model values ∓ 2, so $p_1 = (2, 0)$, and $f(2, -1) = 14$: the model predicts a decrease of 2 and f increases by 14; $\rho_1 < \frac{1}{4}$ divides Δ by 4. $k = 2$: $f(0.5, -1) = -0.4375$. $k = 3$: $g_3 = (-\frac{1}{2}, \frac{1}{2})$, $B_3 = \left(\begin{smallmatrix} 3 & 1 \\ 1 & 2 \end{smallmatrix}\right) \succ 0$, tolerance $\min\{\frac{1}{2}, \sqrt{0.7071}\} \cdot 0.7071 = 0.3536$; the first CG step gives $z_1 = (\frac{1}{3}, -\frac{1}{3})$ and $r_1 = (\frac{1}{6}, \frac{1}{6})$ with $\|r_1\| = 0.2357 < 0.3536$, so CG stops before the Newton step $(0.3, -0.4)$. From $k = 5$ the tolerance is tight enough for two CG steps, the steps are Newton steps and convergence is quadratic ($\rho_k \to 1$).

Figure.



Level curves of $f = x_1^4 + x_1x_2 + (1 + x_2)^2$; the red band $|x_1| < 1/\sqrt{24}$ is where $\nabla^2 f \not\succ 0$. Dotted circles: trust regions of radius 1 (at x_0) and 2, 1, 0.5 (at x_1). Red arrows: rejected steps. The step at $k = 1$ follows the zero-curvature direction $-g$ to the edge of a region that is too large, where the x_1^4 term the model does not see dominates; after shrinking Δ the model becomes reliable again, and from x_4 (positive definite Hessian) the steps are Newton steps.

X 28 PROPOSED EXERCISE *ch5, p. 26 (exercise 9)*

Setting. Let $m_k(p) = f(x_k) + g_k^{\mathsf{T}} p + \frac{1}{2} p^{\mathsf{T}} B_k p$ with $g_k = \nabla f(x_k)$ and B_k symmetric, and the subproblem $\min\{m_k(p) : \|p\|_2 \le \Delta_k\}$. p_k^C is the Cauchy point (minimizer of m_k on $\{-\alpha g_k : 0 \le \alpha \le \Delta_k / \|g_k\|\}$).

Definitions.

- *Acceptance and radius rule of the notes:* with $\rho_k = \dfrac{f(x_k) - f(x_k + p_k)}{m_k(0) - m_k(p_k)}$ and $0 < \eta_1 \le \eta_2 < 1$: if $\rho_k < \eta_1$ the step is rejected ($x_{k+1} = x_k$) and $\Delta_{k+1} = \frac{1}{2}\Delta_k$; if $\rho_k \ge \eta_1$ it is accepted ($x_{k+1} = x_k + p_k$), with $\Delta_{k+1} = \min(2\Delta_k, \Delta_{\max})$ if moreover $\rho_k > \eta_2$ and $\|p_k\| = \Delta_k$, and $\Delta_{k+1} = \Delta_k$ otherwise.

Hypotheses. f of class C^2 with $\nabla^2 f$ bounded, f bounded below, $\|B_k\| \le M$, and the rule above. The global convergence theorem (T48) gives $\lim_k \|\nabla f(x_k)\| = 0$ when $m_k(0) - m_k(p_k) \ge m_k(0) - m_k(p_k^C)$.

Problem. Prove that the conclusion still holds if the steps only satisfy $m_k(0) - m_k(p_k) \ge c_1\big(m_k(0) - m_k(p_k^C)\big)$ for a fixed $c_1 \in (0, 1]$.

Q21 Exam problem *added, 4 pts*

Why the radius cannot shrink to zero. Let f be of class C^2, bounded below, with $\|\nabla^2 f(x)\| \le \beta$ for all x. A11 is run with $B_k = \nabla^2 f(x_k)$ and steps satisfying $\|p_k\| \le \Delta_k$ and the Cauchy bound $m_k(0) - m_k(p_k) \ge \tfrac{1}{2}\|g_k\| \min\{\Delta_k, \|g_k\|/\beta\}$.

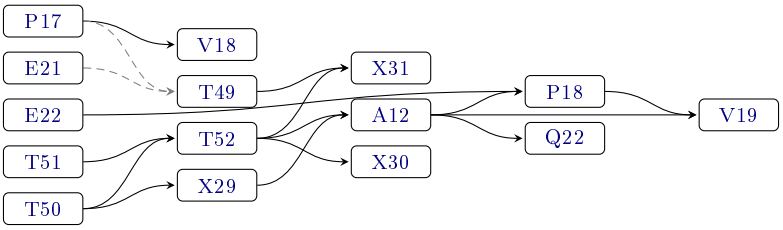
a) Prove that $|\mathrm{ared}_k - \mathrm{pred}_k| \le \beta \Delta_k^2$, and deduce that there is $\bar\Delta > 0$, depending only on ϵ and β, such that $\|g_k\| \ge \epsilon$ and $\Delta_k \le \bar\Delta$ imply $\rho_k > \tfrac{3}{4}$.

b) Deduce that if $\|g_k\| \ge \epsilon$ for all k, then $\Delta_k \ge \min\{\Delta_0, \bar\Delta/4\}$ for all k, and obtain a contradiction with f bounded below. Conclude $\liminf_k \|g_k\| = 0$.

c) Run one iteration by hand on $f(x) = x_1^2 + 10x_2^2$ from $x_0 = (1,1)$ with $\Delta_0 = 0.5$ and the Cauchy point as step: compute p_0, pred, ared, ρ_0, x_1 and Δ_1. Why is $\rho_0 = 1$?

U18. Conjugate directions and linear conjugate gradient

↑Index



E21 Numerical example *ch6, p. 10 (solved exercise 3)*

Nonzero vectors $\{p_0, \dots, p_k\}$ are A-conjugate if $p_i^{\mathsf T} A p_j = 0$ for all $i \ne j$. Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 1.5 \end{pmatrix}$, $p_0 = (1,1)^{\mathsf T}$ and $p_1 = (1,-2)^{\mathsf T}$.

a) Decide whether p_0 and p_1 are A-conjugate.

b) Replace p_1 by $\tilde p_1 = p_1 - \dfrac{p_1^{\mathsf T} A p_0}{p_0^{\mathsf T} A p_0} p_0$ (one Gram–Schmidt step in the inner product $\langle u, v\rangle_A = u^{\mathsf T} A v$) and verify that $\tilde p_1 = (\tfrac{15}{11}, -\tfrac{18}{11})^{\mathsf T}$ is A-conjugate to p_0.

P17 Practice *added*

Why conjugacy matters. Let $\varphi(x) = \tfrac{1}{2} x^{\mathsf T} A x - b^{\mathsf T} x$ with $A = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}$, $b = (1,2)^{\mathsf T}$, $x^* = A^{-1} b = (\tfrac{1}{11}, \tfrac{7}{11})^{\mathsf T}$ and $\varphi(x^*) = -\tfrac{15}{22}$. In every case below, $x_{k+1} = x_k + \alpha_k p_k$ with the exact step $\alpha_k = -\dfrac{(A x_k - b)^{\mathsf T} p_k}{p_k^{\mathsf T} A p_k}$, from $x_0 = (0,0)^{\mathsf T}$.

a) Cyclic coordinate directions $p_0 = e_1$, $p_1 = e_2$, $p_2 = e_1$, $p_3 = e_2$: compute $x_1, \dots, x_4$ and the gap $\varphi(x_k) - \varphi(x^*)$. Show that x^* is not reached in finitely many steps.

b) Directions $p_0 = e_1$ and $p_1 = e_2 - \dfrac{e_2^{\mathsf T} A e_1}{e_1^{\mathsf T} A e_1} e_1 = (-\tfrac{1}{4}, 1)^{\mathsf T}$: verify $p_0^{\mathsf T} A p_1 = 0$ and that $x_2 = x^*$.

c) Steepest descent, $p_k = -(A x_k - b)$: compute $x_1, \dots, x_4$ and the gap, and verify $p_{k+1} \perp p_k$.

d) Explain the difference between a) and b) in terms of the expansion $x^* - x_0 = \sum_i \sigma_i p_i$.

V18 VALIDATION TABLE *solves P17*

$A = \left(\begin{smallmatrix} 4 & 1 \\ 1 & 3 \end{smallmatrix}\right)$, $b = (1, 2)$, $x_0 = 0$, exact step along each direction; gap $= \varphi(x_k) - \varphi(x^*)$ with $\varphi(x^*) = -\frac{15}{22}$.

k	x_k	p_k	α_k	gap
	a) cyclic coordinate directions			
0	$(0,\ 0)$	e_1	$\frac{1}{4}$	$\frac{15}{22}$
1	$\left(\frac{1}{4},\ 0\right)$	e_2	$\frac{7}{12}$	$\frac{49}{88}$
2	$\left(\frac{1}{4},\ \frac{7}{12}\right)$	e_1	$-\frac{7}{48}$	$\frac{49}{1056}$
3	$\left(\frac{5}{48},\ \frac{7}{12}\right)$	e_2	$\frac{7}{144}$	$\frac{49}{12672}$
4	$\left(\frac{5}{48},\ \frac{91}{144}\right)$			$\frac{49}{152064}$
	b) A-conjugate directions			
0	$(0,\ 0)$	e_1	$\frac{1}{4}$	$\frac{15}{22}$
1	$\left(\frac{1}{4},\ 0\right)$	$\left(-\frac{1}{4},\ 1\right)$	$\frac{7}{11}$	$\frac{49}{88}$
2	$\left(\frac{1}{11},\ \frac{7}{11}\right)$			0
	c) steepest descent			
0	$(0,\ 0)$	$(1,\ 2)$	$\frac{1}{4}$	$\frac{15}{22}$
1	$\left(\frac{1}{4},\ \frac{1}{2}\right)$	$\left(-\frac{1}{2},\ \frac{1}{4}\right)$	$\frac{1}{3}$	$\frac{5}{88}$
2	$\left(\frac{1}{12},\ \frac{7}{12}\right)$	$\left(\frac{1}{12},\ \frac{1}{6}\right)$	$\frac{1}{4}$	$\frac{5}{1056}$
3	$\left(\frac{5}{48},\ \frac{5}{8}\right)$	$\left(-\frac{1}{24},\ \frac{1}{48}\right)$	$\frac{1}{3}$	$\frac{5}{12672}$
4	$\left(\frac{13}{144},\ \frac{91}{144}\right)$			$\frac{5}{152064}$

What it shows. An exact line search along directions that are not conjugate (coordinates, gradients) only shrinks the gap by a constant factor ($\frac{1}{12}$ here) per step; along A-conjugate directions the same exact step reaches x^* in $n = 2$ steps, because each step removes one term of $x^* - x_0 = \sum_i \sigma_i p_i$ for good.

T49 PROOF *ch6, Theorem 2.1, pp. 2–3*

Setting. Let $\varphi(x) = \frac{1}{2} x^\mathsf{T} A x - b^\mathsf{T} x$ with $A \in \mathbb{R}^{n \times n}$ symmetric positive definite, $x^* = A^{-1} b$, and residual $r_k = A x_k - b = \nabla \varphi(x_k)$.

Problem. Let $p_0, \ldots, p_{n-1}$ be nonzero and A-conjugate ($p_i^\mathsf{T} A p_j = 0$ for $i \neq j$), $x_0 \in \mathbb{R}^n$ arbitrary, and $x_{k+1} = x_k + \alpha_k p_k$ with $\alpha_k = -\dfrac{r_k^\mathsf{T} p_k}{p_k^\mathsf{T} A p_k}$.

a) Prove that $p_0, \ldots, p_{n-1}$ are linearly independent.

b) Prove that α_k is the exact minimizer of $\alpha \mapsto \varphi(x_k + \alpha p_k)$.

c) Prove that $r_k^\mathsf{T} p_i = 0$ for all $i < k$, and that $x_n = x^*$.

Note. The notes use $\{p_i\}$ as a basis without proving linear independence (part a), which is what makes the expansion $x^* - x_0 = \sum_i \sigma_i p_i$ possible.

T50 PROOF *ch6, Lemma 2.1, p. 3*

Setting. Let $\varphi(x) = \frac{1}{2}x^\mathsf{T}Ax - b^\mathsf{T}x$ with $A \in \mathbb{R}^{n\times n}$ symmetric positive definite, $x^* = A^{-1}b$, and residual $r_k = Ax_k - b = \nabla\varphi(x_k)$. $x_k \in \mathbb{R}^n$, $p_k \neq 0$, α_k the minimizer of $\alpha \mapsto \varphi(x_k + \alpha p_k)$, and $x_{k+1} = x_k + \alpha_k p_k$.

Problem.

a) Prove that $\alpha_k = -\dfrac{r_k^\mathsf{T}p_k}{p_k^\mathsf{T}Ap_k}$.

b) Prove that $r_{k+1} = r_k + \alpha_k Ap_k$ and $r_{k+1}^\mathsf{T}p_k = 0$.

X29 PROPOSED EXERCISE *ch6, p. 12 (exercise 2)*

Setting. Let $\varphi(x) = \frac{1}{2}x^\mathsf{T}Ax - b^\mathsf{T}x$ with $A \in \mathbb{R}^{n\times n}$ symmetric positive definite, $x^* = A^{-1}b$, and residual $r_k = Ax_k - b = \nabla\varphi(x_k)$.

Definitions.

- *Linear CG:* $r_0 = Ax_0 - b$, $p_0 = -r_0$ and, for $k \geq 0$, $\alpha_k = \dfrac{r_k^\mathsf{T}r_k}{p_k^\mathsf{T}Ap_k}$, $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k + \alpha_k Ap_k$, $\beta_{k+1} = \dfrac{r_{k+1}^\mathsf{T}r_{k+1}}{r_k^\mathsf{T}r_k}$, $p_{k+1} = -r_{k+1} + \beta_{k+1}p_k$.

Problem. Prove that the step α_k minimizing $\varphi(x_k + \alpha p_k)$ is exactly $\alpha_k = \dfrac{r_k^\mathsf{T}r_k}{p_k^\mathsf{T}Ap_k}$. You may use $r_k^\mathsf{T}p_{k-1} = 0$ (T50).

T51 PROOF *ch6, Lemma 2.3, pp. 5–6*

Setting. Let $\varphi(x) = \frac{1}{2}x^\mathsf{T}Ax - b^\mathsf{T}x$ with $A \in \mathbb{R}^{n\times n}$ symmetric positive definite, $x^* = A^{-1}b$, and residual $r_k = Ax_k - b = \nabla\varphi(x_k)$.

Definitions.

- *Linear CG:* $r_0 = Ax_0 - b$, $p_0 = -r_0$ and, for $k \geq 0$, $\alpha_k = \dfrac{r_k^\mathsf{T}r_k}{p_k^\mathsf{T}Ap_k}$, $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k + \alpha_k Ap_k$, $\beta_{k+1} = \dfrac{r_{k+1}^\mathsf{T}r_{k+1}}{r_k^\mathsf{T}r_k}$, $p_{k+1} = -r_{k+1} + \beta_{k+1}p_k$.

Problem. Assume $r_0, r_1 \neq 0$. Prove that $p_0^\mathsf{T}Ap_1 = 0$.

Note. The notes write $r_0^\mathsf{T}Ap_0 = \frac{r_0^\mathsf{T}r_0}{\alpha_0}$; the correct value is $r_0^\mathsf{T}Ap_0 = \frac{r_0^\mathsf{T}r_1 - r_0^\mathsf{T}r_0}{\alpha_0} = -\frac{r_0^\mathsf{T}r_0}{\alpha_0}$. With the right sign, $p_0^\mathsf{T}Ap_0 = -r_0^\mathsf{T}Ap_0 = \frac{\|r_0\|^2}{\alpha_0}$ and $p_0^\mathsf{T}Ap_1 = -\frac{\|r_1\|^2}{\alpha_0} + \beta_1\frac{\|r_0\|^2}{\alpha_0} = 0$.

T52 PROOF *ch6, Lemma 2.2 and Theorem 2.2, pp. 3–6*

Setting. Let $\varphi(x) = \frac{1}{2}x^\mathsf{T}Ax - b^\mathsf{T}x$ with $A \in \mathbb{R}^{n\times n}$ symmetric positive definite, $x^* = A^{-1}b$, and residual $r_k = Ax_k - b = \nabla\varphi(x_k)$.

Definitions.

- *Linear CG:* $r_0 = Ax_0 - b$, $p_0 = -r_0$ and, for $k \geq 0$, $\alpha_k = \dfrac{r_k^\mathsf{T}r_k}{p_k^\mathsf{T}Ap_k}$, $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k + \alpha_k Ap_k$, $\beta_{k+1} = \dfrac{r_{k+1}^\mathsf{T}r_{k+1}}{r_k^\mathsf{T}r_k}$, $p_{k+1} = -r_{k+1} + \beta_{k+1}p_k$.

Problem. Assume $r_0, \ldots, r_{k-1} \neq 0$. Prove by a single induction on $k \geq 1$ that

(i) $r_k^\mathsf{T}p_i = 0$ for $i < k$;

(ii) $r_k^\mathsf{T}r_i = 0$ for $i < k$;

(iii) $p_k^\mathsf{T}Ap_i = 0$ for $i < k$;

(iv) $\operatorname{span}\{p_0,\dots,p_{k-1}\} = \operatorname{span}\{r_0,\dots,r_{k-1}\}$.

Derive along the way the formulas $\alpha_{k-1} = \dfrac{\|r_{k-1}\|^2}{p_{k-1}^{\mathsf{T}} A p_{k-1}}$ and $\beta_k = \dfrac{\|r_k\|^2}{\|r_{k-1}\|^2}$ from the conditions $r_k^{\mathsf{T}} p_{k-1} = 0$ and $p_k^{\mathsf{T}} A p_{k-1} = 0$.

Note. In the notes, Lemma 2.2 uses the A-conjugacy of the previous directions as an induction hypothesis, but conjugacy is proved afterwards in Theorem 2.2 using Lemma 2.2. The two inductions must be merged into one, as above, to avoid circularity. The divisions by α_i need $\alpha_i \neq 0$, which holds because $r_i \neq 0$; and the formula for α_{k-1} uses $p_{k-1}^{\mathsf{T}} r_{k-1} = -\|r_{k-1}\|^2$, which itself follows from (i).

X30 Proposed exercise *ch6, p. 11 (exercise 1)*

Setting. Let $\varphi(x) = \frac{1}{2} x^{\mathsf{T}} A x - b^{\mathsf{T}} x$ with $A \in \mathbb{R}^{n\times n}$ symmetric positive definite, $x^* = A^{-1} b$, and residual $r_k = A x_k - b = \nabla\varphi(x_k)$.

Definitions.

- *Linear CG:* $r_0 = A x_0 - b$, $p_0 = -r_0$ and, for $k \geq 0$, $\alpha_k = \dfrac{r_k^{\mathsf{T}} r_k}{p_k^{\mathsf{T}} A p_k}$, $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k + \alpha_k A p_k$, $\beta_{k+1} = \dfrac{r_{k+1}^{\mathsf{T}} r_{k+1}}{r_k^{\mathsf{T}} r_k}$, $p_{k+1} = -r_{k+1} + \beta_{k+1} p_k$.

Problem. Prove formally by induction that the residuals satisfy $r_i^{\mathsf{T}} r_j = 0$ for all $i \neq j$ (as long as they are nonzero).

X31 Proposed exercise *ch6, p. 12 (exercise 3)*

Setting. Let $\varphi(x) = \frac{1}{2} x^{\mathsf{T}} A x - b^{\mathsf{T}} x$ with $A \in \mathbb{R}^{n\times n}$ symmetric positive definite, $x^* = A^{-1} b$, and residual $r_k = A x_k - b = \nabla\varphi(x_k)$.

Definitions.

- *Linear CG:* $r_0 = A x_0 - b$, $p_0 = -r_0$ and, for $k \geq 0$, $\alpha_k = \dfrac{r_k^{\mathsf{T}} r_k}{p_k^{\mathsf{T}} A p_k}$, $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k + \alpha_k A p_k$, $\beta_{k+1} = \dfrac{r_{k+1}^{\mathsf{T}} r_{k+1}}{r_k^{\mathsf{T}} r_k}$, $p_{k+1} = -r_{k+1} + \beta_{k+1} p_k$.

Problem. Prove that linear CG reaches the exact solution x^* of $Ax = b$ in at most n iterations (exact arithmetic).

E22 Numerical example *ch6, p. 9 (solved exercise 1)*

Definitions.

- *Linear CG:* $r_0 = A x_0 - b$, $p_0 = -r_0$ and, for $k \geq 0$, $\alpha_k = \dfrac{r_k^{\mathsf{T}} r_k}{p_k^{\mathsf{T}} A p_k}$, $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k + \alpha_k A p_k$, $\beta_{k+1} = \dfrac{r_{k+1}^{\mathsf{T}} r_{k+1}}{r_k^{\mathsf{T}} r_k}$, $p_{k+1} = -r_{k+1} + \beta_{k+1} p_k$.

Data. $A = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}$, $b = (1,2)^{\mathsf{T}}$, $x_0 = (0,0)^{\mathsf{T}}$.

Problem. Compute r_0, p_0, α_0, x_1 and r_1, and verify $r_1^{\mathsf{T}} r_0 = 0$.

A12 Pseudocode *preconditioned conjugate gradient (ch6, p. 7; N&W, Alg. 5.3)*

LinearCG$(A,\ b,\ x_0;\ M, \varepsilon, k_{\max})$ A symmetric positive definite, preconditioner $M \succ 0$ ($M = I$: plain CG)

1: $x \leftarrow x_0$; $r \leftarrow Ax_0 - b$; $y \leftarrow M^{-1}r$; $p \leftarrow -y$; $\delta \leftarrow r^{\mathsf{T}}y$
2: **for** $k = 0, 1, \ldots, k_{\max}$ **do**
3: **if** $\|r\| \le \varepsilon\|b\|$ **then return** x (relative residual)
4: $w \leftarrow Ap$ (the only matrix–vector product of the iteration)
5: $\alpha \leftarrow \delta/(p^{\mathsf{T}}w)$
6: $x \leftarrow x + \alpha p$; $r \leftarrow r + \alpha w$
7: $y \leftarrow M^{-1}r$; $\delta_{\text{new}} \leftarrow r^{\mathsf{T}}y$; $\beta \leftarrow \delta_{\text{new}}/\delta$; $\delta \leftarrow \delta_{\text{new}}$ (one solve with M)
8: $p \leftarrow -y + \beta p$
9: **return** x with a warning (iteration limit)

With $M = I$ this is the CG of the notes ($y = r$, $\delta = r^{\mathsf{T}}r$). Storage: five vectors (x, r, y, p, w); A is only needed through products Ap and M through solves $My = r$. The preconditioner is chosen so that $M^{-1}A$ has clustered eigenvalues and $My = r$ is cheap: Jacobi $M = \operatorname{diag}(A)$, incomplete Cholesky $M = \tilde{L}\tilde{L}^{\mathsf{T}}$, or a multigrid cycle; the bound of S2 then holds with $\kappa(M^{-1}A)$ instead of $\kappa(A)$. In floating point the recursively updated r drifts away from $Ax - b$, so codes recompute $r \leftarrow Ax - b$ every so often or before stopping. Exact arithmetic: at most n iterations (X31). This is `scipy.sparse.linalg.cg(A, b, rtol, M=...)` and MATLAB's `pcg`; tables V19–V19 use $M = I$.

P18 Practice *added*

Definitions.

- *Linear CG:* $r_0 = Ax_0 - b$, $p_0 = -r_0$ and, for $k \ge 0$, $\alpha_k = \dfrac{r_k^{\mathsf{T}}r_k}{p_k^{\mathsf{T}}Ap_k}$, $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k + \alpha_k Ap_k$, $\beta_{k+1} = \dfrac{r_{k+1}^{\mathsf{T}}r_{k+1}}{r_k^{\mathsf{T}}r_k}$, $p_{k+1} = -r_{k+1} + \beta_{k+1}p_k$.

Problem. Linear CG by hand. Same data as E22: $A = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}$, $b = (1, 2)^{\mathsf{T}}$, $x_0 = (0, 0)^{\mathsf{T}}$.

a) Complete the second iteration: β_1, p_1, Ap_1, α_1, x_2, r_2.

b) Verify $r_1^{\mathsf{T}}p_0 = 0$, $p_1^{\mathsf{T}}Ap_0 = 0$, $r_2 = 0$ and $x_2 = A^{-1}b$.

c) Steepest descent with exact step from the same x_0 takes the same first step. Explain which ingredient of CG makes the second step reach x^* (compare with V18 c).

V19 Validation table *A12; solves P18 and E22*

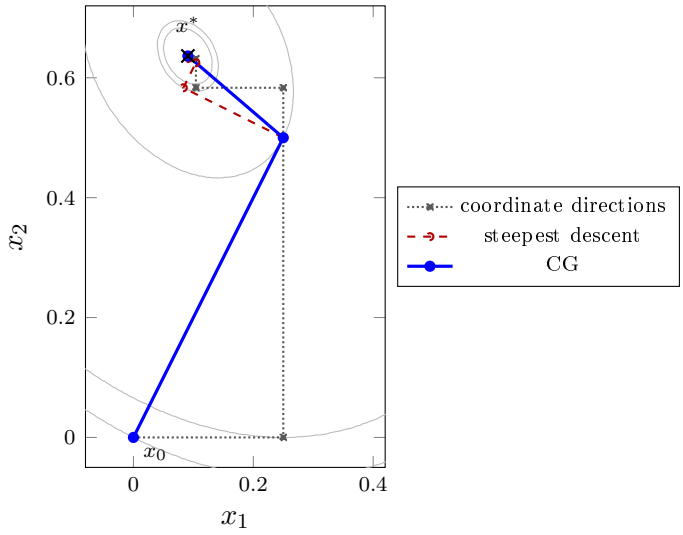
$A = \left(\begin{smallmatrix} 4 & 1 \\ 1 & 3 \end{smallmatrix}\right)$, $b = (1, 2)$, $x_0 = (0, 0)$, $x^* = (\frac{1}{11}, \frac{7}{11})$.

k	x_k	r_k	p_k	Ap_k	α_k	β_{k+1}	$\varphi(x_k)$
0	$(0,\ 0)$	$(-1,\ -2)$	$(1,\ 2)$	$(6,\ 7)$	$\frac{1}{4}$	$\frac{1}{16}$	0
1	$(\frac{1}{4},\ \frac{1}{2})$	$(\frac{1}{2},\ -\frac{1}{4})$	$(-\frac{7}{16},\ \frac{3}{8})$	$(-\frac{11}{8},\ \frac{11}{16})$	$\frac{4}{11}$	0	$-\frac{5}{8}$
2	$(\frac{1}{11},\ \frac{7}{11})$	$(0,\ 0)$					$-\frac{15}{22}$

Checks: $r_0^{\mathsf{T}}r_0 = 5$, $p_0^{\mathsf{T}}Ap_0 = 20$, $r_1^{\mathsf{T}}r_1 = \frac{5}{16}$, $p_1^{\mathsf{T}}Ap_1 = \frac{55}{64}$; $r_1^{\mathsf{T}}r_0 = 0$, $r_1^{\mathsf{T}}p_0 = 0$, $p_1^{\mathsf{T}}Ap_0 = 0$.

What it shows. CG builds the conjugate direction $p_1 = -r_1 + \beta_1 p_0$ from r_1 and p_0 alone ($\beta_1 = \frac{1}{16}$), without a Gram–Schmidt sweep over all previous directions. The first step coincides with steepest descent; the correction $\beta_1 p_0$ is what turns the second step into one that reaches x^*.

Figure.



Level curves of $\varphi(x) = \frac{1}{2}x^\mathsf{T}Ax - b^\mathsf{T}x$, $A = \left(\begin{smallmatrix} 4 & 1 \\ 1 & 3 \end{smallmatrix}\right)$, $b = (1, 2)$, through the iterates. All three methods use the exact step along their direction. Steepest descent and CG take the same first step; CG then moves along p_1, which is A-conjugate to p_0 and points at the centre of the ellipse, so it stops at x^*. Steepest descent zigzags with $g_{k+1} \perp g_k$, and cyclic coordinate directions (not conjugate since A is not diagonal) also never reach x^* in finitely many steps.

Q22 EXAM PROBLEM *added, 4 pts*

Conjugate gradient by hand and finite termination. Let $A = \mathrm{diag}(1, 2, 3)$, $b = (1, 1, 1)^\mathsf{T}$ and $\phi(x) = \frac{1}{2}x^\mathsf{T}Ax - b^\mathsf{T}x$.

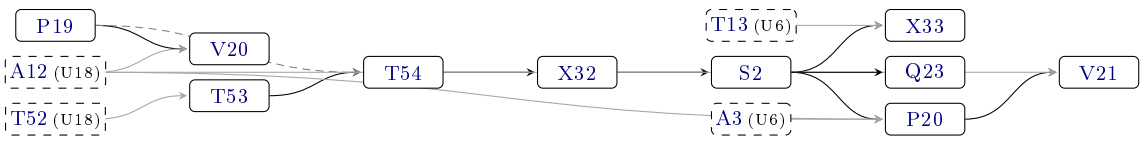
a) Run linear CG (A12 with $M = I$) from $x_0 = 0$ with exact fractions, recording r_k, p_k, α_k, β_k and x_k until $r_k = 0$.

b) Verify that $r_2 \perp r_0, r_1$ and $p_2^\mathsf{T}Ap_0 = p_2^\mathsf{T}Ap_1 = 0$, and that x_2 minimizes ϕ over $x_0 + \mathcal{K}_2(A, r_0)$.

c) Prove that if $A \succ 0$ has only d distinct eigenvalues, CG terminates in at most d iterations (use the polynomial characterization of x_k). Check it with $A = \mathrm{diag}(1, 1, 3)$ and the same b.

U19. Krylov subspaces and rate of convergence of CG

↑Index



P19 PRACTICE *added*

Definitions.

- *Linear CG:* $r_0 = Ax_0 - b$, $p_0 = -r_0$ and, for $k \geq 0$, $\alpha_k = \dfrac{r_k^\mathsf{T} r_k}{p_k^\mathsf{T} A p_k}$, $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k + \alpha_k A p_k$, $\beta_{k+1} = \dfrac{r_{k+1}^\mathsf{T} r_{k+1}}{r_k^\mathsf{T} r_k}$, $p_{k+1} = -r_{k+1} + \beta_{k+1} p_k$.

Problem. Few distinct eigenvalues. Let $A = \mathrm{diag}(1, 1, 10, 10)$, $b = (1, 1, 1, 1)^\mathsf{T}$, $x_0 = 0$, so $x^* = (1, 1, \frac{1}{10}, \frac{1}{10})^\mathsf{T}$, and $\|v\|_A = \sqrt{v^\mathsf{T} A v}$.

a) Run CG by hand and show that $r_2 = 0$, although $n = 4$.

b) Run steepest descent with exact step $\alpha_k = \frac{g_k^\mathsf{T} g_k}{g_k^\mathsf{T} A g_k}$, $g_k = Ax_k - b$, and show that $\|x_{k+1} - x^*\|_A / \|x_k - x^*\|_A = \frac{9}{11} = \frac{\kappa - 1}{\kappa + 1}$ at every step ($\kappa = 10$).

c) Explain a) through the dimension of $\mathcal{K}_k(A, r_0) = \mathrm{span}\{r_0, Ar_0, \ldots, A^{k-1} r_0\}$.

V20 VALIDATION TABLE *solves P19*

$A = \mathrm{diag}(1, 1, 10, 10)$, $b = (1, 1, 1, 1)$, $x_0 = 0$; $\|e_k\|_A^2 = (x_k - x^*)^\mathsf{T} A (x_k - x^*)$.

CG						
k	x_k	r_k	p_k	α_k	β_{k+1}	$\|e_k\|_A^2$
0	$(0, 0, 0, 0)$	$(-1, -1, -1, -1)$	$(1, 1, 1, 1)$	$\frac{2}{11}$	$\frac{81}{121}$	$\frac{11}{5}$
1	$\frac{2}{11}(1, 1, 1, 1)$	$\frac{9}{11}(-1, -1, 1, 1)$	$\frac{1}{121}(180, 180, -18, -18)$	$\frac{11}{20}$	0	$\frac{81}{55}$
2	$(1, 1, \frac{1}{10}, \frac{1}{10})$	0				0
Steepest descent (exact step $\alpha_k = \frac{2}{11}$ at every k)						
$\|e_k\|_A^2$ for $k = 0, 1, 2, 3$: $\frac{11}{5}$, $\frac{81}{55}$, $\frac{6561}{6655}$, $\frac{531441}{805255}$ (ratio $\frac{81}{121} = (\frac{9}{11})^2$ each step)						

What it shows. CG terminates after as many steps as A has distinct eigenvalues, because $\mathcal{K}_k(A, r_0)$ cannot grow beyond dimension 2; steepest descent, from this x_0, contracts by exactly the worst-case factor at every step.

T53 PROOF *ch6, Theorem 2.3, pp. 7–8*

Setting. Let $\varphi(x) = \frac{1}{2} x^\mathsf{T} A x - b^\mathsf{T} x$ with $A \in \mathbb{R}^{n \times n}$ symmetric positive definite, $x^* = A^{-1} b$, and residual $r_k = Ax_k - b = \nabla \varphi(x_k)$.

Definitions.

- *Linear CG:* $r_0 = Ax_0 - b$, $p_0 = -r_0$ and, for $k \geq 0$, $\alpha_k = \dfrac{r_k^\mathsf{T} r_k}{p_k^\mathsf{T} A p_k}$, $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k + \alpha_k A p_k$, $\beta_{k+1} = \dfrac{r_{k+1}^\mathsf{T} r_{k+1}}{r_k^\mathsf{T} r_k}$, $p_{k+1} = -r_{k+1} + \beta_{k+1} p_k$.

Problem. Let $\mathcal{K}_k(A, r_0) = \mathrm{span}\{r_0, Ar_0, \ldots, A^{k-1} r_0\}$. Assume $r_0, \ldots, r_{k-1} \neq 0$. Prove:

a) $r_k^\mathsf{T} r_i = 0$ for all $i < k$;

b) $\mathrm{span}\{r_0, \ldots, r_{k-1}\} = \mathrm{span}\{p_0, \ldots, p_{k-1}\} = \mathcal{K}_k(A, r_0)$;

c) x_k minimizes φ over the affine set $x_0 + \mathcal{K}_k(A, r_0)$.

You may use T52.

Note. The induction in the notes only shows $\mathrm{span}\{r_0, \ldots, r_k\} \subset \mathcal{K}_{k+1}(A, r_0)$. Equality needs a dimension count: $r_0, \ldots, r_k$ are nonzero and mutually orthogonal, so they span a space of dimension $k + 1$, while $\dim \mathcal{K}_{k+1} \leq k + 1$. Item 3 also needs $\{p_0, \ldots, p_{k-1}\}$ to be a basis of $\mathcal{K}_k$ (linear independence, T49 a).

T54 PROOF *ch6, Observation, p. 8 (no proof in the notes)*

Setting. Let $\varphi(x) = \frac{1}{2}x^{\mathsf{T}}Ax - b^{\mathsf{T}}x$ with $A \in \mathbb{R}^{n\times n}$ symmetric positive definite, $x^* = A^{-1}b$, and residual $r_k = Ax_k - b = \nabla\varphi(x_k)$.

Definitions.

- *Linear CG:* $r_0 = Ax_0 - b$, $p_0 = -r_0$ and, for $k \geq 0$, $\alpha_k = \dfrac{r_k^{\mathsf{T}}r_k}{p_k^{\mathsf{T}}Ap_k}$, $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k + \alpha_k Ap_k$, $\beta_{k+1} = \dfrac{r_{k+1}^{\mathsf{T}}r_{k+1}}{r_k^{\mathsf{T}}r_k}$, $p_{k+1} = -r_{k+1} + \beta_{k+1}p_k$.

Problem. Let $\|v\|_A = \sqrt{v^{\mathsf{T}}Av}$ and $\mathcal{P}_k^1$ the polynomials P of degree $\leq k$ with $P(0) = 1$.

a) Using T53 (3), prove that $\|x_k - x^*\|_A = \min_{P\in\mathcal{P}_k^1}\|P(A)(x_0 - x^*)\|_A$, and hence $\|x_k - x^*\|_A \leq \min_{P\in\mathcal{P}_k^1}\max_{\lambda\in\sigma(A)}|P(\lambda)|\,\|x_0 - x^*\|_A$.

b) Deduce that if A has only m distinct eigenvalues, CG terminates in at most m iterations.

X32 PROPOSED EXERCISE *ch6, p. 13 (exercise 7)*

Setting. Let $\varphi(x) = \frac{1}{2}x^{\mathsf{T}}Ax - b^{\mathsf{T}}x$ with $A \in \mathbb{R}^{n\times n}$ symmetric positive definite, $x^* = A^{-1}b$, and residual $r_k = Ax_k - b = \nabla\varphi(x_k)$. $\kappa = \lambda_{\max}(A)/\lambda_{\min}(A)$, $\|v\|_A = \sqrt{v^{\mathsf{T}}Av}$.

Definitions.

- *Linear CG:* $r_0 = Ax_0 - b$, $p_0 = -r_0$ and, for $k \geq 0$, $\alpha_k = \dfrac{r_k^{\mathsf{T}}r_k}{p_k^{\mathsf{T}}Ap_k}$, $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k + \alpha_k Ap_k$, $\beta_{k+1} = \dfrac{r_{k+1}^{\mathsf{T}}r_{k+1}}{r_k^{\mathsf{T}}r_k}$, $p_{k+1} = -r_{k+1} + \beta_{k+1}p_k$.

Tools.

- Chebyshev polynomials: $T_k(t) = \cos(k\arccos t)$ for $|t| \leq 1$ (so $|T_k(t)| \leq 1$), and $T_k(t) = \frac{1}{2}\big[(t + \sqrt{t^2-1})^k + (t - \sqrt{t^2-1})^k\big]$ for $|t| > 1$.
- The scaled polynomial
$$P_k^*(\lambda) = \frac{T_k\big(\frac{\kappa+1-2\lambda/\lambda_{\min}}{\kappa-1}\big)}{T_k\big(\frac{\kappa+1}{\kappa-1}\big)}$$
satisfies $P_k^*(0) = 1$ and $\max_{\lambda\in[\lambda_{\min},\lambda_{\max}]}|P_k^*(\lambda)| = 1/T_k\big(\frac{\kappa+1}{\kappa-1}\big)$.
- (T54 a) $\|x_k - x^*\|_A \leq \min_P \max_{\lambda\in\sigma(A)}|P(\lambda)|\,\|x_0 - x^*\|_A$, over polynomials P of degree $\leq k$ with $P(0) = 1$.

Problem. Prove that $\|x_k - x^*\|_A \leq 2\Big(\dfrac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\Big)^k\|x_0 - x^*\|_A$.

S2 STATEMENT *ch6, Theorem 2.4, p. 8 (no proof in the notes)*

Setting. Let $\varphi(x) = \frac{1}{2}x^{\mathsf{T}}Ax - b^{\mathsf{T}}x$ with $A \in \mathbb{R}^{n\times n}$ symmetric positive definite, $x^* = A^{-1}b$, and residual $r_k = Ax_k - b = \nabla\varphi(x_k)$. $\kappa = \lambda_{\max}(A)/\lambda_{\min}(A)$, $\|v\|_A = \sqrt{v^{\mathsf{T}}Av}$.

Definitions.

- *Linear CG:* $r_0 = Ax_0 - b$, $p_0 = -r_0$ and, for $k \geq 0$, $\alpha_k = \dfrac{r_k^{\mathsf{T}}r_k}{p_k^{\mathsf{T}}Ap_k}$, $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k + \alpha_k Ap_k$, $\beta_{k+1} = \dfrac{r_{k+1}^{\mathsf{T}}r_{k+1}}{r_k^{\mathsf{T}}r_k}$, $p_{k+1} = -r_{k+1} + \beta_{k+1}p_k$.

Statement. The CG iterates satisfy

$$\|x_k - x^*\|_A \leq 2\Big(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\Big)^k\|x_0 - x^*\|_A.$$

Problem. Be able to state it, and compare it with the steepest-descent factor $\frac{\kappa-1}{\kappa+1}$ per step in the same norm (T13). The proof is X32.

X33 Proposed exercise *ch6, p. 13 (exercise 9)*

Setting. $\varphi(x) = \frac{1}{2}x^{\mathsf{T}}Ax - b^{\mathsf{T}}x$ with $\kappa(A) = 1000$, and the error bounds

- steepest descent with exact line search: $\|x_{k+1} - x^*\|_A \le \frac{\kappa-1}{\kappa+1}\|x_k - x^*\|_A$;
- linear CG: $\|x_k - x^*\|_A \le 2\big(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\big)^k \|x_0 - x^*\|_A$ (S2).

Problem.

a) Compare both bounds analytically.

b) Estimate how many iterations each bound needs to reduce $\|x_k - x^*\|_A$ by a factor 10^{-6}.

P20 Practice *added*

Bound against actual behaviour.

Setting. $A = \operatorname{diag}(1, 12, 23, \dots, 100)$ ($n = 10$, $\lambda_i = 1 + 11(i-1)$, $\kappa = 100$), $b = (1, \dots, 1)^{\mathsf{T}}$, $x_0 = 0$, $\|v\|_A = \sqrt{v^{\mathsf{T}}Av}$.

Problem.

a) Run linear CG (A12) and steepest descent with exact step $\alpha_k = \frac{g_k^{\mathsf{T}} g_k}{g_k^{\mathsf{T}} A g_k}$ (A3), and tabulate $\|x_k - x^*\|_A / \|x_0 - x^*\|_A$ for both (table V21).

b) Add the bounds $2\big(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\big)^k = 2(\frac{9}{11})^k$ and $\big(\frac{\kappa-1}{\kappa+1}\big)^k = (\frac{99}{101})^k$ to the table. Which bound is tight?

c) Explain why CG stops at $k = 10$.

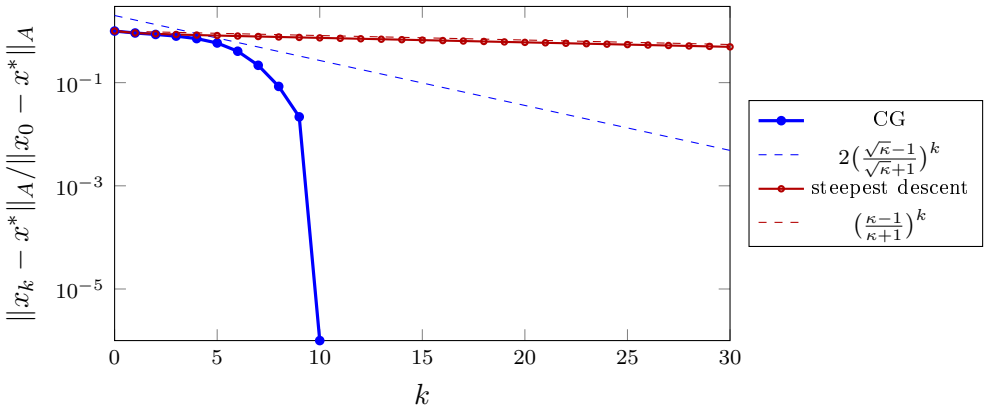
V21 Validation table *solves P20*

$A = \operatorname{diag}(1, 12, \dots, 100)$, $b = (1, \dots, 1)$, $x_0 = 0$. Columns are $\|x_k - x^*\|_A / \|x_0 - x^*\|_A$ and the corresponding bounds.

k	CG	$2(\frac{9}{11})^k$	steepest descent	$(\frac{99}{101})^k$
0	1	2	1	1
1	0.9171	1.6364	0.9171	0.9802
2	0.8595	1.3388	0.8825	0.9608
3	0.7980	1.0954	0.8576	0.9418
4	0.7144	0.8963	0.8368	0.9231
5	0.5869	0.7333	0.8180	0.9048
6	0.4065	0.6000	0.8006	0.8869
7	0.2169	0.4909	0.7840	0.8694
8	8.454×10^{-2}	0.4016	0.7681	0.8521
9	2.171×10^{-2}	0.3286	0.7526	0.8353
10	$\approx 5 \times 10^{-16}$	0.2689	0.7376	0.8187
20			0.6037	0.6703
40			0.4046	0.4493

What it shows. The first CG step is a steepest-descent step (same value at $k = 1$); after that CG pulls away. The $\sqrt{\kappa}$ bound is pessimistic here because it ignores the finite spectrum, and CG ends at $k = n = 10$; steepest descent stays close to its own bound.

Figure.



$A = \mathrm{diag}(1, 12, 23, \ldots, 100)$ ($n = 10$, $\kappa = 100$), $b = (1, \ldots, 1)$, $x_0 = 0$. Steepest descent stays close to its bound, which only gives a factor ≈ 0.98 per step. The CG bound gives $9/11 \approx 0.82$ per step, and the actual CG error falls faster than the bound and reaches 0 (rounding level) at $k = 10 = n$, as it must with 10 distinct eigenvalues.

Q23 EXAM PROBLEM *added, 4 pts*

Condition number, iterations and preconditioning. CG satisfies $\|x_k - x^*\|_A \le 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k \|x_0 - x^*\|_A$, with $\kappa = \kappa(A)$.

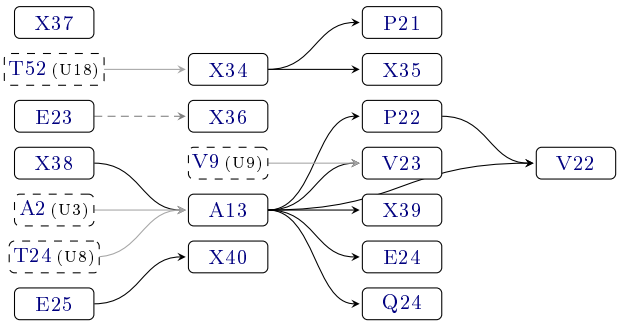
a) Compute the number of iterations that the bound needs to reduce $\|x_k - x^*\|_A$ by 10^{-6} for $\kappa = 10^2$ and $\kappa = 10^4$, and compare with steepest descent with exact step (factor $\frac{\kappa-1}{\kappa+1}$ per iteration in the same norm).

b) For the tridiagonal matrix $A = \mathrm{tridiag}(-1, 2, -1) \in \mathbb{R}^{n\times n}$, with eigenvalues $2 - 2\cos\frac{j\pi}{n+1}$, show that $\kappa(A) \approx \frac{4(n+1)^2}{\pi^2}$, and deduce how the number of CG iterations grows with n.

c) With preconditioner $M \succ 0$, the bound holds with $\kappa(M^{-1}A)$. Show that the Jacobi preconditioner $M = \mathrm{diag}(A)$ does not change κ for the matrix in b), and state which property a useful M must have.

U20. Nonlinear conjugate gradient and L-BFGS

↑Index



ch6 develops only linear CG; nonlinear CG and L-BFGS appear in its solved and proposed exercises (they are developed in N&W, Sections 5.2 and 7.2).

E23 Numerical example *ch6, p. 10 (solved exercise 2)*

Definitions.

- *Nonlinear CG:* $p_0 = -g_0$, $p_k = -g_k + \beta_k p_{k-1}$ with $g_k = \nabla f(x_k)$ and either $\beta_k^{FR} = \dfrac{g_k^{\mathsf{T}} g_k}{g_{k-1}^{\mathsf{T}} g_{k-1}}$ (Fletcher–Reeves) or $\beta_k^{PR} = \dfrac{g_k^{\mathsf{T}}(g_k - g_{k-1})}{g_{k-1}^{\mathsf{T}} g_{k-1}}$ (Polak–Ribière).

Problem. Let $g_{k-1} = (1, 2)^{\mathsf{T}}$ and $g_k = (0.5, 1)^{\mathsf{T}}$.

a) Compute β_k^{FR} and β_k^{PR}.

b) With $p_{k-1} = -g_{k-1}$, compute p_k for both choices and decide whether each is a descent direction ($g_k^{\mathsf{T}} p_k < 0$).

Note. Here $g_k = \frac{1}{2} g_{k-1}$, so $g_k^{\mathsf{T}} p_{k-1} \neq 0$: these gradients cannot come from an exact line search along $p_{k-1} = -g_{k-1}$. The example only exercises the formulas.

X34 Proposed exercise *ch6, p. 12 (exercise 4)*

Definitions.

- *Nonlinear CG:* $p_0 = -g_0$, $p_k = -g_k + \beta_k p_{k-1}$ with $g_k = \nabla f(x_k)$ and either $\beta_k^{FR} = \dfrac{g_k^{\mathsf{T}} g_k}{g_{k-1}^{\mathsf{T}} g_{k-1}}$ (Fletcher–Reeves) or $\beta_k^{PR} = \dfrac{g_k^{\mathsf{T}}(g_k - g_{k-1})}{g_{k-1}^{\mathsf{T}} g_{k-1}}$ (Polak–Ribière).

Setting. $f(x) = \frac{1}{2} x^{\mathsf{T}} A x - b^{\mathsf{T}} x$ with A symmetric positive definite; α_k the exact line-search step; $r_k = g_k = \nabla f(x_k)$.

Problem. Prove that $\beta_k^{FR} = \beta_k^{PR} = \dfrac{r_k^{\mathsf{T}} r_k}{r_{k-1}^{\mathsf{T}} r_{k-1}}$, so that both methods generate the same iterates as linear CG.

P21 Practice *added*

Definitions.

- *Nonlinear CG:* $p_0 = -g_0$, $p_k = -g_k + \beta_k p_{k-1}$ with $g_k = \nabla f(x_k)$ and either $\beta_k^{FR} = \dfrac{g_k^{\mathsf{T}} g_k}{g_{k-1}^{\mathsf{T}} g_{k-1}}$ (Fletcher–Reeves) or $\beta_k^{PR} = \dfrac{g_k^{\mathsf{T}}(g_k - g_{k-1})}{g_{k-1}^{\mathsf{T}} g_{k-1}}$ (Polak–Ribière).

Setting. $f(x) = \frac{1}{2} x^{\mathsf{T}} A x - b^{\mathsf{T}} x$ with $A = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}$, $b = (1, 2)^{\mathsf{T}}$, $x_0 = 0$, exact line search.

Problem.

a) Compute g_0, x_1, g_1, β_1^{FR} and β_1^{PR}.

b) Check that both equal the linear CG value $\beta_1 = \frac{1}{16}$ (V19).

c) Which term of β^{PR} vanishes, and why?

X35 Proposed exercise *ch6, p. 12 (exercise 5)*

Definitions.

- *Nonlinear CG:* $p_0 = -g_0$, $p_k = -g_k + \beta_k p_{k-1}$ with $g_k = \nabla f(x_k)$ and either $\beta_k^{FR} = \dfrac{g_k^{\mathsf T} g_k}{g_{k-1}^{\mathsf T} g_{k-1}}$ (Fletcher–Reeves) or $\beta_k^{PR} = \dfrac{g_k^{\mathsf T}(g_k - g_{k-1})}{g_{k-1}^{\mathsf T} g_{k-1}}$ (Polak–Ribière).

Problem. Explain the theoretical reason why nonlinear CG should be restarted with $p_k = -g_k$ every n iterations.

Hint. On a strictly convex quadratic with exact line search, FR and PR coincide with linear CG (X34), whose directions are mutually conjugate only when the sequence starts with $p_0 = -g_0$; near a minimizer with $\nabla^2 f \succ 0$, f is close to such a quadratic.

X36 Proposed exercise *ch6, p. 13 (exercise 10)*

Definitions.

- *Nonlinear CG:* $p_0 = -g_0$, $p_k = -g_k + \beta_k p_{k-1}$ with $g_k = \nabla f(x_k)$ and either $\beta_k^{FR} = \dfrac{g_k^{\mathsf T} g_k}{g_{k-1}^{\mathsf T} g_{k-1}}$ (Fletcher–Reeves) or $\beta_k^{PR} = \dfrac{g_k^{\mathsf T}(g_k - g_{k-1})}{g_{k-1}^{\mathsf T} g_{k-1}}$ (Polak–Ribière).

Problem.

a) Build an example where $\beta_k^{PR} < 0$ and interpret the resulting direction p_k geometrically (compare with E23).

b) Explain why the variant $\beta_k^{PR+} = \max\{\beta_k^{PR}, 0\}$ is used in practice.

X37 Proposed exercise *ch6, p. 13 (exercise 8)*

Definitions.

- *Nonlinear CG:* $p_0 = -g_0$, $p_k = -g_k + \beta_k p_{k-1}$ with $g_k = \nabla f(x_k)$ and either $\beta_k^{FR} = \dfrac{g_k^{\mathsf T} g_k}{g_{k-1}^{\mathsf T} g_{k-1}}$ (Fletcher–Reeves) or $\beta_k^{PR} = \dfrac{g_k^{\mathsf T}(g_k - g_{k-1})}{g_{k-1}^{\mathsf T} g_{k-1}}$ (Polak–Ribière).

Problem. A problem has $n = 50\,000$ variables and is solved with Fletcher–Reeves. Write the one-dimensional line-search subproblem that determines α_0, and give the first direction p_0 if $g_0 = (2, -1, 3, \dots)^{\mathsf T}$.

Note. The first direction does not depend on the choice of β: it is $p_0 = -g_0$.

X38 Proposed exercise *ch6, p. 14 (exercise 12)*

Definitions.

- *L-BFGS* with memory m applies the inverse BFGS update $H_{k+1}^{\mathrm{BFGS}} = (I - \rho_k s_k y_k^{\mathsf T}) H_k (I - \rho_k y_k s_k^{\mathsf T}) + \rho_k s_k s_k^{\mathsf T}$, $\rho_k = 1/(y_k^{\mathsf T} s_k)$ m times to an initial matrix $H_k^{(0)} = \gamma_k I$, $\gamma_k = \dfrac{s_{k-1}^{\mathsf T} y_{k-1}}{y_{k-1}^{\mathsf T} y_{k-1}}$, using only the last m pairs (s_i, y_i), $s_i = x_{i+1} - x_i$, $y_i = \nabla f(x_{i+1}) - \nabla f(x_i)$; it never forms H_k.

Problem. Show why the scaling $\gamma_k = \dfrac{s_{k-1}^{\mathsf T} y_{k-1}}{y_{k-1}^{\mathsf T} y_{k-1}}$ makes the size of the step $-H_k g_k$ close to that of Newton's step.

Hint. For quadratic f, $y_{k-1} = A s_{k-1}$: relate γ_k to A^{-1} along the recent direction, and show $\gamma_k \in [1/\lambda_{\max}(A), 1/\lambda_{\min}(A)]$.

A13 Pseudocode *L-BFGS: two-loop recursion with weak Wolfe search (N&W, Alg. 7.4–7.5)*

LBFGS$(f,\ \nabla f,\ x_0;\ m, \varepsilon, k_{\max})$

1: $x \leftarrow x_0$; $g \leftarrow \nabla f(x)$; pair list $\mathcal{S} \leftarrow \emptyset$ (at most m pairs (s_i, y_i))
2: **for** $k = 0, 1, \ldots, k_{\max}$ **do**
3: **if** $\|g\| < \varepsilon$ **then return** x
4: $q \leftarrow g$; **for** $(s_i, y_i) \in \mathcal{S}$ from newest to oldest **do** $\rho_i \leftarrow 1/(y_i^{\mathsf{T}} s_i)$; $a_i \leftarrow \rho_i s_i^{\mathsf{T}} q$; $q \leftarrow q - a_i y_i$
5: **if** $\mathcal{S} \neq \emptyset$ **then** $\gamma \leftarrow s^{\mathsf{T}} y / (y^{\mathsf{T}} y)$ for the newest pair **else** $\gamma \leftarrow 1$; $r \leftarrow \gamma q$
6: **for** $(s_i, y_i) \in \mathcal{S}$ from oldest to newest **do** $b \leftarrow \rho_i y_i^{\mathsf{T}} r$; $r \leftarrow r + s_i(a_i - b)$
7: $p \leftarrow -r$ $(r = H_k g)$
8: $\alpha \leftarrow$ StrongWolfe$(\phi_k, \phi_k';\ \alpha_1, 10^{-4}, 0.9)$, $\alpha_1 = \min\{1, 1/\|g\|\}$ if $k = 0$, else $\alpha_1 = 1$; **if** failure **then return** x
9: $s \leftarrow \alpha p$; $x_{\text{new}} \leftarrow x + s$; $g_{\text{new}} \leftarrow \nabla f(x_{\text{new}})$; $y \leftarrow g_{\text{new}} - g$
10: **if** $s^{\mathsf{T}} y > 10^{-10}\, y^{\mathsf{T}} y$ **then** append (s, y) to $\mathcal{S}$, and drop the oldest pair if $|\mathcal{S}| > m$
11: $x \leftarrow x_{\text{new}}$; $g \leftarrow g_{\text{new}}$

StrongWolfe is A2 and $\phi_k(\alpha) = f(x + \alpha p)$. Storage: $2m$ vectors of length n plus a few work vectors; no $n \times n$ matrix is formed. Defaults: $m = 5$ to 20 (SciPy's L-BFGS-B uses $m = 10$). With a Wolfe search $s^{\mathsf{T}} y > 0$ always holds (T19), so line 10 only discards a pair when $s^{\mathsf{T}} y$ is tiny compared with $y^{\mathsf{T}} y$, which would make γ and ρ unreliable in floating point. Since ρ_i does not change, codes store it with the pair instead of recomputing it (line 4). This is the L-BFGS of Nocedal (Fortran `LBFGS`, liblbfgs, Julia `Optim.LBFGS`); L-BFGS-B adds bounds with a projected search, as in A6.

P22 Practice *added*

Definitions.

- *L-BFGS* with memory m applies the inverse BFGS update $H_{k+1}^{\text{BFGS}} = (I - \rho_k s_k y_k^{\mathsf{T}}) H_k (I - \rho_k y_k s_k^{\mathsf{T}}) + \rho_k s_k s_k^{\mathsf{T}}$, $\rho_k = 1/(y_k^{\mathsf{T}} s_k)$ m times to an initial matrix $H_k^{(0)} = \gamma_k I$, $\gamma_k = \dfrac{s_{k-1}^{\mathsf{T}} y_{k-1}}{y_{k-1}^{\mathsf{T}} y_{k-1}}$, using only the last m pairs (s_i, y_i), $s_i = x_{i+1} - x_i$, $y_i = \nabla f(x_{i+1}) - \nabla f(x_i)$; it never forms H_k.
- *Two-loop recursion* for $r = H_k g_k$: $q \leftarrow g_k$; for $i = k-1, \ldots, k-m$: $\rho_i = 1/(y_i^{\mathsf{T}} s_i)$, $a_i \leftarrow \rho_i s_i^{\mathsf{T}} q$, $q \leftarrow q - a_i y_i$; then $r \leftarrow \gamma_k q$; for $i = k-m, \ldots, k-1$: $b \leftarrow \rho_i y_i^{\mathsf{T}} r$, $r \leftarrow r + s_i(a_i - b)$.

Problem. Two-loop recursion by hand. Take $m = 1$ and the single pair $s_0 = (\frac{5}{4}, \frac{5}{8})^{\mathsf{T}}$, $y_0 = (\frac{25}{16}, \frac{15}{8})^{\mathsf{T}}$, with current gradient $g_1 = (-\frac{7}{16}, \frac{7}{8})^{\mathsf{T}}$. (These come from one exact-step iteration on $f(x) = \frac{5}{8}x_1^2 + \frac{3}{2}x_2^2 - 2x_1 - x_2$ from $x_0 = 0$ with $H_0 = I$.)

a) Run it (A13) and obtain $r = H_1 g_1$.

b) Form $H_1 = (I - \rho_0 s_0 y_0^{\mathsf{T}})\gamma_1 I (I - \rho_0 y_0 s_0^{\mathsf{T}}) + \rho_0 s_0 s_0^{\mathsf{T}}$ explicitly and check $H_1 g_1 = r$ and $H_1 y_0 = s_0$.

c) Compare $p_1 = -H_1 g_1$ with the full BFGS direction $(\frac{105}{128}, -\frac{175}{256})^{\mathsf{T}}$ obtained with $H_0 = I$. Explain why the first loop gives $a_0 = 0$.

V22 VALIDATION TABLE *A13, two loops with $m = 1$; solves P22*

$s_0 = (\frac{5}{4}, \frac{5}{8})$, $y_0 = (\frac{25}{16}, \frac{15}{8})$, $g_1 = (-\frac{7}{16}, \frac{7}{8})$.

quantity	value
$\rho_0 = 1/(y_0^\mathsf{T} s_0)$	$\frac{8}{25}$
first loop: $a_0 = \rho_0 s_0^\mathsf{T} g_1$, $q = g_1 - a_0 y_0$	$a_0 = 0$, $q = \left(-\frac{7}{16},\ \frac{7}{8}\right)$
$\gamma_1 = s_0^\mathsf{T} y_0/(y_0^\mathsf{T} y_0)$, $r = \gamma_1 q$	$\gamma_1 = \frac{32}{61}$, $r = \left(-\frac{14}{61},\ \frac{28}{61}\right)$
second loop: $b = \rho_0 y_0^\mathsf{T} r$, $r \leftarrow r + s_0(a_0 - b)$	$b = \frac{49}{305}$, $r = \left(-\frac{105}{244},\ \frac{175}{488}\right)$
explicit H_1	$\begin{pmatrix} \frac{53}{61} & -\frac{7}{122} \\ -\frac{7}{122} & \frac{93}{244} \end{pmatrix}$, $H_1 y_0 = \left(\frac{5}{4},\ \frac{5}{8}\right) = s_0$
$p_1 = -H_1 g_1$	$\left(\frac{105}{244},\ -\frac{175}{488}\right) = \frac{32}{61}\left(\frac{105}{128},\ -\frac{175}{256}\right)$

What it shows. The two loops reproduce the BFGS update of $\gamma_1 I$ using only inner products and vector updates. Because the previous step was exact, $s_0^\mathsf{T} g_1 = 0$ and the first loop does nothing; the result is the full BFGS direction scaled by γ_1, which an exact line search then compensates.

V23 VALIDATION TABLE *A13 with $m = 2$; compare with V9*

$f(x) = x_1^4 + x_1 x_2 + (1 + x_2)^2$, $\nabla f(x) = (4x_1^3 + x_2,\ x_1 + 2(1 + x_2))^\mathsf{T}$, $x^* \approx (0.695884, -1.347942)^\mathsf{T}$; $x_0 = (1, 1)^\mathsf{T}$, $m = 2$, StrongWolfe with $c_1 = 10^{-4}$, $c_2 = 0.9$; $\varepsilon = 10^{-6}$; $\gamma_0 = 1$. "eval." is the number of steps tried in A2 and $e_k = \|x_k - x^*\|$.

k	x_k	$f(x_k)$	$\|g_k\|$	γ_k	α_k	eval.	e_k
0	(1, 1)	6	7.0711	1	0.141421	1	2.3676
1	(0.292893, 0.292893)	1.764719	2.9054	0.184962	1	1	1.6896
2	(0.164425, −0.518820)	0.146958	1.2332	0.397230	1	1	0.98483
3	(0.293358, −1.069984)	−0.301585	0.98106	0.408205	1	1	0.48917
4	(0.805038, −1.531180)	−0.530490	0.61243	0.388844	1	1	0.21329
5	(0.601529, −1.305369)	−0.561041	0.43484	0.247065	1	1	0.10352
6	(0.678855, −1.344882)	−0.581659	9.4129×10^{-2}	0.227167	1	1	1.7302×10^{-2}
7	(0.698753, −1.349916)	−0.582423	1.4803×10^{-2}	0.178105	1	1	3.4816×10^{-3}
8	(0.695869, −1.348594)	−0.582445	1.5128×10^{-3}	0.184621	1	1	6.5211×10^{-4}
9	(0.695891, −1.348345)	−0.582445	8.7866×10^{-4}	0.335610	1	1	4.0243×10^{-4}
10	(0.695885, −1.347942)	−0.582445	1.7900×10^{-6}	0.412529	0.511129	2	4.3134×10^{-7}
11	(0.695884, −1.347942)	−0.582445	1.03×10^{-7}				4.34×10^{-8}

Iterations $k = 0, 1$ coincide with V9: the first step is the same, and with one pair L-BFGS computes H_1 as the BFGS update of $\gamma_1 I$, which is exactly what A5 does after rescaling H_0 (compare V22). They separate at $k = 2$, because L-BFGS restarts from $\gamma_2 I$ with $\gamma_2 = 0.397230$, while BFGS keeps the accumulated H_2. With $m = 1$ the run also needs 11 iterations.

What it shows. L-BFGS is not BFGS even with $m = n$, but on this problem it needs the same number of iterations (11) as BFGS in V9 while storing only $2m$ vectors. The unit step is accepted at almost every iteration, which is what the scaling γ_k is for (X38).

X39 PROPOSED EXERCISE *ch6, p. 12 (exercise 6)*

Definitions.

- *L-BFGS* with memory m applies the inverse BFGS update $H_{k+1}^{\mathrm{BFGS}} = (I - \rho_k s_k y_k^{\mathsf{T}})H_k(I - \rho_k y_k s_k^{\mathsf{T}}) + \rho_k s_k s_k^{\mathsf{T}}$, $\rho_k = 1/(y_k^{\mathsf{T}} s_k)$ m times to an initial matrix $H_k^{(0)} = \gamma_k I$, $\gamma_k = \dfrac{s_{k-1}^{\mathsf{T}} y_{k-1}}{y_{k-1}^{\mathsf{T}} y_{k-1}}$, using only the last m pairs (s_i, y_i), $s_i = x_{i+1} - x_i$, $y_i = \nabla f(x_{i+1}) - \nabla f(x_i)$; it never forms H_k.
- *Two-loop recursion* for $r = H_k g_k$: $q \leftarrow g_k$; for $i = k-1, \dots, k-m$: $\rho_i = 1/(y_i^{\mathsf{T}} s_i)$, $a_i \leftarrow \rho_i s_i^{\mathsf{T}} q$, $q \leftarrow q - a_i y_i$; then $r \leftarrow \gamma_k q$; for $i = k-m, \dots, k-1$: $b \leftarrow \rho_i y_i^{\mathsf{T}} r$, $r \leftarrow r + s_i(a_i - b)$.

Problem. Write detailed pseudocode for the complete L-BFGS method with a line search satisfying the Wolfe conditions $f(x_k + \alpha_k p_k) \le f(x_k) + c_1 \alpha_k \nabla f(x_k)^{\mathsf{T}} p_k$ and $\nabla f(x_k + \alpha_k p_k)^{\mathsf{T}} p_k \ge c_2 \nabla f(x_k)^{\mathsf{T}} p_k$, with $0 < c_1 < c_2 < 1$, including the storage and update of the pairs. (A13 is one answer.)

E24 NUMERICAL EXAMPLE *ch6, p. 11 (solved exercise 4)*

Definitions.

- *L-BFGS* with memory m applies the inverse BFGS update $H_{k+1}^{\mathrm{BFGS}} = (I - \rho_k s_k y_k^{\mathsf{T}})H_k(I - \rho_k y_k s_k^{\mathsf{T}}) + \rho_k s_k s_k^{\mathsf{T}}$, $\rho_k = 1/(y_k^{\mathsf{T}} s_k)$ m times to an initial matrix $H_k^{(0)} = \gamma_k I$, $\gamma_k = \dfrac{s_{k-1}^{\mathsf{T}} y_{k-1}}{y_{k-1}^{\mathsf{T}} y_{k-1}}$, using only the last m pairs (s_i, y_i), $s_i = x_{i+1} - x_i$, $y_i = \nabla f(x_{i+1}) - \nabla f(x_i)$; it never forms H_k.
- *Two-loop recursion* for $r = H_k g_k$: $q \leftarrow g_k$; for $i = k-1, \dots, k-m$: $\rho_i = 1/(y_i^{\mathsf{T}} s_i)$, $a_i \leftarrow \rho_i s_i^{\mathsf{T}} q$, $q \leftarrow q - a_i y_i$; then $r \leftarrow \gamma_k q$; for $i = k-m, \dots, k-1$: $b \leftarrow \rho_i y_i^{\mathsf{T}} r$, $r \leftarrow r + s_i(a_i - b)$.

Problem. Count the floating-point operations of the two-loop recursion with m pairs $(s_i, y_i) \in \mathbb{R}^n \times \mathbb{R}^n$, and show that the cost is linear in n.

Note. The notes announce "at most $4mn + O(m)$" but their own count gives $8mn + O(n)$. Both are linear in n; the difference is the convention: counting each inner product and each vector update as $2n$ flops gives $8mn$, while N&W's $4mn$ counts multiplications only.

E25 NUMERICAL EXAMPLE *ch6, p. 11 (solved exercise 5)*

Definitions.

- *L-BFGS* with memory m applies the inverse BFGS update $H_{k+1}^{\mathrm{BFGS}} = (I - \rho_k s_k y_k^{\mathsf{T}})H_k(I - \rho_k y_k s_k^{\mathsf{T}}) + \rho_k s_k s_k^{\mathsf{T}}$, $\rho_k = 1/(y_k^{\mathsf{T}} s_k)$ m times to an initial matrix $H_k^{(0)} = \gamma_k I$, $\gamma_k = \dfrac{s_{k-1}^{\mathsf{T}} y_{k-1}}{y_{k-1}^{\mathsf{T}} y_{k-1}}$, using only the last m pairs (s_i, y_i), $s_i = x_{i+1} - x_i$, $y_i = \nabla f(x_{i+1}) - \nabla f(x_i)$; it never forms H_k.

Problem. A problem has $n = 10^5$ variables and L-BFGS is used with $m = 5$. Compute the memory needed to store the pairs (s_i, y_i) in 64-bit floating point, and compare with storing a dense $n \times n$ matrix H_k.

X40 PROPOSED EXERCISE *ch6, p. 14 (exercise 11)*

Definitions.

- *L-BFGS* with memory m applies the inverse BFGS update $H_{k+1}^{\mathrm{BFGS}} = (I - \rho_k s_k y_k^{\mathsf{T}})H_k(I - \rho_k y_k s_k^{\mathsf{T}}) + \rho_k s_k s_k^{\mathsf{T}}$, $\rho_k = 1/(y_k^{\mathsf{T}} s_k)$ m times to an initial matrix $H_k^{(0)} = \gamma_k I$, $\gamma_k = \dfrac{s_{k-1}^{\mathsf{T}} y_{k-1}}{y_{k-1}^{\mathsf{T}} y_{k-1}}$, using only the last m pairs (s_i, y_i), $s_i = x_{i+1} - x_i$, $y_i = \nabla f(x_{i+1}) - \nabla f(x_i)$; it never forms H_k.

Problem. A problem with $n = 10^6$ variables is solved with L-BFGS on a machine with 4 GB of RAM. What is the largest memory parameter m that can be used?

Note. State whether 4 GB means $4 \cdot 10^9$ or $4 \cdot 2^{30}$ bytes, and whether the vectors x, g, p and the work vectors of the two loops are counted; the answer changes slightly.

Q24 Exam problem *added, 4 pts*

Descent of Fletcher–Reeves under strong Wolfe. Nonlinear CG: $p_0 = -g_0$, $p_{k+1} = -g_{k+1} + \beta_{k+1} p_k$ with $\beta_{k+1}^{FR} = \|g_{k+1}\|^2 / \|g_k\|^2$, and α_k satisfying the strong Wolfe conditions with $0 < c_1 < c_2 < \tfrac{1}{2}$.

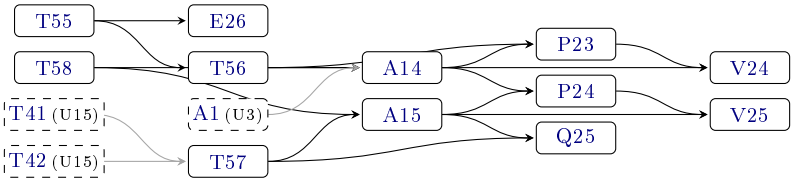
a) Prove by induction that $-\sum_{j=0}^{k} c_2^j \le \dfrac{g_k^{\mathsf T} p_k}{\|g_k\|^2} \le -2 + \sum_{j=0}^{k} c_2^j$, and deduce that every p_k is a descent direction.

b) Show by an example (or explain) why the conclusion can fail for $c_2 \ge \tfrac{1}{2}$, and why the Polak–Ribière choice $\beta^{PR+} = \max\{0,\ g_{k+1}^{\mathsf T}(g_{k+1} - g_k)/\|g_k\|^2\}$ is preferred in practice.

c) Compare the memory and the work per iteration of nonlinear CG and L-BFGS with m pairs for $n = 10^6$, and explain when each one is preferable.

U21. Nonlinear least squares: Gauss–Newton and Levenberg–Marquardt

↑Index



Syllabus week 6 (N&W, Ch. 10; Bertsekas, §1.5). Not yet covered in ch1–ch6; all items are added.

T55 Proof *added (N&W, §10.1–10.3)*

Setting. $r : \mathbb{R}^n \to \mathbb{R}^m$ of class C^2 ($m \ge n$), residuals r_j, Jacobian $J(x) \in \mathbb{R}^{m \times n}$ with rows $\nabla r_j(x)^{\mathsf T}$, and $f(x) = \tfrac{1}{2}\|r(x)\|^2$.

Problem.

a) Prove that $\nabla f(x) = J(x)^{\mathsf T} r(x)$ and

$$\nabla^2 f(x) = J(x)^{\mathsf T} J(x) + S(x), \qquad S(x) = \sum_{j=1}^{m} r_j(x) \nabla^2 r_j(x).$$

b) Let $J = J(x)$ have full column rank. Prove that the Gauss–Newton step $p^{GN} = -(J^{\mathsf T}J)^{-1} J^{\mathsf T} r$ is the minimizer of the linear least-squares problem $\min_p \tfrac{1}{2}\|Jp + r\|^2$, and that $\nabla f(x)^{\mathsf T} p^{GN} = -\|J p^{GN}\|^2 < 0$ whenever $\nabla f(x) \neq 0$.

c) Interpret Gauss–Newton as Newton's method with $\nabla^2 f$ replaced by $J^{\mathsf T} J$. For which problems is this replacement exact, and why does it need no second derivatives?

E26 Numerical example *added*

Setting. Model $\varphi(t; x) = x_1 e^{x_2 t}$, data $(t_i, y_i) = (0, 2), (1, 1), (2, \tfrac{1}{2})$, residuals $r_i(x) = x_1 e^{x_2 t_i} - y_i$ and $f = \tfrac{1}{2}\|r\|^2$. The exact fit is $x^* = (2, -\ln 2)$ (zero residual).

Problem.

a) At $x_0 = (1, 0)^{\mathsf T}$ compute $r(x_0)$, $J(x_0)$, $J^{\mathsf T}J$ and $\nabla f(x_0) = J^{\mathsf T} r$.

b) Compute the Gauss–Newton step p_0 and $x_1 = x_0 + p_0$, and compare $f(x_1)$ with $f(x_0)$.

T56 Proof *added (N&W, §10.3, eq. 10.30)*

Setting. r of class C^2 near a stationary point x^* of $f = \tfrac{1}{2}\|r\|^2$, with $J^* = J(x^*)$ of full column rank and $S^* = S(x^*) = \sum_j r_j(x^*)\nabla^2 r_j(x^*)$. Gauss–Newton: $x_{k+1} = x_k - (J_k^{\mathsf{T}} J_k)^{-1} J_k^{\mathsf{T}} r_k$.

Problem.

a) Prove that $x_{k+1} - x^* = -(J^{*\mathsf{T}} J^*)^{-1} S^*(x_k - x^*) + O(\|x_k - x^*\|^2)$.

b) Deduce: quadratic convergence for zero-residual problems ($r(x^*) = 0$); linear convergence with rate $\approx \|(J^{*\mathsf{T}} J^*)^{-1} S^*\|$ when this number is < 1; and possible failure when it is > 1.

c) For $r(x) = (x + 1,\ \lambda x^2 + x - 1)$, $x \in \mathbb{R}$, check that $x^* = 0$ is stationary and that the rate in b) is exactly λ. For which λ is $x^* = 0$ a local minimizer?

T57 Proof *added (N&W, §10.3)*

Setting. $J \in \mathbb{R}^{m\times n}$, $r \in \mathbb{R}^m$, $\mu > 0$, and the Levenberg–Marquardt step $p(\mu) = -(J^{\mathsf{T}} J + \mu I)^{-1} J^{\mathsf{T}} r$.

Tool. (T41) p^* solves $\min\{g^{\mathsf{T}} p + \tfrac{1}{2} p^{\mathsf{T}} B p :\ \|p\| \le \Delta\}$ if and only if $\|p^*\| \le \Delta$ and there is $\lambda \ge 0$ with $(B + \lambda I)p^* = -g$, $\lambda(\Delta - \|p^*\|) = 0$, $B + \lambda I \succeq 0$.

Problem.

a) Prove that $p(\mu)$ solves $\min \tfrac{1}{2}\|Jp + r\|^2$ subject to $\|p\| \le \Delta$, with $\Delta = \|p(\mu)\|$: Levenberg–Marquardt is a trust-region method whose radius is set implicitly by μ.

b) Prove that $\mu \mapsto \|p(\mu)\|$ is decreasing, that $p(\mu) \to p^{GN}$ as $\mu \to 0^+$ when J has full column rank, and that $\mu\, p(\mu) \to -J^{\mathsf{T}} r = -\nabla f$ as $\mu \to \infty$ (a short steepest-descent step).

c) Prove that $p(\mu)$ is a descent direction for $f = \tfrac{1}{2}\|r\|^2$ for every $\mu > 0$, even when J is rank deficient.

T58 Proof *added (N&W, §10.2)*

Setting. $J \in \mathbb{R}^{m\times n}$ of full column rank, $r \in \mathbb{R}^m$, thin QR factorization $J = QR$ ($Q^{\mathsf{T}} Q = I$, R upper triangular and invertible).

Problem.

a) Prove that $\kappa_2(J^{\mathsf{T}} J) = \kappa_2(J)^2$.

b) Prove that the solution of $\min\|Jp + r\|$ is the solution of $Rp = -Q^{\mathsf{T}} r$.

c) Prove that the LM step $p(\mu)$ is the least-squares solution of $\begin{pmatrix} J \\ \sqrt{\mu}\, I \end{pmatrix} p \approx -\begin{pmatrix} r \\ 0 \end{pmatrix}$.

d) For $J = \begin{pmatrix} 1 & 1 \\ \delta & 0 \\ 0 & \delta \end{pmatrix}$ with $\delta = 10^{-8}$, show that $J^{\mathsf{T}} J$ rounded to double precision ($u \approx 1.1\times 10^{-16}$) is singular, while J has full rank and its QR factorization is well conditioned enough to solve the problem.

A14 Pseudocode *damped Gauss–Newton (N&W, §10.3)*

GaussNewton$(r,\ J,\ x_0;\ \varepsilon, k_{\max})$ $\quad f(x)=\tfrac{1}{2}\|r(x)\|^2$

1: **for** $k=0,1,\dots,k_{\max}$ **do**
2: $\quad r_k \leftarrow r(x_k);\quad J_k \leftarrow J(x_k);\quad g_k \leftarrow J_k^{\mathsf T} r_k;\quad$ **if** $\|g_k\| \le \varepsilon$ **then return** x_k
3: $\quad (Q,R) \leftarrow$ thin QR factorization of J_k $\qquad (J_k = QR,\ R \in \mathbb{R}^{n\times n}$ upper triangular)
4: $\quad$ **if** $\min_i |r_{ii}| \le 10^{-12}\max_i |r_{ii}|$ **then return** x_k with "rank deficient: use LM"
5: $\quad$ solve $Rp_k = -Q^{\mathsf T} r_k$ $\qquad$ (least-squares solution of $J_k p \approx -r_k$; $J_k^{\mathsf T}J_k$ is never formed)
6: $\quad \alpha_k \leftarrow$ Backtracking$(\phi_k;\ 1, 10^{-4}, 0.1, 0.5, 30)$(A1, $\phi_k(\alpha)=\tfrac{1}{2}\|r(x_k+\alpha p_k)\|^2$, $\phi_k'(0)=g_k^{\mathsf T}p_k$)
7: $\quad x_{k+1} \leftarrow x_k + \alpha_k p_k$

The QR route solves the linear least-squares subproblem with the conditioning of J_k instead of that of $J_k^{\mathsf T}J_k$ (T58). The line search makes the method globally convergent when the J_k stay uniformly of full rank (Zoutendijk, since p_k is a descent direction with $\cos\theta_k$ bounded away from 0). This is the default algorithm of R's `nls` (Gauss–Newton with step halving). When J_k is rank deficient or the residuals are large, production codes switch to Levenberg–Marquardt (A15).

P23 Practice *added*

Gauss–Newton with zero and nonzero residuals.

Problem.

a) Continue E26 with A14 from $x_0=(1,0)^{\mathsf T}$ until $\|\nabla f\| < 10^{-8}$. Record x_k, $f(x_k)$, α_k and $e_k=\|x_k-x^*\|$, and check that e_{k+1}/e_k^2 stays bounded (table V24 a).

b) For $r(x)=(x+1,\ \lambda x^2+x-1)$ from $x_0=1$, run pure Gauss–Newton for $\lambda=0.1$, 0.5 and 2. Compare the observed ratios $|x_{k+1}-\bar x|/|x_k-\bar x|$ with the rate of T56 (table V24 b).

c) For $\lambda=2$, find all stationary points of f and explain why Gauss–Newton converges to $\bar x=\tfrac14$ and not to the zero-residual solution $x=-1$.

V24 Validation table *A14; solves P23*

a) Exponential fit (E26), zero residual. $\varphi(t;x)=x_1e^{x_2t}$, data $(0,2)$, $(1,1)$, $(2,\tfrac12)$, $x^*=(2,-0.693147)$. A1 accepted $\alpha_k=1$ at every iteration.

k	x_k	$f(x_k)$	$\|\nabla f(x_k)\|$	e_k	e_{k+1}/e_k^2
0	(1, 0)	0.625	1.1180	1.21674	0.068
1	(1.916667, −0.75)	1.0566×10^{-2}	0.20629	0.10088	0.502
2	(1.999825, −0.688045)	2.5580×10^{-5}	1.0818×10^{-2}	5.1057×10^{-3}	0.757
3	(1.999998, −0.693128)	3.6456×10^{-10}	4.0164×10^{-5}	1.9721×10^{-5}	0.730
4	(2.000000, −0.693147)	7.5×10^{-20}	5.8×10^{-10}	2.8×10^{-10}	

First step: $r(x_0)=(-1,0,\tfrac12)$, $J(x_0)=\left(\begin{smallmatrix}1&0\\1&1\\1&2\end{smallmatrix}\right)$, $J^{\mathsf T}J=\left(\begin{smallmatrix}3&3\\3&5\end{smallmatrix}\right)$, $J^{\mathsf T}r=(-\tfrac12,1)$, $p_0=(\tfrac{11}{12},-\tfrac34)$.

b) $r(x)=(x+1,\ \lambda x^2+x-1)$, *pure Gauss–Newton from* $x_0=1$. Limit $\bar x$; rate of T56: $|S/J^{\mathsf T}J|$ at $\bar x$.

	$\lambda = 0.1$ ($\bar{x} = 0$, rate 0.1)		$\lambda = 0.5$ ($\bar{x} = 0$, rate 0.5)		$\lambda = 2$ ($\bar{x} = \frac{1}{4}$, rate 0.5)	
k	x_k	ratio	x_k	ratio	x_k	ratio
0	1	0.131	1	0.400	1	0.385
1	0.131148	0.104	0.4	0.432	0.538462	0.395
2	0.013635	0.100	0.172973	0.464	0.363825	0.435
3	0.001369	0.100	0.080190	0.481	0.299539	0.466
4	0.000137	0.100	0.038611	0.491	0.273086	0.483
5	1.37×10^{-5}		0.018947	0.495	0.261147	0.491
6			0.009385		0.255477	

For $\lambda = 2$: $f'(x) = 0$ at $x = -1$ ($f = 0$, global minimum), $x = 0$ ($f''(0) = 2 - 2\lambda = -2$: local maximum) and $x = \frac{1}{4}$ ($f = 0.976563$, local minimum). Starting from $x_0 = 1$, both Gauss–Newton and LM (A15) descend into the basin of $\frac{1}{4}$.

What it shows. With zero residual the neglected term $S(x^*)$ vanishes and the convergence is quadratic; with nonzero residual it is linear with the rate predicted by $\|(J^\mathsf{T}J)^{-1}S\|$, and Gauss–Newton (like any descent method) finds a local, not the global, minimizer.

A15 Pseudocode *Levenberg–Marquardt (Madsen–Nielsen–Tingleff, Alg. 3.16)*

LevenbergMarquardt$(r,\ J,\ x_0;\ \tau, \varepsilon_1, \varepsilon_2, k_{\max})$ $\quad F(x) = \frac{1}{2}\|r(x)\|^2$

$A \leftarrow J(x_0)^\mathsf{T}J(x_0)$; $\quad g \leftarrow J(x_0)^\mathsf{T}r(x_0)$; $\quad \mu \leftarrow \tau \max_i a_{ii}$; $\quad \nu \leftarrow 2$; $\quad x \leftarrow x_0$
for $k = 0, 1, \ldots, k_{\max}$ **do**
  if $\|g\|_\infty \le \varepsilon_1$ **then return** x
  solve $(A + \mu I)h = -g$ (Cholesky; or QR of $\begin{pmatrix} J \\ \sqrt{\mu} I \end{pmatrix}$, T58 c)
  if $\|h\| \le \varepsilon_2(\|x\| + \varepsilon_2)$ **then return** x
  $\rho \leftarrow \dfrac{F(x) - F(x+h)}{\frac{1}{2}h^\mathsf{T}(\mu h - g)}$ (denominator = model decrease $L(0) - L(h) > 0$)
  if $\rho > 0$ **then**
    $x \leftarrow x + h$; $\quad A \leftarrow J(x)^\mathsf{T}J(x)$; $\quad g \leftarrow J(x)^\mathsf{T}r(x)$ (accept)
    $\mu \leftarrow \mu \cdot \max\{\frac{1}{3},\ 1 - (2\rho - 1)^3\}$; $\quad \nu \leftarrow 2$
  else
    $\mu \leftarrow \mu\nu$; $\quad \nu \leftarrow 2\nu$ (reject; increase damping faster each time)

Defaults: $\tau = 10^{-3}$ (use $\tau \approx 1$ if x_0 is far from the solution), $\varepsilon_1 = 10^{-8}$, $\varepsilon_2 = 10^{-10}$. The damping update is smooth in ρ: μ decreases by up to a factor 3 when the model is good ($\rho \approx 1$) and increases when ρ is small, which is the radius update of a trust region (T57) expressed through μ. This is the algorithm of `levmar` and of many curve-fitting libraries. MINPACK's `lmder` (SciPy `least_squares(method='lm')` and `curve_fit`) implements Moré's variant: an explicit radius Δ, μ found from it as in A8, and the scaling $\|Dp\| \le \Delta$ with $D = \mathrm{diag}(\|J_{:,i}\|)$, which makes the method invariant to the units of each parameter.

P24 Practice *added*

When Gauss–Newton needs help.

Setting. The exponential fit of E26 ($\varphi(t; x) = x_1 e^{x_2 t}$, data $(0, 2), (1, 1), (2, \frac{1}{2})$), now from the poor start $x_0 = (0.1,\ 1)^\mathsf{T}$.

Problem.

a) Compute $J(x_0)^\mathsf{T}J(x_0)$ and its condition number, and the pure Gauss–Newton steps p_0, p_1. What happens to x_2?

b) Run A14 (with line search) and record α_k; at which iteration does the line search act?

c) Run A15 with $\tau = 10^{-3}$, recording μ, h, ρ and the decision at each iteration (table V25).

d) Explain with T57 why LM never takes the huge step of pure Gauss–Newton.

V25 Validation table *A14, A15; solves P24*

$x_0 = (0.1, 1)$: $r(x_0) = (-1.9,\ -0.728172,\ 0.238906)$, $J^{\mathsf{T}}J = \left(\begin{smallmatrix} 62.987206 & 11.658536 \\ 11.658536 & 2.257817 \end{smallmatrix}\right)$ with $\kappa \approx 675$, $\nabla f(x_0) = (-2.114089,\ 0.155120)$, $f(x_0) = 2.098660$.

Pure Gauss–Newton. $p_0 = (1.046024, -5.469990)$, $x_1 = (1.146024, -4.469990)$, $f(x_1) = 0.976529$; $p_1 = (0.853913,\ 75.312623)$, $x_2 = (1.999937,\ 70.842633)$ with $f(x_2) \approx 2.3 \times 10^{123}$: the model $J^{\mathsf{T}}J$ is nearly singular along x_2 and the step explodes.

A14 (Gauss–Newton with A1). $k = 0$: $\alpha = 1$; $k = 1$: the same direction p_1 is cut to $\alpha = 0.05$ (3 trials), giving $x_2 = (1.188720, -0.704358)$; then $\alpha = 1$, and $\|\nabla f\| < 10^{-8}$ at $k = 5$.

A15 (Levenberg–Marquardt, $\tau = 10^{-3}$).

k	x_k	$F(x_k)$	μ	h	ρ	next μ
0	(0.1, 1)	2.098660	0.062987	(0.645307, −3.308533)	0.587	0.062649
1	(0.745307, −2.308533)	1.337130	0.062649	(1.253276, 0.971995)	1.204	0.020883
2	(1.998583, −1.336538)	0.178278	0.020883	(0.003950, 0.935209)	0.239	0.023851
3	(2.002533, −0.401329)	0.136958	0.023851	(−0.012567, −0.227757)	0.974	0.007950
4	(1.989966, −0.629087)	4.0439×10^{-3}	0.007950	(0.009508, −0.060940)	0.998	0.002650
5	(1.999475, −0.690027)	8.7274×10^{-6}	0.002650	(0.000521, −0.003108)	1.000	0.000883
6	(1.999995, −0.693135)	1.2402×10^{-10}	0.000883			

All steps are accepted; LM stops at $k = 8$ with $F \approx 7 \times 10^{-24}$.

What it shows. Near-singular $J^{\mathsf{T}}J$ gives a huge Gauss–Newton step; a line search can repair it after the fact, while LM bounds the step from the start ($\|h\|$ decreases with μ). As $\rho \to 1$, $\mu \to 0$ and LM becomes Gauss–Newton, keeping its fast local convergence.

Q25 Exam problem *added, 4 pts*

Fitting an exponential growth rate.

Setting. Model $y = e^{xt}$ with one parameter $x \in \mathbb{R}$; data $(t_1, y_1) = (1, 2)$, $(t_2, y_2) = (2, 4)$; residuals $r_i(x) = e^{xt_i} - y_i$ and $f(x) = \frac{1}{2}(r_1^2 + r_2^2)$.

Problem.

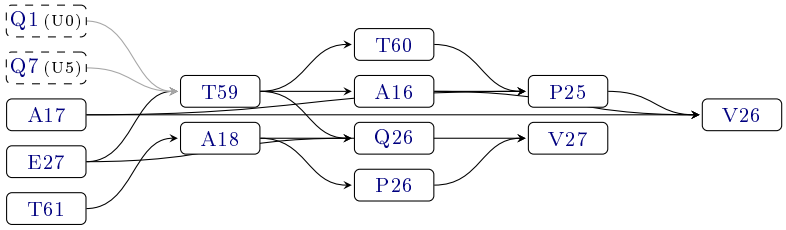
a) Write $J(x)$, $\nabla f(x) = J^{\mathsf{T}}r$ and $\nabla^2 f(x) = J^{\mathsf{T}}J + \sum_i r_i \nabla^2 r_i$. Show that $x^* = \ln 2$ is a zero-residual solution and that Gauss–Newton converges quadratically near it.

b) From $x_0 = 0$, compute the Gauss–Newton step and x_1. Compare $f(x_1)$ with $f(x_0)$ and comment on the overshoot.

c) Compute the Levenberg–Marquardt step $p(\mu) = -J^{\mathsf{T}}r/(J^{\mathsf{T}}J + \mu)$ from $x_0 = 0$ with $\mu = J^{\mathsf{T}}J$, and the ratio ρ of actual to predicted reduction. Explain, using the trust-region interpretation, how the update of μ in A15 reacts to ρ.

U22. Stochastic gradient methods

↑Index



Syllabus week 7 (Bertsekas, §1.7; Bottou–Curtis–Nocedal, SIAM Review 2018). Not yet covered in ch1–ch6; all items are added.

E27 Numerical example *added*

Setting. Finite sum $f(x) = \frac{1}{m}\sum_{i=1}^{m} f_i(x)$; at each step an index i is drawn uniformly from $\{1,\dots,m\}$ and $g = \nabla f_i(x)$ is used as a stochastic gradient.

Problem.

a) Show that $\mathbb{E}[\nabla f_i(x)] = \nabla f(x)$ (unbiasedness).

b) For $m = 4$, $f_i(x) = \frac{1}{2}(x - b_i)^2$ with $b = (-1, 0, 1, 4)$: find x^* and f^*, and compute the variance $\sigma^2 = \mathbb{E}[(\nabla f_i(x^*))^2]$ at the solution. Why is it not zero?

c) Show that the average of a mini-batch of b indices drawn independently (with replacement) has variance σ^2/b.

T59 Proof *added (Bottou–Curtis–Nocedal, Thm. 4.6)*

Setting. f of class C^1, μ-strongly convex with minimum value f^*, ∇f Lipschitz with constant L. SGD: $x_{k+1} = x_k - \alpha g_k$ with constant $\alpha \in (0, \frac{1}{L}]$, where $\mathbb{E}[g_k \mid x_k] = \nabla f(x_k)$ and $\mathbb{E}[\|g_k - \nabla f(x_k)\|^2 \mid x_k] \le \sigma^2$.

Tools. Descent lemma $f(y) \le f(x) + \nabla f(x)^{\mathsf{T}}(y - x) + \frac{L}{2}\|y - x\|^2$; and $\|\nabla f(x)\|^2 \ge 2\mu(f(x) - f^*)$ (Q7 b).

Problem.

a) Prove $\mathbb{E}[f(x_{k+1}) \mid x_k] \le f(x_k) - \frac{\alpha}{2}\|\nabla f(x_k)\|^2 + \frac{L\alpha^2\sigma^2}{2}$.

b) Deduce $\mathbb{E}[f(x_{k+1}) - f^*] \le (1 - \alpha\mu)\,\mathbb{E}[f(x_k) - f^*] + \frac{L\alpha^2\sigma^2}{2}$.

c) Conclude $\mathbb{E}[f(x_k) - f^*] \le (1-\alpha\mu)^k\big(f(x_0) - f^*\big) + \frac{\alpha L\sigma^2}{2\mu}$: linear convergence to a neighbourhood whose size is proportional to α.

T60 Proof *added (Bottou–Curtis–Nocedal, Thm. 4.7)*

Setting. As in T59, but with diminishing steps $\alpha_k = \dfrac{\beta}{\gamma + k}$, where $\beta > \frac{1}{\mu}$, $\gamma > 0$ and $\alpha_0 = \beta/\gamma \le \frac{1}{L}$.

Problem.

a) Let $\nu = \max\Big\{\dfrac{\beta^2 L\sigma^2}{2(\beta\mu - 1)},\ \gamma\big(f(x_0) - f^*\big)\Big\}$. Prove by induction, using b) of T59 with $\alpha = \alpha_k$, that $\mathbb{E}[f(x_k) - f^*] \le \dfrac{\nu}{\gamma + k}$ for all k.

b) Check that α_k satisfies the Robbins–Monro conditions $\sum_k \alpha_k = \infty$, $\sum_k \alpha_k^2 < \infty$, and explain the role of each one.

c) Compare with T59: why does no constant step give $\mathbb{E}[f(x_k) - f^*] \to 0$ when $\sigma > 0$?

A16 PSEUDOCODE *mini-batch SGD with momentum (as in PyTorch `optim.SGD`)*

SGD$(\{\nabla f_i\},\ x_0;\ \alpha_0, \text{schedule}, b, \beta, E)$ $\quad f = \frac{1}{m}\sum_i f_i$

$x \leftarrow x_0$; $\quad v \leftarrow 0$; $\quad \alpha \leftarrow \alpha_0$; $\quad t \leftarrow 0$
for epoch $e = 1, \dots, E$ **do**
  shuffle $\{1, \dots, m\}$ and split it into mini-batches of size b (random reshuffling)
  for each mini-batch $\mathcal{B}$ **do**
    $g \leftarrow \frac{1}{|\mathcal{B}|}\sum_{i\in\mathcal{B}} \nabla f_i(x)$
    if $t = 0$ **then** $v \leftarrow g$ **else** $v \leftarrow \beta v + g$ (heavy-ball momentum; $\beta = 0$: plain SGD)
    $x \leftarrow x - \alpha v$; $\quad t \leftarrow t + 1$
  $\alpha \leftarrow \text{schedule}(\alpha_0, e)$ (e.g. step decay $\alpha_0 \cdot 0.1^{\lfloor e/E_d \rfloor}$, or $\alpha_0/(1 + e/\tau)$)
return x (or the average of the iterates of the last epoch)

Typical values: $b = 32$ to 256, $\beta = 0.9$, α_0 chosen by a short trial run. The full gradient is never computed, so there is no test $\|\nabla f\| \le \varepsilon$: runs stop after a budget of epochs, or when f on a held-out sample stops improving. Random reshuffling (each sample once per epoch) is what libraries do; the theory of T59–T60 assumes independent sampling with replacement.

A17 PSEUDOCODE *SVRG: stochastic variance-reduced gradient (Johnson–Zhang, 2013)*

SVRG$(\{\nabla f_i\},\ x_0;\ \alpha, T, S)$ $\quad f = \frac{1}{m}\sum_i f_i$

$x \leftarrow x_0$
for $s = 0, 1, \dots, S-1$ **do**
  $\tilde{x} \leftarrow x$; $\quad \tilde{g} \leftarrow \nabla f(\tilde{x}) = \frac{1}{m}\sum_{i=1}^{m} \nabla f_i(\tilde{x})$ (one full pass: the snapshot)
  for $t = 1, \dots, T$ **do**
    pick i uniformly in $\{1, \dots, m\}$
    $x \leftarrow x - \alpha\big(\nabla f_i(x) - \nabla f_i(\tilde{x}) + \tilde{g}\big)$ (unbiased, and variance $\to 0$ as $x, \tilde{x} \to x^*$)
return x

Typical values: $T = 2m$ inner steps, constant $\alpha \approx \frac{1}{10 L_{\max}}$ with $L_{\max} = \max_i L_i$. Because the variance of the corrected gradient vanishes at the solution, a constant step gives linear convergence to x^* itself (compare T59). SAGA, a variant that stores the last gradient of each f_i instead of taking snapshots, is one of the solvers of scikit-learn's logistic regression.

P25 PRACTICE *added*

Noise floor and variance reduction.

Setting. $f(x) = \frac{1}{4}\sum_{i=1}^{4} \frac{1}{2}(x - b_i)^2$ with $b = (-1, 0, 1, 4)$ (E27), $x_0 = 5$, and the sampled indices $i_0, \dots, i_{11} = 3, 1, 4, 2,\ 4, 1, 2, 3,\ 1, 4, 3, 2$.

Problem.

a) Run SGD with constant $\alpha = \frac{1}{2}$ (A16 with $b = 1$, $\beta = 0$) for 12 steps (table V26).

b) Run SGD with $\alpha_k = \frac{1}{k+1}$ and prove that x_{k+1} is the average of $b_{i_0}, \dots, b_{i_k}$.

c) Run SVRG (A17) with $\alpha = \frac{1}{2}$ and a new snapshot every $T = 4$ steps. Prove that here the corrected gradient equals $\nabla f(x)$ exactly, and explain why.

d) For constant α, compute the exact stationary value of $\mathbb{E}[f(x_k) - f^*]$, namely $\frac{\alpha\sigma^2}{2(2-\alpha)}$, and compare it with the bound $\frac{\alpha L \sigma^2}{2\mu}$ of T59.

V26 Validation table *A16, A17; solves P25*

$x^* = 1$, $f^* = \frac{7}{4}$, $\sigma^2 = 3.5$, $L = \mu = 1$.

k	i_k	SGD $\alpha = \frac{1}{2}$	$f(x_k) - f^*$	SGD $\alpha_k = \frac{1}{k+1}$	SVRG $\alpha = \frac{1}{2}$
0	3	5	8	5	5
1	1	3	2	1	3
2	4	1	0	0	2
3	2	2.5	1.125	1.333333	1.5
4	4	1.25	0.03125	1	1.25
5	1	2.625	1.320313	1.6	1.125
6	2	0.8125	0.017578	1.166667	1.0625
7	3	0.40625	0.176270	1	1.03125
8	1	0.703125	0.044067	1	1.015625
9	4	−0.148438	0.659454	0.777778	1.007813
10	3	1.925781	0.428535	1.1	1.003906
11	2	1.462891	0.107134	1.090909	1.001953
12		0.731445	0.036061	1	1.000977

SVRG: since every f_i has Hessian 1, $\nabla f_i(x) - \nabla f_i(\tilde{x}) = x - \tilde{x}$, so the corrected gradient is $x - \tilde{x} + (\tilde{x} - 1) = x - 1 = \nabla f(x)$ and $x_k - 1 = 4 \cdot 2^{-k}$. Stationary SGD value with $\alpha = \frac{1}{2}$: $\frac{0.5 \cdot 3.5}{2 \cdot 1.5} = 0.583$, against the bound 0.875.

What it shows. A constant step stalls at a noise floor proportional to α; diminishing steps remove it at the price of slow $O(1/k)$ progress; variance reduction removes the noise and keeps a constant step.

T61 Proof *added (Kingma–Ba, 2015)*

Setting. Adam's moment estimates, coordinatewise: $m_k = \beta_1 m_{k-1} + (1-\beta_1) g_k$, $v_k = \beta_2 v_{k-1} + (1-\beta_2) g_k^2$, $m_0 = v_0 = 0$, $\hat{m}_k = m_k/(1-\beta_1^k)$, $\hat{v}_k = v_k/(1-\beta_2^k)$, step $x_{k+1} = x_k - \alpha\, \hat{m}_k/(\sqrt{\hat{v}_k} + \epsilon)$.
Problem.

a) Prove that $m_k = (1-\beta_1) \sum_{j=1}^{k} \beta_1^{k-j} g_j$, and that if $\mathbb{E}[g_j] = \bar{g}$ for all j, then $\mathbb{E}[\hat{m}_k] = \bar{g}$ (likewise $\mathbb{E}[\hat{v}_k] = \mathbb{E}[g^2]$). Explain why, without the correction, the first steps would be too small.

b) Show that the first step is $x_1 - x_0 = -\alpha\, g_1/(|g_1| + \epsilon) \approx -\alpha \operatorname{sign}(g_1)$ in every coordinate.

c) With $\epsilon = 0$, show that the iterates are unchanged if every gradient coordinate i is multiplied by a constant $c_i > 0$. Interpret this for $f(x) = \frac{1}{2} \sum_i d_i x_i^2$.

A18 Pseudocode *Adam and RMSprop (as in PyTorch `optim.Adam`, `optim.RMSprop`)*

Adam(stochastic gradient $g(\cdot)$, x_0; $\alpha, \beta_1, \beta_2, \epsilon, K$) all operations coordinatewise

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$m \leftarrow 0$; $v \leftarrow 0$; $x \leftarrow x_0$
for $k = 1, \dots, K$ **do**
  $g \leftarrow g(x)$ (mini-batch gradient, as in A16)
  $m \leftarrow \beta_1 m + (1-\beta_1) g$; $v \leftarrow \beta_2 v + (1-\beta_2) g \odot g$
  $\hat{m} \leftarrow m/(1-\beta_1^k)$; $\hat{v} \leftarrow v/(1-\beta_2^k)$ (bias correction, T61)
  $x \leftarrow x - \alpha\, \hat{m}/(\sqrt{\hat{v}} + \epsilon)$
return x

Defaults: $\alpha = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$. *RMSprop* is the same loop with $\beta_1 = 0$ (no momentum) and no bias correction: $v \leftarrow \rho v + (1-\rho) g \odot g$, $x \leftarrow x - \alpha\, g/(\sqrt{v} + \epsilon)$, defaults $\alpha = 10^{-2}$,

$\rho = 0.99$. Weight decay is added as $g \leftarrow g + \lambda x$ (Adam) or applied directly to x (AdamW). Each coordinate gets its own step $\alpha/\sqrt{\hat{v}_i}$, a diagonal rescaling estimated from the gradients; this helps when coordinates have very different scales, but Adam with constant α need not converge, and it is usually combined with a decaying schedule.

P26 Practice *added*

Adam on a badly scaled quadratic.

Setting. $f(x) = \frac{1}{2}(100x_1^2 + x_2^2)$, exact gradients (no noise), $x_0 = (1, 1)^\mathsf{T}$.

Problem.

a) Run A18 with $\alpha = 0.1$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$ for 8 steps: record g_k, $\hat{m}_k$, $\hat{v}_k$ and the step (table V27).

b) Run gradient descent with the largest safe fixed step $\alpha = 1/L = 0.01$ for the same number of steps.

c) Explain why both coordinates of Adam move identically (T61 c), and why gradient descent solves x_1 in one step but needs about 100 times more steps for x_2.

V27 Validation table *A18; solves P26*

$f(x) = \frac{1}{2}(100x_1^2 + x_2^2)$, $x_0 = (1, 1)$. Adam: both coordinates are equal at every iteration, so one value is listed for each.

k	x_k (Adam)	$f(x_k)$	$\hat{m}_k$	$\hat{v}_k$	step	x_k (GD, $\alpha = 0.01$)
0	(1, 1)	50.5	(100, 1)	(10^4, 1)	0.1	(1, 1)
1	(0.9, 0.9)	40.905	(94.737, 0.947)	(9049.5, 0.905)	0.099588	(0, 0.99)
2	(0.800412, ·)	32.353	(89.314, 0.893)	(8167.7, 0.817)	0.098826	(0, 0.9801)
3	(0.701586, ·)	24.857	(83.744, 0.837)	(7355.1, 0.736)	0.097647	(0, 0.970299)
4	(0.603939, ·)	18.419	(78.042, 0.780)	(6612.1, 0.661)	0.095975	(0, 0.960596)
5	(0.507964, ·)	13.030	(72.227, 0.722)	(5938.4, 0.594)	0.093727	(0, 0.950990)
6	(0.414236, ·)	8.665	(66.323, 0.663)	(5333.4, 0.533)	0.090816	(0, 0.941480)
7	(0.323421, ·)	5.282	(60.356, 0.604)	(4795.6, 0.480)	0.087157	(0, 0.932065)
8	(0.236264, ·)	2.819				(0, 0.922745)

GD: $f(x_8) = 0.425729$, against 2.819 for Adam, but GD's error in x_2 shrinks only by the factor 0.99 per step. Adam's steps stay close to $\alpha = 0.1$ in both coordinates while $\hat{m}$ and $\sqrt{\hat{v}}$ shrink together; it does not converge with a constant α in the long run (it oscillates around 0 with amplitude of order α).

What it shows. Adam's per-coordinate normalization removes the bad scaling (both coordinates progress at the same rate), but its step length is set by α, not by the curvature; plain gradient descent is limited by the largest curvature.

Q26 Exam problem *added, 4 pts*

Choosing the step and the batch size in SGD.

Setting. $f = \frac{1}{m}\sum_i f_i$ is μ-strongly convex with L-Lipschitz gradient; a single-sample stochastic gradient is unbiased with variance at most σ^2. SGD uses mini-batches of size b (drawn with replacement) and a constant step $\alpha \le \frac{1}{L}$.

Problem.

a) Using T59 with variance σ^2/b, show that $\mathbb{E}[f(x_k) - f^*] \le (1 - \alpha\mu)^k (f(x_0) - f^*) + \frac{\alpha L \sigma^2}{2\mu b}$.

b) Given a target accuracy ε, choose α so that the noise term is at most $\varepsilon/2$, and estimate the number of iterations k needed for the first term to be at most $\varepsilon/2$. Express the total number of gradient evaluations kb and discuss whether increasing b pays off.

c) For $\mu = 1$, $L = 10$, $\sigma^2 = 4$, $f(x_0) - f^* = 10$ and $\varepsilon = 10^{-2}$, compute α and k for $b = 1$ and for $b = 10$.